
Vector Bundles and K-Theory

— *EYNTKA* Series —

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K -theory begins with a deceptively simple idea: rather than study a space X directly, study the vector bundles that can be laid over it. A vector bundle is a continuously varying family of vector spaces parametrized by X , and the collection of all of them, taken up to isomorphism, already remembers a great deal about the shape of X . Under direct sum these bundles form a commutative monoid; completing that monoid to a group, exactly as the integers are built from the natural numbers, produces the Grothendieck group $K^0(X)$. This single construction turns the geometry of bundles into an algebraic invariant that is computable, functorial, and sensitive to phenomena that ordinary cohomology cannot detect.

What makes topological K -theory more than a convenient bookkeeping device is *Bott periodicity*: the groups $K^n(X)$ repeat with period two, and this periodicity promotes K^0 and K^1 into a full generalized cohomology theory, one satisfying every Eilenberg–Steenrod axiom except the dimension axiom ([5]). A reader who knows ordinary cohomology will find in K -theory a close structural parallel, with its own long exact sequences, products, and computational machinery, but with answers ordinary cohomology cannot see. The same circle of ideas produces the *characteristic classes* (Stiefel–Whitney, Chern, Euler, Pontryagin), cohomology classes attached to a bundle that obstruct sections, detect orientability, and measure twisting, turning qualitative geometry into numbers.

This book develops the topological theory from vector bundles through Bott periodicity to characteristic classes. It is the topological member of a family. Its algebraic counterpart, where rings replace spaces and projective modules replace bundles, is [2]; its operator-algebraic counterpart, where the Serre–Swan theorem reincarnates bundles as projections over $C(X)$, is [3]. The non-periodicity of the algebraic theory and the 2-periodicity shared by the topological and operator theories are two sides of one story, and the reader is encouraged to read across the three.

We assume the cohomology, classifying spaces, and generalized-cohomology framework of [5], and use the connections and curvature of [7], where characteristic classes are computed by the Chern–Weil theory of the Riemannian-geometry volume `NateRiemannianGeometry`. Beyond these, the prerequisites are point-set topology and linear algebra.

Vector Bundles

1.0.1 Topological Convention We are interested in topological space with “geometric” properties. Hence, we shall limit ourselves to paracompact topological space. If we are working with manifolds, they shall be Hausdorff. When working with schemes, we shall not assume Hausdorffness (for obvious reasons^a)

Many of the result covered apply to paracompact topological spaces (every open cover has a locally finite refinement). We require the paracompact condition so that many of the constructions we shall want to have will work, namely we have good behaviour of sheaves and sufficient conditions for homotopy theory and differential topology.

^asee [1] why only trivial schemes are Hausdorff

1.1 Fiber Bundles, Principal Bundles, Vector Bundles

Definition 1.1.1: Fiber Bundle

Let B be a paracompact topological space. Then a quadruple (B, E, π, F) where E is a topological space called the *total space*, $\pi : E \rightarrow B$ is a continuous surjection called the *bundle projection*, and B is called the *base space*, and F is the fiber is called *fiber bundle* if:

1. (regular fibers) each $\pi^{-1}(b)$ for $b \in B$ is isomorphic to F , or in particular

$$\pi^{-1}(b) \cong b \times F$$

2. (local trivialization) For each $b \in B$, there exists an open set $U \subseteq B$ containing b such that the following diagram commutes

$$\begin{array}{ccc} \pi^{-1}(U) & \xrightarrow{\sim} & U \times F \\ \pi \downarrow & \swarrow \pi_1 & \\ U & & \end{array}$$

When clear from context, the fiber bundle is usually denoted as E .

Definition 1.1.2: Vector Bundle

If $F = V$ is a vector space and each fiber-morphism in the local trivialization is a linear isomorphism, then is called a *vector bundle* (over B). If the vector space is real or complex, the vector bundle can be called a *real vector bundle* *complex vector bundle*.

Definition 1.1.3: Principal Bundle

If $F = G$ is a topological and $P \rightarrow X$ is a fiber bundle over X group where there is a free and transitive continuous right group action $P \times G \rightarrow P$ that preserves the fibers of P , then it is called a *principal G -bundle* (or *principal bundle* if G is clear from context).

Each fiber is called a *G -torsor* (which is a space that where the action $G \curvearrowright F_x$ has trivial stabilizer, hence “is” the group G but does not intrinsically have a notion of identity element)

Note that these conditions make each fiber homeomorphic to G . As a consequence the Orbit space P/G is homeomorphic to X .

Example 1.1.4: Fiber Bundle

1. Given a topological B and fiber F , then $E = B \times F$ is the *trivial F -fiber bundle* or simply

the *trivial bundle*. Such a bundle has a *global trivialization*, namely by letting $U = B$:

$$\begin{array}{ccc} B \times F & \xrightarrow{\text{id}} & B \times F \\ \pi \downarrow & \swarrow \pi_1 & \\ B & & \end{array}$$

2. If M is a n -manifold, then there are various interesting vector bundles defined on it:

- (a) Tangent bundle TM
- (b) cotangent bundle T^*M
- (c) differential forms $\Omega^k(M)$ for $k \leq n$

If M has more structure, ex. if is a Riemann manifold, a Khäler manifold, has a complex manifold, etc, then we can define more vector bundles that capture the extra structure (ex. Normal bundles, holomorphic bundles, spin bundles, flat bundles, etc.)

- 3. Let $p : E \rightarrow B$ be a fiber bundles and $A \subseteq B$. Then $p : p^{-1}(A) \rightarrow A$ is certainly a fiber bundle. This is called the *restriction of E over A* . The same goes through for vector-bundles.
- 4. Sub-bundle shave compliments: if $E \rightarrow X$ is a bundle (on a paracompact topological space) and F is a subbundle, then there exists a bundle $F^\perp$ such that $E = F \oplus F^\perp$.
- 5. the Tensor product is well-defined: Given a vector bundle $E \rightarrow X$ and $F \rightarrow X$, there exists a vector-bundle $E \otimes F \rightarrow X$ whose fibers over x are $E_x \otimes F_x$. From this, we get our usual more general bundles such as the *determinant bundle* $\det(E) = \Lambda^n E$ (given E is n -dimensional).
- 6. Let $E = I \times \mathbb{R} / \sim$ where $(0, t) \sim (1, -t)$. Then the projection map $I \times \mathbb{R} \rightarrow I$ produces a quotient $p : E \rightarrow S^1$ known as the Möbius bundle
- 7. Let TS^n be the tangent bundle of S^n . Then notice that this can be taken as a sub-bundle of $T\mathbb{R}^{n+1}$.
- 8. Another important bundle on S^n is the normal bundle which is a line bundle that that is where we take the one dimensional vector space panned by the line perpendicular to each point $p \in S^n$. We shall see that this bundle is isomorphic to the trivial bundle later. If NS^n is the normal bundle, we shall see that $TS^n \oplus NS^n$ is a trivial bundle.
- 9. Let $\mathbb{RP}^n$ and $\mathbb{CP}^n$ be real and complex projective space respectively. Then there is a *canonical line bundle* $p : E \rightarrow \mathbb{RP}^n$ and $p : E \rightarrow \mathbb{CP}^n$ where E is the subspace of $\mathbb{RP}^n \times \mathbb{R}^{n+1}$ consisting of pairs (ℓ, v) where $v \in \ell$ and $p(\ell, v) = \ell$. Note that the canonical line bundle over $\mathbb{RP}^n$ has an orthogonal compliment

$$E^\perp = \{(\ell, v) \in \mathbb{RP}^n \times \mathbb{R}^{n+1} \mid v \perp \ell\}$$

- 10. An important bundle for the future is the infinite real/complex projective space $\mathbb{RP}^\infty$ and $\mathbb{CP}^\infty$ taking by $\varinjlim_n \mathbb{RP}^n$ and $\varinjlim_n \mathbb{CP}^n$.
- 11. The Grassmann manifold $G_k(\mathbb{R}^n)$ was shown in [7] to be the space of all k -dimensional planes through the origin of $\mathbb{R}^n$.

Definition 1.1.5: Morphism of Fiber Bundles

If E_1, E_2 are two vector bundles, $\varphi : E_1 \rightarrow E_2$ is a vector-bundle morphism is a continuous map where this diagram commutes:

$$\begin{array}{ccc} E_1 & \xrightarrow{\varphi} & E_2 \\ & \searrow & \swarrow \\ & B & \end{array}$$

and it respects local trivialization; this includes a linear homomorphism on each fiber. An isomorphism would have φ be a homeomorphism and is a linear isomorphism on each fiber.

For each dimension n , let $\text{Vect}_k^n(B)$ be the isomorphism classes of vector bundles over B (with field k).

For a topological group G , let $b_G(B)$ be the isomorphism classes of principal G -bundles over B .

Often, if $k = \mathbb{R}$, we shall just write $\text{Vect}^n(B)$. Note that $\text{Vect}_k^n(B)$ and $b_G(B)$ are contravariant functors by applying the pullback map, for example:

$$f : X \rightarrow Y \quad \Rightarrow \quad f^* : b_G(Y) \rightarrow b_G(X)$$

Lemma 1.1.6: Morphism is Fiber-local

Let $p_1 : E_1 \rightarrow B, p_2 : E_2 \rightarrow B$ be two vector bundles over B and $\varphi : E_1 \rightarrow E_2$ a continuous. Then φ is an isomorphism if and only if it takes each fiber $p_1^{-1}(b)$ to the corresponding fiber $p_2^{-1}(b)$ via a linear isomorphism.

Proof:
exercise.

Example 1.1.7: Morphism of Bundles

We can see that the normal bundle NS^n as a subspace of $\mathbb{R}^1$ is isomorphic to $S^n \times \mathbb{R}$ via the map:

$$(x, tx) \rightarrow (x, t)$$

We shall later see that only S^1, S^3, S^7 have trivial line bundles (this shall correspond to the existence of division algebra's over $\mathbb{R}$ existing only for dimension 2, 4, 8, see [ref:HERE](#)). For S^1 , the isomorphism is simple enough as we are comfortable with complex analysis:

$$\varphi : TS^1 \rightarrow S^1 \times \mathbb{R} \quad (e^{i\theta}, ite^{i\theta}) \mapsto (e^{i\theta}, t)$$

Another important example is that the Möbius bundle is isomorphic to the canonical line bundle over $\mathbb{RP}^1$ (note that $\mathbb{RP}^1 \cong S^1$).

Definition 1.1.8: Section

Let $p : E \rightarrow B$ be a fiber bundle. Then a *section* is a continuous map $s : B \rightarrow E$ such that $p \circ s = \text{id}$.

We shall focus on sections of vector bundles. We immediately get the following

Definition 1.1.9: Zero Section

Let $p : E \rightarrow B$ be a vector bundle. Then the *zero section* is the section $z : B \rightarrow E$ given pointwise by $z(p) = 0 \in p^{-1}(b)$.

As each vector-bundle morphism on fibers is linear, the zero section is taken to the zero section. This can be taken advantage of: for example the Möbius bundle is not isomorphic to $S^1 \times \mathbb{R}$ since the compliment of the zero section is connected in the Möbius bundle but not for the product bundle. The other type of section is a *non-vanishing section* where no elements of the section are zero. Famously, by the Hairy-ball theorem (or Hedgehog theorem), there does not exist a section of TS^2 (i.e a vector-field) that is everywhere nonzero (see [5]). We shall see in [ref:HERE](#) that S^n has a nonvanishing vector field if and only if n is odd, and so we shall see that the tangent bundle of S^n cannot be isomorphic to the trivial bundle if $n > 0$ is even. The existence of zero's shall be really important, for example it shall be key in how to interpret the Euler characteristic (see [ref:HERE](#)).

Lemma 1.1.10: Characterizing Trivial Bundles using Sections

Let $p : E \rightarrow B$ be an n -dimensional vector bundle. Then p is trivial if and only if it has n sections $s_1, \dots, s_n$ such that $(s_1(b), \dots, s_n(b))$ are linearly independent on each fiber $p^{-1}(b)$.

Proof:

A trivial n -dimensional vector bundle certainly respects these properties. Conversely, given such n section $s_1, \dots, s_n$, define:

$$h : B \times \mathbb{R}^n \rightarrow E \quad h(b, t_1, \dots, t_n) \rightarrow \sum_i t_i s_i(b)$$

this is certainly a linear isomorphism on each fiber, hence by lemma 1.1.6 it is an isomorphism, completing the proof.

Note that another way of characterizing a trivial real bundle is to say there is a projection map $E \rightarrow \mathbb{R}^n$ that is a linear isomorphism on each fiber.

Using this lemma, we can show that S^3 has a trivial line bundle by relying on the Quaternions. Recall that any quaternion $z \in \mathbb{H}$ is given by:

$$z = x_1 + ix_2 + jx_3 + kx_4$$

$$i^2 = j^2 = k^2 = -1, \quad ij = k, \quad jk = i, \quad ki = j, \quad ji = -k, \quad kj = -i, \quad ik = -j$$

By identifying $\mathbb{H}$ with $\mathbb{R}^4$ via the coordinates (x_1, x_2, x_3, x_4) , then the unit sphere S^3 can be given

by the three sections of the tangent bundle:

$$\begin{array}{lll} z \mapsto iz & \text{or} & (x_1, x_2, x_3, x_4) \mapsto (-x_2, x_1, -x_4, x_3) \\ z \mapsto jz & \text{or} & (x_1, x_2, x_3, x_4) \mapsto (-x_3, x_4, x_1, -x_2) \\ z \mapsto kz & \text{or} & (x_1, x_2, x_3, x_4) \mapsto (-x_4, -x_3, x_2, x_1) \end{array}$$

The key idea is that the quaternions $1, i, j, k$ form the standard orthonormal basis for $\mathbb{R}^3$, multiplying by an arbitrary unit quaternion $z \in S^3$ gets another orthonormal basis z, iz, jz, kz , and since $|zw| = |z| \cdot |w|$ (where $|\cdot|$ is the usual norm in $\mathbb{R}^3$), we get that multiplication by a quaternion in the unit sphere S^3 is an isometry of $\mathbb{H}$. A similar process can be done for the Cayley Octonions by thinking of $\mathbb{R}^8$ as $\mathbb{H} \times \mathbb{H}$ and multiplying the Octonions by:

$$(z_1, z_2)(w_1, w_2) = (z_1 w_2 - \overline{w_2} z_2, z_2 \overline{w_1} + w_2 z_1)$$

This also defines a division algebra, and also satisfies $|zw| = |z||w|$, giving 7 orthogonal tangent vectors fields on S^7 that are nowhere vanishing, and hence S^7 is also trivial. We shall show in [ref:HERE](#) that these are the only ones.

Definition 1.1.11: Direct Sum of Vector Bundles

Let $p_1 : E_1 \rightarrow B$, $p_2 : E_2 \rightarrow B$ be vector bundles over B . Then the *direct sum* or *coproduct* (historically called *Whitney sum*) of E_1, E_2 is:

$$E_1 \oplus E_2 = \{(v_1, v_2) \in E_1 \times E_2 \mid p_1(v_1) = p_2(v_2)\}$$

Certainly the fibers of the direct sum is the direct sum of the fibers. An interesting way of thinking of the direct sum is by taking the product of the two vector bundles (including the base space)

$$E_1 \times E_2 \rightarrow B \times B$$

Then the restriction to the product to the diagonal $\{(b, b) \in B \times B\}$ is $E_1 \times E_2$. Note that just like for vector spaces, the (categorical) product and coproduct are isomorphic, however the notation $E_1 \times E_2$ is sometimes be used in literature to also mean the multiplication of the base-space, a convention we shall void.

Definition 1.1.12: Stably Trivial

Let E be a vector bundle. Then E is *stably trivial* if there exists a vector bundle F such that $E \oplus F$ is the trivial bundle.

We saw that the tangent bundle S^n is stably trivial, with NS^n being the compliment. Similarly, $E \oplus E^\perp$ where E is the canonical line bundle with its orthogonal compliment is isomorphic to $\mathbb{R}P^n \times \mathbb{R}^{n+1}$ via

$$(\ell, v, w) \mapsto (\ell, v + w)$$

In the $n = 1$ case, $E^\perp$ is isomorphic to E by rotating each vector in the plane by $\pi/2$ degrees. As mention earlier, the canonical line bundle E in the $n = 1$ case is isomorphic to the Möbius bundle. Hence, the direct sum of the Möbius bundle with itself is the trivial bundle! Take a moment to think about this one as the “untwisting” is not an isometry and will have self-intersection.

Example 1.1.13: Stable Triviality of $\mathbb{RP}^n$

By the above discussion, let $S^n \times \mathbb{R}^{n+1} \cong TS^n \oplus NS^n$. Define the equivalence (i.e. quotient by)

$$(x, v) \sim (-x, -v)$$

on both sides of the isomorphism. Then when applied to TS^n , we get $T\mathbb{RP}^n$ (see [7], or even [1]). This is not preserved by the isomorphism (general quotient usually doesn't), however we do get the closest next thing: what we will show is a special case of the *Direct Sum decomposition for compact space* that will show that $T\mathbb{RP}^n$ is a direct sum of nontrivial line bundles. We will show this by simply directly computing the two sides.

In NS^n , the identification $(x, v) \sim (-x, -v)$ is explicitly given by $(x, tx) \sim (-x, t(-x))$. This yields the product bundle $\mathbb{RP}^n \times \mathbb{R}$ as $x \mapsto (-x, -x)$ is well-defined in the quotient. Now the identification $(x, v) \sim (-x, -v)$ in $S^n \times \mathbb{R}^{n+1}$ respects the coordinate macros of $\mathbb{R}^{n+1}$, so the quotient is the direct sum of $n+1$ copies of line bundles E over $\mathbb{RP}^n$ given by $(x, t) \sim (-x, -t)$ in $S^n \times \mathbb{R}$. It is now easy to see that E is the canonical line bundle over $\mathbb{RP}^n$: identifying $S^n \times \mathbb{R}$ via the isomorphism $(x, t) \rightarrow (x, tx)$, we have $(-x, -t) \mapsto ((-x, (-t)(-x)) = (-x, tx)$ and so we have the gluing in NS^n that is the canonical line bundle over $\mathbb{RP}^n$.

Hence, the direct sum of the tangent bundle $T\mathbb{RP}^n$ with a trivial line bundle is isomorphic to $n+1$ copies of the canonical bundle over $\mathbb{RP}^n$. When $n=3$, $T\mathbb{RP}^3$ is trivial (as shown earlier), and so we get four copies of the canonical line bundle over $\mathbb{RP}^3$ is trivial (and similarly for the canonical line bundle over $\mathbb{RP}^7$). We shall fully explore these properties in section [ref:HERE](#).

As we are working with vector bundles, we can define an inner product:

$$\langle -, - \rangle_{E \oplus E} \rightarrow \mathbb{R}$$

where E is a vector bundle over B . As we are over a paracompact space, we always have such an inner product

Proposition 1.1.14: Existence of Inner Product

Let $p : E \rightarrow B$ be a (real) vector bundle (over a paracompact space). Then there exists an inner product $\langle -, - \rangle$ on E

the complex inner product (also named Hermitian inner product) can be defined similarly.

Proof:

This is the same proof as for the existence of a Riemann metric on a differentiable manifold. Let $\{U_\alpha\}$ be an open cover of B given by local trivializations. These can then pull-back the standard inner product on $\mathbb{R}^n$, and then by setting:

$$\langle v, w \rangle = \sum_{\beta} \varphi_{\beta} p(v) \langle v, w \rangle_{\alpha(\beta)}$$

where $\{\varphi_{\beta}\}$ is a partition of unity subordinate to $\{U_{\alpha}\}$ with support of φ_{β} contained in $U_{\alpha(\beta)}$.

Vector bundles are nice enough so satisfy the complementary property:

Proposition 1.1.15: Vector Bundles Satisfy Complimentary Property

Let $E \rightarrow B$ be a vector bundle (over a paracompact space B) and let $E_0 \subseteq E$ be a vector subbundle. Then there exists a vector subbundle $E_0^\perp \subseteq E$ such that

$$E_0 \oplus E_0^\perp \cong E$$

Proof:

Let E have dimension n . Choose an inner product on E . Let $E_0^\perp$ be the subspace of E which in each fiber is the orthogonal to vectors in E_0 . First, the natural projection $E_0^\perp \rightarrow B$ is a vector bundle. If this is so, then $E_0 \oplus E_0^\perp$ is isomorphic to E via the map $(v, w) \mapsto v + w$ by lemma 1.1.10.

To see that $E_0^\perp$ satisfies the local triviality condition, note that we may assume E is the product $B \times \mathbb{R}^n$ as the question is local in B . As E_0 is a vector bundle, it has some $m \leq n$ independent local sections $b \mapsto (b, s_i(b))$ near each point $b_0 \in B$. This set of m independent local sections of E_0 can be enlarged to a set of n independent local sections $b \mapsto (b, s_i(b))$ of E by choosing $s_{m+1}, \dots, s_n$ first in the fiber $p^{-1}(b_0)$, then taking the same vectors for all nearby fibers, since if $s_1, \dots, s_m, s_{m+1}, \dots, s_n$ are independent at b_0 , they will be independent near b by continuity of the determinant function. Applying the Gram-Schmidt orthogonalization process to $s_1, \dots, s_m, s_{m+1}, \dots, s_n$ in each fiber we obtain new sections s'_i . Note that each s'_i is continuous as they were obtained via the explicit formulas for the Gram-Schmidt process which is certainly a continuous process, and the first m of them are a basis for E_0 in each fiber. The sections s'_i allow us to define a local trivialization $h : p^{-1}(U) \rightarrow U \times \mathbb{R}^n$ with $h(b, s'_i(b))$ equal to the i^{th} standard basis vector of $\mathbb{R}^n$. Now, this h carries E_0 to $U \times \mathbb{R}^m$ and $E_0^\perp$ to $U \times \mathbb{R}^{n-m}$, so $h|_{E_0^\perp}$ is a local trivialization of $E_0^\perp$, as we sought to show.

When $E_0 = E$, then this proof also shows that local trivializations are also isometries. When B is compact, then we get that vector bundles are “projective”

Proposition 1.1.16: Vector Bundle Over Compact Are Projective

Let $E \rightarrow B$ be a vector bundle over a compact Hausdorff space B . Then there exists a vector bundle $E' \rightarrow B$ such that $E \oplus E'$ is the trivial bundle

Note this fails for non-compact spaces, we shall show this for $\mathbb{R}P^n$ in example 3.2.5(3).

Proof:

Suppose first this holds, so E is a subbundle of a trivial bundle $B \times \mathbb{R}^N$. Composing the inclusion of E into this product with the projection of the product onto $\mathbb{R}^N$ yields a map $E \rightarrow \mathbb{R}^N$ that is a linear injection on each fiber. We shall show such a map exists by first constructing a map $E \rightarrow \mathbb{R}^N$ that is a linear injection on each fiber, then showing that this gives an embedding of E in $B \times \mathbb{R}^N$ as a direct summand.

To that end, each point $x \in B$ has an open neighborhood U_x over which E is trivial. By Urysohn’s Lemma there is a map $\varphi_x : B \rightarrow [0, 1]$ that is nonzero at x and has support contained in U_x . Letting x vary, the sets $\varphi_x^{-1}(0, 1]$ form an open cover of B . By compactness, there exists finite subcover. Relabel the corresponding U_x ’s and φ_x ’s as U_i and φ_i . Define $g_i : E \rightarrow \mathbb{R}^n$ by

$g_i(v) = \varphi_i(p(v))(\pi_i h_i(v))$ where p is the projection $E \rightarrow B$ and $\pi_i h_i$ is the composition of a local trivialization $h_i : p^{-1}(U_i) \rightarrow U_i \times \mathbb{R}^n$ with the projection π_i to $\mathbb{R}^n$. Then g_i is a linear injection on each fiber over $\varphi_i^{-1}(0, 1]$, so making the various g_i 's the coordinates of a map $g : E \rightarrow \mathbb{R}^N$ with $\mathbb{R}^N$ a product of copies of $\mathbb{R}^n$, then g is a linear injection on each fiber.

The map g is the second coordinate of a map $f : E \rightarrow B \times \mathbb{R}^N$ with first coordinate p . The image of f is a subbundle of the product $B \times \mathbb{R}^N$ since projection of $\mathbb{R}^N$ onto the i th $\mathbb{R}^n$ factor gives the second coordinate of a local trivialization over $\varphi_i^{-1}(0, 1]$. Thus, E isomorphic to a subbundle of $B \times \mathbb{R}^N$ so by the preceding proposition there is a complementary subbundle E' with $E \oplus E'$ isomorphic to $B \times \mathbb{R}^N$, as we sought to show.

Next, we define the tensor product.

Definition 1.1.17: Tensor Product of Vector Bundles

Let $p_1 : E_1 \rightarrow B$, $p_2 : E_2 \rightarrow B$ be vector bundles over B . Then *tensor product* is a vector bundle where as the set

$$E_1 \otimes E_2 = \coprod_{x \in B} p_1^{-1}(x) \otimes p_2^{-1}(x)$$

and for the topology, choose an open cover $\{U_\alpha\}$ with local trivializations $h_i : p_i^{-1}(U_\alpha) \rightarrow U_\alpha \times \mathbb{R}^{n_i}$, then the topology $\mathcal{T}_U$ on $p_1^{-1}(U) \otimes p_2^{-1}(U)$ is given by letting the fiberwise tensor product map:

$$h_1 \otimes h_2 : p_1^{-1}(U) \otimes p_2^{-1}(U) \rightarrow U \times (\mathbb{R}^{n_1} \times \mathbb{R}^{n_2})$$

be a homeomorphism.

This topology is independent of the choice of local trivialization cover as any other choice is will be obtained by composing with isomorphism of $U \times \mathbb{R}^{n_i}$ of the form:

$$(x, v) \mapsto (x, g_i(x)(v))$$

where $g_i : U \rightarrow \text{GL}_{n_i}(\mathbb{R})$. Thus, $h_1 \otimes h_2$ changes by composing with analogous isomorphism of $U \times (\mathbb{R}^{n_1} \otimes \mathbb{R}^{n_2})$ whose second coordinate $g_1 \otimes g_2$ are continuous maps

$$U \rightarrow \text{GL}_{n_1 n_2}(\mathbb{R})$$

If the reader is familiar with the fibered product of schemes, or is comfortable with the more “categorical” approach, then we may also define the tensor product by first defining the vector-bundle as the gluing of $\coprod_\alpha (U_\alpha \times \mathbb{R}^n)$ where on intersections we have the gluing condition:

$$g_{\beta\alpha} : U_\alpha \cap U_\beta \rightarrow \text{GL}_n(\mathbb{R})$$

that satisfy the cocycle condition on $U_\alpha \cap U_\beta \cap U_\gamma$

$$g_{\gamma\beta} g_{\beta\alpha} = g_{\gamma\alpha}$$

Then the tensor product between $E_1 \rightarrow B$ and $E_2 \rightarrow B$ is given by choosing an open cover $\{U_\alpha\}$ that is locally trivial for both E_1, E_2 and getting gluing functions $g_{\beta\alpha}^i : U_\alpha \cap U_\beta \rightarrow \text{GL}_{n_i}(\mathbb{R})$ for each E_i . Then the tensor bundle $E_1 \otimes E_2$ is the tensor product of the functions $g_{\beta\alpha}^1 \otimes g_{\beta\alpha}^2$ assigning to each $x \in U_\alpha \cap U_\beta$ the tensor product of the matrices.

The key is that it is a fiberwise product. Hence, over a fixed base we get that it is commutative, associate, has an identity element (is the trivial bundle), and distributive with direct sum. If we

only look at line bundles, then the tensor product of two line bundles is a line bundle. In that case, $\text{Vect}^1(B)$ (the isomorphism classes of line bundles) is an abelian group with respect to the tensor product; the inverse is giving by inverses matrices $g_{\beta\alpha}(x) \in \text{GL}_1(\mathbb{R})$ and the cocycle condition is preserved as 1 by 1 matrices commute. Giving E an inner product, we can re-scale the local trivialization h_α to be isometries, which then the vectors in fibers of E go to vectors in $\mathbb{R}^1$ of the same length. Thus, all values of the gluing functions $g_{\beta\alpha}$ are ± 1 . Then the gluing function for $E \otimes E$ are the squares of these $g_{\beta\alpha}$, and hence are identically 1; thus every element of $\text{Vect}^1(B)$ is its own inverse. This should remind the reader of the cup product in algebraic topology, and indeed in [ref:HERE](#) we shall see that $\text{Vect}^1(B)$ is isomorphic to the cohomology group $H^1(B; \mathbb{Z})$ when B is homotopy equivalent to a CW complex.

The same idea goes through for complex vector bundles but with some different behaviour: the gluing functions $g_{\beta\alpha}$ will have norm 1 (not just ± 1 , so it is not $E \otimes E$ need not be trivial. We shall see that $\text{Vect}_{\mathbb{C}}^1(B) \cong H^2(B; \mathbb{Z})$ when B is homotopy equivalent to a CW complex. However, if we take the *conjugate bundle* $\bar{E} \rightarrow B$ defined in the natural way, then as $z\bar{z} = 1$ when $|z| = 1$, $E \otimes \bar{E}$ is the trivial line bundle.

With the tensor product, we can then also define the symmetric and alternating bundle, which also gives us the determinant bundle.

1.2 Classifying Vector Bundles

Proposition 1.2.1: Pullback of Vector bundle

Let $f : A \rightarrow B$ be a continuous map and $p : E \rightarrow B$ a vector bundle. Then there exists a vector bundle $p' : E' \rightarrow A$ with a map $f' : E' \rightarrow E$ taking the fibers of E' over each $a \in A$ isomorphically onto the fibers of E over $f(a)$. The vector bundle E' is unique up to isomorphism.

Thus, we get the pullback map $f^* : \text{Vect}(B) \rightarrow \text{Vect}(A)$ mapping isomorphism classes of E to isomorphism classes of $f^*(E) = E'$.

Proof:

Let

$$E' = \{(a, v) \in A \times E \mid f(a) = p(v)\}$$

This subspace of $A \times E$ fits into the following commutative diagram

$$\begin{array}{ccc} E' & \xrightarrow{f'} & E \\ p' \downarrow & & \downarrow p \\ A & \xrightarrow{f} & B \end{array}$$

where $p'(a, v) = a$ and $f'(a, v) = v$. Let Γ_f denote the graph of f in $A \times B$. Then p' factors as the composition $E' \rightarrow \Gamma_f \rightarrow A$, $(a, v) \mapsto (a, p(v)) = (a, f(a)) \mapsto a$. The first of these two maps is the restriction of the vector bundle $\mathbb{K} \times p : A \times E \rightarrow A \times B$ over the graph Γ_f , so it is a vector bundle, and the second map is a homeomorphism, so their composition $p' : E' \rightarrow A$ is a vector bundle. The map f' obviously takes the fiber E' over a isomorphically onto the fiber of E over $f(a)$, showing existence.

For uniqueness, construct an isomorphism from an arbitrary E' satisfying the conditions in the proposition to the particular one just constructed by sending $v' \in E'$ to the pair $(p'(v'), f'(v'))$. This map takes each fiber of E' to the corresponding fiber of $f^*(E)$ by a vector space isomorphism, so by lemma 1.1.10 it is an isomorphism of vector bundles, completing the proof.

When restricting to Vect^1 , notice the trivial bundle pulls back to the trivial bundle and the tensor product is well-behaved; hence this pullback becomes a group homomorphism.

Example 1.2.2: Vector Bundle Pullbacks

1. Let $A \subseteq B$ so that $\iota : A \hookrightarrow B$. Then the restriction of E over B to A is the pullback $\iota^*(E)$.
2. Let $f : A \rightarrow B$ be a constant map: $f(A) = b$. Then

$$f^*(E) = A \times p^{-1}(b)$$

3. The tangent bundle TS^n is the pullback of $T\mathbb{RP}^n$ via the map $S^n \rightarrow \mathbb{RP}^n$.
4. Consider the Möbius bundle $E \rightarrow S^1$, and take $f : S^1 \rightarrow S^1$ given by $z \mapsto z^2$. By thinking of the Möbius bundle as $S^1 \times \mathbb{R}$ under $(z, t) \sim (-z, -t)$, then $f^*(E)$ is the trivial bundle (think about this for awhile!). If $f(z) = z^n$, then E_n is the trivial bundle if n is even, and the Möbius bundle if n is odd.
5. Let $F(E)$ be the flag bundle of n -dimensions (rest here; [8, p.20]).

Proposition 1.2.3: Properties of Vector-bundle Pullback

Let E be a vector bundle and f, g continuous maps, everything over the appropriate domains. Then:

1. $(fg)^*(E) \cong g^*(f^*(E))$
2. $\text{id}^*(E) \cong E$
3. $f^*(E_1 \oplus E_2) \cong f^*(E_1) \oplus f^*(E_2)$
4. $f^*(E_1 \otimes E_2) \cong f^*(E_1) \otimes f^*(E_2)$

Proof:
exercise

Let us next show that isomorphism classes of vector bundles are invariant under homotopy. We first need a technical lemma:

Lemma 1.2.4: Homotopy Invariant Vector Bundle

Let $E \rightarrow X \times I$ be a vector bundle where X is paracompact. Then the restriction over $X \times \{0\}$ and $X \times \{1\}$ is isomorphic.

Proof:

First two smaller proofs:

1. A vector bundle $p : E \rightarrow X \times [a, b]$ is trivial if its restrictions over $X \times [a, c]$ and $X \times [c, b]$ are both trivial for some $c \in (a, b)$. Indeed, let these restrictions be $E_1 = p^{-1}(X \times [a, c])$ and $E_2 = p^{-1}(X \times [c, b])$, and let $h_1 : E_1 \rightarrow X \times [a, c] \times \mathbb{R}^n$ and $h_2 : E_2 \rightarrow X \times [c, b] \times \mathbb{R}^n$ be isomorphisms. These isomorphisms may not agree on $p^{-1}(X \times \{c\})$, but they can be made to agree by replacing h_2 by its composition with the isomorphism $X \times [c, b] \times \mathbb{R}^n \rightarrow X \times [c, b] \times \mathbb{R}^n$ which on each slice $X \times \{x\} \times \mathbb{R}^n$ is given by $h_1 h_2^{-1} : X \times \{c\} \times \mathbb{R}^n \rightarrow X \times \{c\} \times \mathbb{R}^n$. Once h_1 and h_2 agree on $E_1 \cap E_2$, they define a trivialization of E .
2. For a vector bundle $p : E \rightarrow X \times I$, there exists an open cover $\{U_\alpha\}$ of X where each restriction $p^{-1}(U_\alpha \times I) \rightarrow U_\alpha \times I$ is trivial. Indeed, for each $x \in X$ find open neighborhoods $U_{x,1}, \dots, U_{x,k}$ in X and a partition $0 = t_0 < t_1 < \dots < t_k = 1$ of $[0, 1]$ such that the bundle is trivial over $U_{x,i} \times [t_{i-1}, t_i]$, using compactness of $[0, 1]$. Then by (1) the bundle is trivial over $U_\alpha \times I$ where $U_\alpha = U_{x,1} \cap \dots \cap U_{x,k}$.

Now for the proposition, by (2), choose an open cover $\{U_\alpha\}$ of X so that E is trivial over each $U_\alpha \times I$. Let's assume for now the simpler case that X is compact Hausdorff. Then a finite number of U_α 's cover X . Relabel these as U_i for $i = 1, 2, \dots, m$. As shown in proposition 1.1.14, there is a corresponding partition of unity by functions φ_i with the support of φ_i contained in U_i . For $i \geq 0$, let $\psi_i = \varphi_1 + \dots + \varphi_i$, so in particular $\psi_0 = 0$ and $\psi_m = 1$. Let X_i be the graph of ψ_i , the subspace of $X \times I$ consisting of points of the form $(x, \psi_i(x))$, and let $p_i : E_i \rightarrow X_i$ be the restriction of the bundle E over X_i . Since E is trivial over $U_i \times I$, the natural projection homeomorphism $X_i \rightarrow X_{i-1}$ lifts to a homeomorphism $h_i : E_i \rightarrow E_{i-1}$ which is the identity outside $p^{-1}(U_i \times I)$ and which takes each fiber of E_i isomorphically onto the corresponding fiber of E_{i-1} . Explicitly, on points in $p^{-1}(U_i \times I) = U_i \times I \times \mathbb{R}^n$ let $h_i(x, \psi_i(x), v) = (x, \psi_{i-1}(x), v)$. The composition $h = h_1 h_2 \dots h_m$ is then an isomorphism from the restriction of E over $X \times \{1\}$ to the restriction over $X \times \{0\}$.

In the general case that X is only paracompact, there is a countable cover $\{V_i\}_{i \geq 1}$ of X and a partition of unity $\{\varphi_i\}$ with φ_i supported in V_i , such that each V_i is a disjoint union of open sets each contained in some U_α (see [7]). Then E is trivial over each $V_i \times I$. As before we let $\psi_i = \varphi_1 + \dots + \varphi_i$ and let $p_i : E_i \rightarrow X_i$ be the restriction of E over the graph of ψ_i . Also as before we construct $h_i : E_i \rightarrow E_{i-1}$ using the fact that E is trivial over $V_i \times I$. The infinite composition $h = h_1 h_2 \dots$ is then a well-defined isomorphism from the restriction of E over $X \times \{1\}$ to the restriction over $X \times \{0\}$ since near each point $x \in X$ only finitely many φ_i 's are nonzero, so there is a neighborhood of x in which all but finitely many h_i 's are the identity.

Theorem 1.2.5: Homotopy Invariance Vector Bundles

Let $p : E \rightarrow B$ a vector bundle and $f_0, f_1 : A \rightarrow B$ be homotopic maps. Then if A is paracompact,

$$f_0^*(E) \cong f_1^*(E)$$

Note this result holds for fiber bundles more generally.

Proof:

Let $F : A \times I \rightarrow B$ be a homotopy from f_0 to f_1 . Then the restriction $f^*(E)$ over $A \times \{0\}$ and $A \times \{1\}$ are $f_0^*(E)$ and $f_1^*(E)$, and we get the result by the above lemma.

Corollary 1.2.6: Homotopy For N-vector Bundles

Let $f : A \rightarrow B$ be a homotopy equivalence of paracompact spaces. Then

$$f^* : \text{Vect}^n(B) \rightarrow \text{Vect}^n(A)$$

is a bijection. In particular, vector bundles over a contractible paracompact base is trivial.

Proof:

Let g be the homotopy inverse of f . Then

$$f^*g^* = \text{id}^* = \text{id} \quad g^*f^* = \text{id}^* = \text{id}$$

completing the proof.

1.2.1 Classifying Vector Bundles on Spheres

Using the tensor product, we can construct some non-trivial bundles on the sphere S^k . The clutching function is a way of creating a bundle on the sphere S^k by gluing a bundle on the upper and lower hemisphere with a compatibility condition on the “equator subspace” (this is the “clutch”). In particular, let D_+^k, D_-^k be the upper and lower hemisphere so that $D_+^k \cap D_-^k \cong S^{k-1}$. Given a map $f : S^{k-1} \rightarrow \text{GL}_n(\mathbb{R})$, we can define:

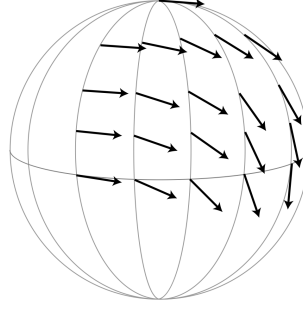
$$E_f = ((D_+^k \times \mathbb{R}^n) \sqcup (D_-^k \times \mathbb{R}^n)) / \sim$$

$$\partial D_-^k \times \mathbb{R}^n \ni (x, v) \sim (x, f(x)(v)) \in \partial D_+^k \times \mathbb{R}^n$$

Then we get a natural n -vector bundle map $f : E_f \rightarrow S^k$. Naturally, replacing $\mathbb{R}$ with $\mathbb{C}$ would require $f : S^{k-1} \rightarrow \text{GL}_n(\mathbb{C})$.

Example 1.2.7: Clutching Function

1. Let us create TS^2 using a clutching function. Let v_+ be a vector field on D_+^2 given by choosing a tangent vector at the north pole and parallel-transporting it along each meridian circle (see [7] for parallel-transports). Here is a visual on only a quarter of the upper hemisphere:

Figure 1.1: v_+ on D_+^2 (from Hatcher [8])

Reflect across the equator to create the vector field v_- on D_-^2 . Then take w_+ and w_- to be v_+ and v_- respectively each rotate by $\pi/2$ degree when viewed from outside the sphere. Then these for vector fields give a trivialization of TS^2 over $D_\pm^2$: the clutching function f read off the coordinates of v_-, w_- in the v_+, w_+ coordinate system, in particular f rotates (v_+, w_+) to (v_-, w_-) . As we go around the equator S^1 counter-clockwise, the angle will go from 0 to 4π .

2. Can you think of how do do this for $\mathbb{CP}^1 = S^2$ (hint: we should get a clutching function $f : S^1 \rightarrow \text{GL}_1(\mathbb{C})$ where $f(z) = (z^{-1})$).

The advantage of working with clutching functions is that we get that if f, g are clutching functions for E_f, E_g we have $E_f \cong E_g$ if and only if f, g are homotopic. When $k = \mathbb{C}$, we get a bijection (we shall show what we need to modify for the bijection when our $\mathbb{R}$, particularly it will have to be dealt with that $\mathbb{R} \setminus \{0\}$ is no longer connected)

Proposition 1.2.8: Clutching Function Vector Bundle Correspondence

The map:

$$\Phi : [S^{k-1}, \text{GL}_n(\mathbb{C})] \rightarrow \text{Vect}_{\mathbb{C}}^n(S^k)$$

sending homotopy classes of clutching functions $f : S^{k-1} \rightarrow \text{GL}_n(\mathbb{C})$ to complex vector bundles E_f is a bijection

Proof:

Let $[X, Y]$ denote the set of homotopy equivalences of functions. If $f \simeq g$, then there is a homotopy $F : S^{k-1} \times I \rightarrow \text{GL}_n(\mathbb{C})$. Then the same sort of clutching construction can be used to produce a vector bundle $E_F \rightarrow S^k \times I$ that restricts to E_f over $S^k \times \{-1\}$ and E_g over $S^k \times \{1\}$, hence $E_f \cong E_g$ by theorem 1.2.5. We thus have a map:

$$\Phi : [S^{k-1}, \text{GL}_n(\mathbb{C})] \rightarrow \text{Vect}_{\mathbb{C}}^n(S^k)$$

Let us find an inverse Ψ . Let $p : E \rightarrow S^k$ be an n -vector bundle, and let E_+, E_- be its restriction to D_+^k and D_-^k . $D_\pm^k$ is contractible, by theorem 1.2.5 these bundles are trivial. Choose trivializations $h_\pm : E_\pm \rightarrow D_\pm^k \times \mathbb{C}^n$. Then we get that $h_+ h_-^{-1}$ defines a map $S^{k-1} \rightarrow \text{GL}_n(\mathbb{C})$ whose homotopy class is by construction

$$\Psi(E) \in [S^{k-1}, \text{GL}_n(\mathbb{C})]$$

Let us make sure $\Psi(E)$ is well-defined, that is the trivialization is unique up to homotopy. First off, any two choices of $h_{\pm}$ differ by a map $D_{\pm}^k \rightarrow \mathrm{GL}_n(\mathbb{C})$, as $D_{\pm}^k$ is contractible, these maps are homotopic to a constant map. As $\mathrm{GL}_n(\mathbb{C})$ is path connected, elementary row operations in $\mathrm{GL}_n(\mathbb{C})$ can be realized by a path in $\mathrm{GL}_n(\mathbb{C})$ (insert a factor of $t \in [0, 1]$ in front of the scalar multiples). As $\mathbb{C}$ is algebraically closed, $M \in \mathrm{GL}_n(\mathbb{C})$ can be diagonalized. As the set of diagonal matrices in $\mathrm{GL}_n(\mathbb{C})$ is path-connected (since it is homeomorphic to the product of n copies of $\mathbb{C} - \{0\}$), we see that h_+ and h_- is unique up to homotopy. Thus $h_+ h_-^{-1} : S^{k-1} \rightarrow \mathrm{GL}_n(\mathbb{C})$ is unique up to homotopy, hence $\Psi : \mathrm{Vect}_{\mathbb{C}}^n(S^k) \rightarrow [S^{k-1}, \mathrm{GL}_n(\mathbb{C})]$ is well-defined, and by construction it is clear it is the inverse, completing the proof.

This correspondence gives some immediate results. For example, every complex vector bundles over S^1 is trivial since every class $\mathrm{Vect}_{\mathbb{C}}^n(S^1)$ corresponds to $[s^0, \mathrm{GL}_n(\mathbb{C})]$ which as a single element (as $\mathrm{GL}_n(\mathbb{C})$ is path connected). Another important result is that the canonical line bundle $H \rightarrow \mathbb{CP}^1$ satisfies the relation:

$$(H \otimes H) \oplus 1 \cong H \oplus H$$

where 1 is the trivial one-dimensional bundle (see [8, p.24]).

For the real case, we have that $\mathrm{GL}_n(\mathbb{R})$ has two homeomorphic path-connected components given by the pre-image of det. However, everything passes through if we consider *oriented (real) vector bundles* (where all the transition maps have the same sign) as we only need the fact the component is path connect:

Proposition 1.2.9: Clutching Function Vector Bundle Correspondence

The map:

$$\Phi : [S^{k-1}, \mathrm{GL}_n^+(\mathbb{R})] \rightarrow \mathrm{Vect}_+^n(S^k)$$

sending homotopy classes of clutching functions $f : S^{k-1} \rightarrow \mathrm{GL}_n^+(\mathbb{R})$ to oriented real vector bundles E_f is a bijection.

Proof:

exercise.

From this, let us figure out some information about $\mathrm{Vect}_{\mathbb{R}}^n(S^k)$ for various n, k . To understand this structure better, let us first define a map $\mathrm{Vect}_{x_0, \mathbb{R}}^n(S^k) \rightarrow \mathrm{Vect}_{\mathbb{R}}^n(S^k)$. The set $\mathrm{Vect}_{x_0, \mathbb{R}}^n(S^k)$ is all isomorphism classes with an orientation specified by the point $x_0 \in S^{k-1}$; by our above results we get that $\mathrm{Vect}_{x_0, \mathbb{R}}^n(S^k)$ is in bijective correspondence with the homotopy classes of maps $S^{k-1} \rightarrow \mathrm{GL}_n(\mathbb{R})$ taking x_0 to $\mathrm{GL}_n^+(\mathbb{R})$. Then we get the map

$$\mathrm{Vect}_{x_0, \mathbb{R}}^n(S^k) \rightarrow \mathrm{Vect}_{\mathbb{R}}^n(S^k)$$

that forgets the orientation over x_0 . This map is certainly surjective. It is 2-to-1 at all points except where a bundle has an automorphism that reverses the orientation of the fiber over x_0 ; on such points it is 1-to-1.

When $k = 1$ and $n \in \mathbb{N}_{>0}$, there is two homotopy classes $S^0 \rightarrow \mathrm{GL}_n(\mathbb{R})$: x_0 goes to $\mathrm{id} \in \mathrm{GL}_n^+(\mathbb{R})$, and then the other point goes to either the identity or the reflection. When $n = 1$, we get the trivial bundle and the Möbius bundle. When $n > 1$, we get that these bundles all have automorphisms reversing the orientations of the fibers. Hence, $\mathrm{Vect}_{\mathbb{R}}^n(S^1)$ always have two elements for each $n \geq 1$, with the nontrivial bundle being nonorientable.

When $k > 1$, then S^{k-1} is path-connected, and hence $S^{k-1} \rightarrow \mathrm{GL}_n(\mathbb{R})$ taking x_0 to $\mathrm{GL}_n^+(\mathbb{R})$ must take all of S^{k-1} to $\mathrm{GL}_n^+(\mathbb{R})$, implying that we get a bijection

$$\mathrm{Vect}_+^n(S^k) \xrightarrow{\sim} \mathrm{Vect}_{x_0, \mathbb{R}}^n(S^k)$$

and so every vector bundle over S^k is orientable and has (exactly) two different orientations. Thus, the map:

$$\mathrm{Vect}_+^n(S^k) \rightarrow \mathrm{Vect}^n(S^k)$$

is surjective, being 2-to-1 on bundles not having an orientation-reversing automorphism, and is injective otherwise.

As we are working over paracompact spaces, and have access to an inner product, we get that we can deformation-retract $\mathrm{GL}_n(\mathbb{R})$ to the orthogonal subgroup $O(n)$ where $O(n) \leq \mathrm{GL}_n(\mathbb{R})$ where:

$$O(n) = \{A \in \mathrm{GL}_n(\mathbb{R}) \mid AA^T = I\}$$

The deformation retract is easy by keeping track the Gram-Schmidt process and applying “paths” to each process (as has been done earlier). Similarly, $\mathrm{GL}_n^+(\mathbb{R})$ is a deformation retract of the special orthogonal group $SO(n)$ where

$$SO(n) = \{A \in O(n) \mid \det(A) = 1\}$$

In the complex case, we get a deformation retract onto the unitary group $U(n)$ where

$$U(n) = \{A \in \mathrm{GL}_n(\mathbb{C}) \mid AA^* = I\}$$

and can be characterized by having column forming an orthonormal basis.

Now, as we have a deformation retract $SO(n) \hookrightarrow \mathrm{GL}_n^+(\mathbb{R})$ and $U(n) \hookrightarrow \mathrm{GL}_n(\mathbb{C})$ we get natural bijections:

$$[S^{k-1}, SO(n)] \xrightarrow{\sim} [S^{k-1}, \mathrm{GL}_n^+(\mathbb{R})] \quad \text{and} \quad [S^{k-1}, U(n)] \xrightarrow{\sim} [S^{k-1}, \mathrm{GL}_n(\mathbb{C})]$$

This can greatly simplify classification efforts as $SO(n)$ is a much smaller Lie groups. For example, $SO(2)$ is a circle (orientation-preserving isometric of $\mathbb{R}^2$ are rotations). Letting $k = 2$ and $n = 2$, we get for each integer $n \in \mathbb{Z}$ a map $f_n : S^1 \rightarrow S^1$ given by $z \mapsto z^n$ which is a clutching function for an oriented 2-dimensional vector bundle $E_n \rightarrow S^2$. Note that E_2 is TS^2 and E_1 is the real vector bundle underlying the canonical complex line bundle over $\mathbb{CP}^1 = S^2$. As every arbitrary continuous map $f : S^1 \rightarrow S^1$ is homotopic to a map $z \mapsto z^n$ with a unique $n \in \mathbb{Z}$ (see [5]), we get:

$$\mathrm{Vect}_+^2(S^2) \cong \mathbb{Z}$$

Note that reversing the orientation of E_n gives E_{-n} , and hence:

$$\mathrm{Vect}_{\mathbb{R}}^2(S^2) \cong \mathbb{N}$$

As $U(1)$ is homeomorphic to S^1 , we similarly get:

$$\mathrm{Vect}_{\mathbb{C}}^2(S^2) \cong \mathbb{Z}$$

For $k > 2$, recall that any two maps $S^{k-1} \rightarrow S^1$ are homotopic if $k > 2$ (such a map lifts to a covering space $\mathbb{R}$ of S^1 and $\mathbb{R}$ is contractible, see [5]). Hence, every 2-dimensional vector bundle over S^k for $k \geq 3$ is trivial:

$$\mathrm{Vect}_{\mathbb{R}}^2(S^k) \cong \{0\} \quad k \geq 3$$

The same logic applies to complex vector bundles:

$$\text{Vect}_{\mathbb{C}}^2(S^k) \cong \{0\} \quad k \geq 3$$

We shall give some more examples after the next proposition. We first point out a couple more important facts. First, for any space X , the set $[X, SO(n)]$ and $[X, U(n)]$ has a natural group structure given by point-wise multiplication. Using this fact, we have that:

$$[S^i, SO(n)] \cong_{\text{Grp}} \pi_i(SO(n)) \quad \text{and} \quad [S^i, U(n)] \cong_{\text{Grp}} \pi(U(n))$$

where homotopies preserve base points.

Proposition 1.2.10: Vector bundles and Suspension

et SX denote the suspension of a paracompact space X . Then given a choice of base-point $x \in X$, there is a 1-1 correspondence between the set $\text{Vect}^n(SX)$ of isomorphism classes of n -dimensional (resp. real, complex, or quaternion) vector bundles over SX and the respective set of based homotopy classes of maps

$$[X, O_n]_*, \quad [X, U_n]_*, \quad \text{or} \quad [X, Sp_n]_*$$

from X to the orthogonal group O_n , unitary group U_n , or symplectic group Sp_n .

Proof:

The proof is a complicated proof, the following is a sketch given from [4]. From [5], we know that SX is the union of two contractible cones C_1 and C_2 whose intersection is X . As every vector bundle on the cones C_i is trivial as it is contractible, and every vector bundle on SX is obtained from an isomorphism of trivial bundles over X via the clutching construction (see [4]; it is the Mayer-Vietoris sequence for this suspension construction). Such an isomorphism is given by a homotopy class of maps from X to GL_n , or equivalently, to the appropriate deformation retract (O_n , U_n , or Sp_n) of GL_n . Then the indeterminacy in the resulting map from $[X, GL_n]$ to classes of vector bundles is eliminated by insisting that the base point of X map to $1 \in GL_n$.

Now, from [6] or [5], we know the computations of the homotopy groups classical Lie groups, which we now take advantage of to find the isomorphism classes of vector bundles on spheres

Example 1.2.11: Vector Bundles On Spheres

1. Let $X = S^1$, and recall that $|\pi_0(O_n)| = 2$ for real vector bundles of dimension n for all $n \geq 1$. This corresponds to the trivial bundle and the möbius bundle. The Whitney direct sum of these two shall give all non-trivial real vector bundles on S^1 .

As $|\pi_0(U_n)| = 1$ for all n , all complex vector bundles are trivial!

2. Let $X = S^2$. Then $\pi_1(O_1) = 0$, and so there is only the trivial real line bundle on S^1 . As $\pi_1(O_2) = \mathbb{Z}$, there are infinitely many 2-dimensional real line bundles of dimension 2 (indexed by the degree of the clutching map mentioned in the above proof). As $\pi_1(O_n) = \mathbb{Z}/2\mathbb{Z}$ for $n \geq 2$, there is only two real line bundles for dimension $n \geq 3$.

For complex line bundles on S^3 , we get the same phenomena as $\pi_1(U_2) = \mathbb{Z}$

3. Recall from [6] that $\pi_2(G) = 0$ for all compact Lie groups. Thus, all vector bundles on S^3 are trivial!
4. From the above, we can conclude that all (real and complex) line bundles on S^4 trivial. as:

$$\begin{array}{lll} \pi_3(O_n) = 3 & \pi_3(U_n) = \mathbb{Z} & \pi_3(Sp_n) = \mathbb{Z} \\ \text{for } n \geq 5 & \text{for } n \geq 2 & \text{for } n \geq 1 \end{array}$$

We get the corresponding infinite family of n -dimensional Real/Complex/Quaternion vector bundles.

1.2.2 Universal Bundles

In this section, we shall show that every vector bundles is the pullback of a canonical vector bundle on the Grassmannian G_n over $\mathbb{R}, \mathbb{C}$, or $\mathbb{H}$. As a quick reminder of the Grassmannian, let V be a finite dimensional vector space and let $\text{Grass}_k(V)$ be all k -dimensional subspaces of V ; i.e. the *Grassmannian manifold* of all sub-spaces of V . Let's define a topology on $\text{Grass}_n(\mathbb{R}^k)$ by first defining the *Stiefel Manifold* $V_n(\mathbb{R}^k)$ which is the space of orthonormal n -frames in $\mathbb{R}^k$. This is a subspace of the product of n copies of the unit sphere S^{k-1} , that is the subspace of orthonormal n -tuples. It is a closed subspace as orthogonality is given by an algebraic equation. Hence, $V_n(\mathbb{R}^k)$ is compact, being the product of spheres. Then there is a natural surjection $V_n(\mathbb{R}^k) \rightarrow \text{Grass}_n(\mathbb{R}^k)$ sending n -frames to the subspace it spans. This gives the quotient topology on $G_n(\mathbb{R}^k)$, and it is easy to verify it is also compact. To show its a manifold, see [7]. The *infinite Grassmannian* Grass_n is the union all $\text{Grass}_n(V)$ as V ranges over all finite dimensional subspace of a fixed infinite dimensional vector space (usually $\mathbb{R}^\infty$, $\mathbb{C}^\infty$, or $\mathbb{H}^\infty$). With a bit of work, we see that it is an infinite-dimensional CW complex (it is the limit of CW complexes respecting the appropriate directed mappings, see 1.2.3). These are rather natural, for example if taking $n = 1$, we get that Grass_1 is $\mathbb{R}P^\infty$, $\mathbb{C}P^\infty$, or $\mathbb{H}P^\infty$ respectively.

Just like for $\mathbb{R}P^n$, $\mathbb{C}P^n$, and $\mathbb{H}P^n$, there a canonical line bundle for Grassmannian: let:

$$E_n(\mathbb{R}^k) = \{(\ell, v) \in \text{Grass}_n(\mathbb{R}^k) \times \mathbb{R}^k \mid v \in \ell\}$$

and similarly for $\mathbb{C}^k, \mathbb{H}^k$. Then the inclusion $\mathbb{R}^k \subseteq \mathbb{R}^{k+1} \subseteq \dots$ induces inclusions $E_n(\mathbb{R}^k) \subseteq E_n(\mathbb{R}^{k+1}) \subseteq \dots$, and hence

$$E_n(\mathbb{R}^\infty) = \cup_k E_n(\mathbb{R}^k)$$

where it is given the colimit topology, and similarly for the complex and quaternion case. Then:

Proposition 1.2.12: Grassmannian Bundle is Vector Bundle

The projection $p : E_n(\mathbb{R}^k) \rightarrow \text{Grass}_n(\mathbb{R}^k)$ where $p(\ell, v) = \ell$ is a vector bundle for finite and infinite k .

Proof:

By proving the finite case, the infinite case follows by local trivializations working over every $k \in \mathbb{N}$, hence we prove the finite case. For $\ell \in G_n(\mathbb{R}^k)$, let $\pi_\ell : \mathbb{R}^k \rightarrow \ell$ be orthogonal projection and let

$$U_\ell = \{\ell' \in G_n(\mathbb{R}^k) \mid \pi_\ell(\ell') \text{ has dimension } n\}.$$

so that $\ell \in U_\ell$. We will show that U_ℓ is open in $G_n(\mathbb{R}^k)$ and that the map $h : p^{-1}(U_\ell) \rightarrow U_\ell \times \ell \cong U_\ell \times \mathbb{R}^n$ defined by $h(\ell', v) = (\ell', \pi_\ell(v))$ is a local trivialization of $E_n(\mathbb{R}^k)$.

For U_ℓ to be open is equivalent to its preimage in $V_n(\mathbb{R}^k)$ being open. This preimage consists of orthonormal frames v_i , $1 \leq i \leq n$, such that $\pi_\ell(v_1), \dots, \pi_\ell(v_n)$ are independent. Let A be the matrix of π_ℓ with respect to the standard basis in the domain $\mathbb{R}^k$ and any fixed basis in the range ℓ . The condition on v_i is then that the $n \times n$ matrix with columns Av_i has nonzero determinant. Since the value of this determinant is a continuous function of v_i , it follows that the frames v_i yielding a nonzero determinant form an open set in $V_n(\mathbb{R}^k)$.

Then it's clear that h is a bijection which is a linear isomorphism on each fiber. Furthermore h is continuous since its two coordinate functions are continuous. To see that h^{-1} is continuous, for $\ell' \in U_\ell$ there is a unique invertible linear map $L_{\ell'} : \mathbb{R}^k \rightarrow \mathbb{R}^k$ restricting to π_ℓ on ℓ' and the identity on $\ell^\perp = \text{Ker } \pi_\ell$. Then $L_{\ell'}$, regarded as a $k \times k$ matrix, depends continuously on ℓ' . Indeed, we can write L_ℓ as a product AB^{-1} where:

- B sends the standard basis to $v_1, \dots, v_n, v_{n+1}, \dots, v_k$ with $v_1, \dots, v_n$ an orthonormal basis for ℓ' and $v_{n+1}, \dots, v_k$ a fixed basis for $\ell^\perp$.
- A sends the standard basis to $\pi_\ell(v_1), \dots, \pi_\ell(v_n), v_{n+1}, \dots, v_k$.

Both A and B depend continuously on $v_1, \dots, v_n$. Since matrix multiplication and matrix inversion are continuous operations, it follows that the product $L_{\ell'} = AB^{-1}$ depends continuously on $v_1, \dots, v_n$. But since $L_{\ell'}$ depends only on ℓ' , not on the basis $v_1, \dots, v_n$ for ℓ' , it follows that $L_{\ell'}$ depends continuously on ℓ' since $G_n(\mathbb{R}^k)$ has the quotient topology from $V_n(\mathbb{R}^k)$. We have $h^{-1}(\ell', w) = (\ell', L_{\ell'}^{-1}(w))$ with $L_{\ell'}^{-1}$ depending continuously on ℓ' since matrix inversion is continuous, so h^{-1} is continuous, and so we have a local trivialization, as we sought to show.

Theorem 1.2.13: Classifying Vector Bundles

Let X be a paracompact space. Then $\text{Vect}^n(X)$ is in correspondence with the set of homotopy classes of maps $X \rightarrow \text{Grass}_n$, denoted $[X, \text{Grass}_n]$. In other words:

$$\text{Vect}^n(X) \cong [X, \text{Grass}_n]$$

In particular, if $E \rightarrow X$ is an n -dimensional vector bundle, it is isomorphic to $f^*(E_n)$ for some $f : X \rightarrow \text{Grass}_n$ where E determines f up to homotopy.

Proof:

We shall prove it for the real case, the complex, quaternion, and orientable real case follow similarly. To shorten notation, let G_n denote the real Grassmannian for n -dimensional subspaces.

The key observation is the following: For an n -dimensional vector bundle $p : E \rightarrow X$, an isomorphism $E \cong f^*(E_n)$ is equivalent to a map $g : E \rightarrow \mathbb{R}^\infty$ that is a linear injection on each fiber. To see this, suppose that we have a map $f : X \rightarrow G_n$ and an isomorphism $E \cong f^*(E_n)$.

Then we have a commutative diagram

$$\begin{array}{ccccccc}
 E & \xrightarrow{\cong} & f^*(E_n) & \xrightarrow{\tilde{f}} & E_n & \xrightarrow{\pi} & \mathbb{R}^\infty \\
 & \searrow p & \downarrow & & \downarrow & & \\
 & & X & \xrightarrow{f} & G_n & &
 \end{array}$$

where $\pi(\ell, v) = v$. The composition across the top row is a map $g : E \rightarrow \mathbb{R}^\infty$ that is a linear injection on each fiber, since both $\tilde{f}$ and π have this property. Conversely, given a map $g : E \rightarrow \mathbb{R}^\infty$ that is a linear injection on each fiber, define $f : X \rightarrow G_n$ by letting $f(x)$ be the n -plane $g(p^{-1}(x))$. This clearly yields the above commutative diagram.

To show surjectivity of the map $[X, G_n] \rightarrow \text{Vect}^n(X)$, suppose $p : E \rightarrow X$ is an n -dimensional vector bundle. Let $\{U_\alpha\}$ be an open cover of X such that E is trivial over each U_α . As X is paracompact, there a countable open cover $\{U_i\}$ of X such that E is trivial over each U_i , and there is a partition of unity $\{\varphi_i\}$ with φ_i supported in U_i . Let $g_i : p^{-1}(U_i) \rightarrow \mathbb{R}^n$ be the composition of a trivialization $p^{-1}(U_i) \rightarrow U_i \times \mathbb{R}^n$ with projection onto $\mathbb{R}^n$. The map $(\varphi_i p)g_i$, $v \mapsto \varphi_i(p(v))g_i(v)$, extends to a map $E \rightarrow \mathbb{R}^n$ that is zero outside $p^{-1}(U_i)$. Near each point of X only finitely many φ_i 's are nonzero, and at least one φ_i is nonzero, so these extended $(\varphi_i p)g_i$'s are the coordinates of a map $g : E \rightarrow (\mathbb{R}^n)^\infty = \mathbb{R}^\infty$ that is a linear injection on each fiber.

For injectivity, if we have isomorphisms $E \cong f_0^*(E_n)$ and $E \cong f_1^*(E_n)$ for two maps $f_0, f_1 : X \rightarrow G_n$, then these give maps $g_0, g_1 : E \rightarrow \mathbb{R}^\infty$ that are linear injections on fibers, as in the first paragraph of the proof. Then g_0 and g_1 are homotopic through maps g_t that are linear injections on fibers. If this is so, then f_0 and f_1 will be homotopic via $f_t(x) = g_t(p^{-1}(x))$.

To show this, a homotopy g_t is constructed. The first step is to compose g_0 with the homotopy $L_t : \mathbb{R}^\infty \rightarrow \mathbb{R}^\infty$ defined by $L_t(x_1, x_2, \dots) = (1-t)(x_1, x_2, \dots) + t(x_1, 0, x_2, 0, \dots)$. For each t this is a linear map whose kernel is easily computed to be 0, so L_t is injective. Composing the homotopy L_t with g_0 moves the image of g_0 into the odd-numbered coordinates. Similarly we can homotope g_1 into the even-numbered coordinates. Still calling the new g 's g_0 and g_1 , let $g_t = (1-t)g_0 + tg_1$. This is linear and injective on fibers for each t since g_0 and g_1 are linear and injective on fibers.

Definition 1.2.14: Universal Bundle

The Grassmannians Grass_n are called the *universal bundles*; for $\mathbb{R}, \mathbb{C}, \mathbb{H}$ they are often denoted BO_n, BU_n , and BSp_n respectively.

As we saw we can put a natural structure group on vector bundles (i.e. O_n, U_n or Sp_n depending on whether the vector bundle is a real, complex, or quaternion bundle respectively). Hence, the infinite Grassmannian is called “classifying space” of O_n, U_n or Sp_n . For this reason, the infinite Grassmannian is denoted BO_n, BU_n , or BSp_n respectively.

1.2.3 *Cell-Structure on Grassmannian

HERE

1.3 Bott Periodicity

The classification of theorem 1.2.13 reduces vector bundles to homotopy classes of maps into the classifying spaces BO_n , BU_n , BSp_n and, in the limit, into BO , BU , BSp . The homotopy of these spaces is governed by the stable groups $O = \varinjlim_n O(n)$, $U = \varinjlim_n U(n)$, $Sp = \varinjlim_n Sp(n)$: the fibrations

$$O(n-1) \hookrightarrow O(n) \rightarrow S^{n-1}, \quad U(n-1) \hookrightarrow U(n) \rightarrow S^{2n-1}$$

make the inclusions $\pi_k(O(n-1)) \rightarrow \pi_k(O(n))$ and $\pi_k(U(n-1)) \rightarrow \pi_k(U(n))$ isomorphisms in a stable range. The resulting stable homotopy groups obey one of the most influential periodicities in topology.

Theorem 1.3.1: Bott Periodicity

The stable homotopy groups of the unitary and orthogonal groups are periodic: $\pi_k(U)$ depends only on $k \bmod 2$ and $\pi_k(O)$ only on $k \bmod 8$, according to

$k \bmod 8$	0	1	2	3	4	5	6	7
$\pi_k(U)$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$	0	$\mathbb{Z}$
$\pi_k(O)$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	0	$\mathbb{Z}$	0	0	0	$\mathbb{Z}$

Equivalently there are homotopy equivalences

$$\mathbb{Z} \times BU \simeq \Omega^2(\mathbb{Z} \times BU), \quad \mathbb{Z} \times BO \simeq \Omega^8(\mathbb{Z} \times BO)$$

so that $\pi_k(U) \cong \pi_{k+2}(U)$ and $\pi_k(O) \cong \pi_{k+8}(O)$.

The two formulations are interchangeable through $\pi_k(U) \cong \pi_{k+1}(BU)$ and the recognition of $\mathbb{Z} \times BU$ as the representing space of complex K -theory developed in chapter 2: the loop-space periodicity $\Omega^2(\mathbb{Z} \times BU) \simeq \mathbb{Z} \times BU$ is the engine that turns K -theory into a 2-periodic cohomology theory.

Proof Sketch:

The full proof is beyond this book's scope and is given completely in [8] (and, via Morse theory, in Milnor's *Morse Theory*); what follows is the idea, with the deferral being to those complete published proofs rather than to an unfilled gap. Bott studies the loop space of a compact symmetric space M (a compact Riemannian manifold carrying a geodesic-reversing isometry at each point) by Morse theory for the energy functional, whose critical points are the geodesics joining two fixed points. An index comparison gives $\pi_k(M^\nu) \cong \pi_{k+1}(M)$ for the spaces M^ν of minimal geodesics, and iterating through the chain of symmetric spaces U , O , U/O , Sp/U , $Sp, \dots$ yields the displayed periodicities, for instance $\pi_k(U) \cong \pi_{k+1}(BU) \cong \pi_{k+2}(U)$, completing the sketch.

The classifying spaces also compute K -theory directly. Write $\tilde{K}(X)$ for the reduced group developed in chapter 2 (the kernel of the rank map); it is captured by maps into the stable classifying space, in each field at once.

Theorem 1.3.2: Stabilization Theorem of Classifying Spaces

Let X be a compact space or a finite-dimensional connected CW complex. Then

$$\tilde{K}(X) \cong \varinjlim_n \mathbf{VB}_n(X) \cong \varinjlim_n [X, \text{Grass}_n]$$

and in particular $\widetilde{KO}(X) \cong \varinjlim_n [X, BO_n]$, $\widetilde{KU}(X) \cong \varinjlim_n [X, BU_n]$, and $\widetilde{KSp}(X) \cong \varinjlim_n [X, BSp_n]$.

Proof:

Over a compact (or finite-dimensional connected CW) base the trivial bundles are cofinal in $\mathbf{VB}(X)$: every bundle E is a summand of a trivial bundle (proposition 1.1.16), so $E \oplus E' \cong T^N$ for some complement E' and some N . Cofinality lets every reduced class be written in the normal form $[E] - n$, and $[E] - n = [F] - n$ holds in $\tilde{K}(X)$ exactly when $E \oplus T^k \cong F \oplus T^k$ for some k , that is, when E and F are stably isomorphic. Hence $\tilde{K}(X) \cong \varinjlim_n \mathbf{VB}_n(X)$, the colimit taken along the stabilization $E \mapsto E \oplus T^1$. The classification of rank- n bundles by homotopy classes into the Grassmannian (theorem 1.2.13) identifies $\mathbf{VB}_n(X) \cong [X, \text{Grass}_n]$ compatibly with stabilization, giving $\varinjlim_n [X, \text{Grass}_n]$; specializing the field to $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$ gives BO_n , BU_n , BSp_n , completing the proof.

The stabilization theorem is what makes K -theory representable: the colimit over n replaces the unstable Grassmannians by the single space BO , BU , or BSp , and chapter 2 restores the rank component to obtain $\tilde{K}(X) \cong [X, \mathbb{Z} \times BU]$.

K-Theory for Vector Bundles

The previous chapter solved the classification problem for vector bundles over a paracompact base: by theorem 1.2.13, the isomorphism classes of n -dimensional bundles over X are the homotopy classes $[X, \text{Grass}_n]$, and by theorem 1.2.5 this set is a homotopy invariant of X . Yet the collection of all vector bundles over X , taken together under the direct sum of definition 1.1.11, carries more structure than a bare set of homotopy classes: it is a commutative monoid. The single defect of this monoid is that it has no inverses, since a nonzero bundle can never sum to the zero bundle. K-theory is what results from repairing exactly this defect. One formally adjoins inverses to the monoid of vector bundles, in the same way that the integers are built from the natural numbers, and the resulting group $K^0(X)$ becomes a powerful and computable invariant of the space.

This chapter develops topological K-theory from that single idea. The group $K^0(X)$ is introduced here as the group completion of the monoid of bundles, with its reduced companion $\tilde{K}^0(X)$ recording bundles up to the addition of trivial summands. The stabilization theorem of the previous chapter (theorem 1.3.2) already hints that this group is representable: stable equivalence classes of bundles are homotopy classes of maps into the infinite Grassmannian. Making this precise identifies $K^0(X)$ with the homotopy functor $[X, \mathbb{Z} \times BU]$ in the complex case, and the tensor product of definition 1.1.17 upgrades the group to a ring. The chapter then turns these structural results into a genuine cohomology theory: Bott periodicity (theorem 1.3.1) supplies the periodicity isomorphism that lets one define negative-degree groups, and the resulting theory satisfies the generalized cohomology axioms. We close with explicit computations, including the K-theory of spheres and projective spaces.

2.1 The Group $K^0(X)$

Throughout this section X denotes a compact Hausdorff space, and $\text{Vect}(X)$ denotes the set of isomorphism classes of (finite-dimensional, complex) vector bundles over X . The direct sum of definition 1.1.11 descends to isomorphism classes: if $E \cong E'$ and $F \cong F'$, then $E \oplus F \cong E' \oplus F'$, so $\oplus$

is a well-defined binary operation on $\text{Vect}(X)$. It is commutative and associative up to isomorphism, and the zero bundle (the rank-0 bundle, whose total space is X itself) is a two-sided identity. Thus $(\text{Vect}(X), \oplus)$ is a commutative monoid. The obstruction to it being a group is total: the only invertible element is 0, because the rank of a direct sum is the sum of the ranks, and ranks are non-negative integers. The construction that follows forces inverses to exist by a purely formal device, exactly the device that produces $\mathbb{Z}$ from $\mathbb{N}$.

Definition 2.1.1: The Grothendieck Group $K^0(X)$

Let X be a compact Hausdorff space and let $(\text{Vect}(X), \oplus)$ be the commutative monoid of isomorphism classes of complex vector bundles over X . The *Grothendieck group* (or *topological K-theory group*) $K^0(X)$ is the abelian group with one generator $[E]$ for each isomorphism class of bundle E and one relation

$$[E \oplus F] = [E] + [F]$$

for each pair of bundles E, F . Equivalently, $K^0(X)$ is the quotient of the free abelian group on isomorphism classes of bundles by the subgroup generated by all elements $[E \oplus F] - [E] - [F]$.

The elements of $K^0(X)$ are *formal differences* $[E] - [F]$ of bundles, called *virtual bundles*, and two such differences $[E] - [F]$ and $[E'] - [F']$ are equal precisely when $E \oplus F' \oplus G \cong E' \oplus F \oplus G$ for some bundle G . The auxiliary bundle G is unavoidable: it encodes the equivalence relation built into the Grothendieck construction, where one identifies pairs that become equal only after adding a common summand. The class $[E]$ of an honest bundle is the image of the monoid element E under the canonical map $\text{Vect}(X) \rightarrow K^0(X)$, and every element of $K^0(X)$ is a difference of two such classes. The universal property that pins down $K^0(X)$ is that any monoid homomorphism from $\text{Vect}(X)$ into an abelian group factors uniquely through $K^0(X)$; this is what makes K-theory the receptacle for every additive invariant of vector bundles.

Example 2.1.2: K^0 of a Point

Let $X = \{*\}$ be the one-point space. A vector bundle over a point is the same data as its single fiber, a finite-dimensional complex vector space, so isomorphism classes of bundles are classified by dimension. Hence $\text{Vect}(\{*\}) \cong \mathbb{N}$ as monoids, with $\oplus$ corresponding to addition of dimensions and $[\mathbb{C}^n]$ corresponding to n . The Grothendieck group of $(\mathbb{N}, +)$ is $\mathbb{Z}$: the formal differences $m - n$ of natural numbers are exactly the integers, with $m - n = m' - n'$ in $\mathbb{Z}$ iff $m + n' = m' + n$, which is the Grothendieck relation with trivial auxiliary term. Therefore

$$K^0(\{*\}) \cong \mathbb{Z}$$

with the class $[\mathbb{C}^n] - [\mathbb{C}^m]$ corresponding to the integer $n - m$. The isomorphism sends a virtual bundle to its rank, the difference of the dimensions of its positive and negative parts.

The rank computation in this example is the universal one. Over a connected base, every bundle has a well-defined rank (a single non-negative integer), and rank is additive under direct sum, so it defines a monoid homomorphism $\text{Vect}(X) \rightarrow \mathbb{N} \hookrightarrow \mathbb{Z}$. By the universal property this extends to a group homomorphism, which is the central tool for splitting off the trivial part of $K^0(X)$.

Proposition 2.1.3: The Rank Homomorphism

Let X be a compact Hausdorff space with connected components $X_1, \dots, X_r$. Assigning to a bundle its rank on each component defines a surjective group homomorphism

$$\mathrm{rk}: K^0(X) \longrightarrow \mathbb{Z}^r, \quad [E] \longmapsto (\mathrm{rk}E|_{X_1}, \dots, \mathrm{rk}E|_{X_r})$$

which is split by the homomorphism $\mathbb{Z}^r \rightarrow K^0(X)$ sending the i -th standard generator to the class of the line bundle that is trivial of rank 1 on X_i and rank 0 elsewhere. In particular, when X is connected, there is a split surjection $\mathrm{rk}: K^0(X) \rightarrow \mathbb{Z}$.

Proof:

On each component X_i the rank of a bundle is a single non-negative integer, because the rank function $X_i \rightarrow \mathbb{N}$ is locally constant by the local triviality of definition 1.1.2 and X_i is connected. Hence $E \mapsto (\mathrm{rk}E|_{X_1}, \dots, \mathrm{rk}E|_{X_r})$ is a well-defined map $\mathrm{Vect}(X) \rightarrow \mathbb{N}^r$, and it is a monoid homomorphism since rank is additive under $\oplus$. By the universal property of definition 2.1.1 it extends uniquely to a group homomorphism $\mathrm{rk}: K^0(X) \rightarrow \mathbb{Z}^r$.

For the splitting, let T_i denote the bundle that restricts to the trivial line bundle on X_i and to the rank-0 bundle on the remaining components; this is a genuine bundle over X since the components are open and closed. Then $\mathrm{rk}[T_i]$ is the i -th standard generator $e_i \in \mathbb{Z}^r$, so the homomorphism $s: \mathbb{Z}^r \rightarrow K^0(X)$ defined on generators by $s(e_i) = [T_i]$ satisfies $\mathrm{rk} \circ s = \mathrm{id}$. This exhibits rk as a split surjection, and surjectivity is immediate from the existence of the splitting, completing the proof.

The kernel of the rank homomorphism is the part of $K^0(X)$ that sees genuine twisting rather than the dimension count. Isolating it gives the reduced theory. For the rest of the section it is convenient to assume X is connected so that rank is a single integer; the general statement over r components differs only by replacing $\mathbb{Z}$ with $\mathbb{Z}^r$.

Definition 2.1.4: Reduced K-Theory $\tilde{K}^0(X)$

Let X be a connected compact Hausdorff space. The *reduced K-theory* of X is the kernel of the rank homomorphism,

$$\tilde{K}^0(X) = \ker(\mathrm{rk}: K^0(X) \rightarrow \mathbb{Z})$$

the subgroup of virtual bundles of rank zero.

Reduced K-theory discards the information that any space, however trivial, already carries: the integer recording the dimension of a bundle. What remains in $\tilde{K}^0(X)$ is the obstruction to a virtual bundle being a difference of trivial bundles. The splitting of proposition 2.1.3 shows that this discarding loses nothing beyond the dimension count, since the full group decomposes as the reduced group together with a copy of the integers.

Corollary 2.1.5: The Splitting $K^0 = \tilde{K}^0 \oplus \mathbb{Z}$

For X a connected compact Hausdorff space there is a natural isomorphism

$$K^0(X) \cong \tilde{K}^0(X) \oplus \mathbb{Z}$$

where the $\mathbb{Z}$ summand is generated by the class $[\mathbb{C}]$ of the trivial line bundle and the projection to $\mathbb{Z}$ is the rank.

Proof:

The rank homomorphism $\text{rk}: K^0(X) \rightarrow \mathbb{Z}$ is split surjective by proposition 2.1.3, with splitting $s: \mathbb{Z} \rightarrow K^0(X)$, $s(1) = [\mathbb{C}]$. For a split short exact sequence of abelian groups

$$0 \rightarrow \tilde{K}^0(X) \rightarrow K^0(X) \xrightarrow{\text{rk}} \mathbb{Z} \rightarrow 0$$

the splitting lemma gives $K^0(X) \cong \ker(\text{rk}) \oplus \mathbb{Z} = \tilde{K}^0(X) \oplus \mathbb{Z}$, the isomorphism being $\xi \mapsto (\xi - s(\text{rk } \xi), \text{rk } \xi)$ with inverse $(\eta, n) \mapsto \eta + n[\mathbb{C}]$. Naturality in X holds because rank, the trivial bundle, and the splitting are all preserved by pullback along continuous maps, completing the proof.

To make $\tilde{K}^0(X)$ usable one needs a concrete description of its elements, and this is where the compactness of X enters decisively. The Grothendieck construction in general produces formal differences $[E] - [F]$ with two genuine bundles in play, and the equivalence relation involves an unknown auxiliary summand G . Over a compact base both of these complications collapse. The mechanism is that trivial bundles are *cofinal* in $\text{Vect}(X)$: every bundle embeds in a trivial one as a direct summand, by proposition 1.1.16. The next two results extract from cofinality first a normal form for elements of $K^0(X)$ and then a clean criterion for two bundles to be equal in the reduced group.

Lemma 2.1.6: Cofinality and the Normal Form $[E] - n$

Let X be a compact Hausdorff space and write $\underline{\mathbb{C}}^n$ for the trivial bundle of rank n .

1. For every bundle F over X there is a bundle F' and an integer n with $F \oplus F' \cong \underline{\mathbb{C}}^n$.
2. Every element of $K^0(X)$ can be written in the form $[E] - n := [E] - [\underline{\mathbb{C}}^n]$ for some bundle E and some $n \geq 0$.

Proof:

Part (1): cofinality. By proposition 1.1.16, for the bundle F over the compact Hausdorff space X there is a complementary bundle F' with $F \oplus F'$ isomorphic to a trivial bundle, say $F \oplus F' \cong \underline{\mathbb{C}}^n$ where $n = \text{rk } F + \text{rk } F'$. This is the cofinality statement: every bundle is a summand of a trivial bundle.

Part (2): normal form. A general element of $K^0(X)$ is a formal difference $[E_0] - [F]$ of two bundle classes, since by definition 2.1.1 the group is generated by the classes $[E]$ under addition and subtraction, and any such combination collapses to a single difference using $[E] + [E'] = [E \oplus E']$.

Apply part (1) to F : choose F' with $F \oplus F' \cong \mathbb{C}^n$. Then in $K^0(X)$,

$$[E_0] - [F] = [E_0] - [F] + ([F'] - [F']) = [E_0] + [F'] - ([F] + [F']) = [E_0 \oplus F'] - [\mathbb{C}^n]$$

using $[F] + [F'] = [F \oplus F'] = [\mathbb{C}^n]$. Setting $E = E_0 \oplus F'$ gives the normal form $[E_0] - [F] = [E] - n$, completing the proof.

The normal form $[E] - n$ replaces a formal difference of two unknown bundles by a single honest bundle minus a trivial summand, which is a substantial simplification: it says every virtual bundle is, up to a shift by trivial bundles, an actual bundle. The same cofinality argument now eliminates the auxiliary summand G from the equivalence relation, pinning down exactly when two normal forms agree. This is the telescoping argument, and it is the heart of the reduced theory.

Definition 2.1.7: Stable Isomorphism and Stable Equivalence

Two vector bundles E and F over X are *stably isomorphic*, written $E \cong_s F$, if there is an integer $k \geq 0$ with

$$E \oplus \mathbb{C}^k \cong F \oplus \mathbb{C}^k$$

(the *same* trivial summand on each side). They are *stably equivalent*, written $E \approx_s F$, if there exist integers $m, n \geq 0$ with

$$E \oplus \mathbb{C}^m \cong F \oplus \mathbb{C}^n$$

(possibly *different* trivial summands). A bundle is *stably trivial* (in the sense of definition 1.1.12) precisely when it is stably equivalent to a trivial bundle.

The two notions differ by exactly the rank: stable isomorphism preserves rank ($\text{rk} E = \text{rk} F$, since adding $\mathbb{C}^k$ to both sides forces equal ranks), whereas stable equivalence allows the ranks to differ. Stable isomorphism is the relation that equality in $K^0(X)$ detects, and stable equivalence is the coarser relation that equality in $\tilde{K}^0(X)$ detects. The next theorem proves both statements, with the cofinality lemma supplying the telescoping that turns the abstract Grothendieck congruence into these concrete bundle isomorphisms.

Theorem 2.1.8: Reduced K-Theory Classifies Stable Equivalence

Let X be a connected compact Hausdorff space, and let E, F be vector bundles over X .

1. In $K^0(X)$ one has $[E] = [F]$ if and only if E and F are stably isomorphic, $E \cong_s F$.
2. Sending a bundle E to the class $[E] - \text{rk} E \in \tilde{K}^0(X)$ induces a bijection between stable equivalence classes of bundles and elements of $\tilde{K}^0(X)$. Under this bijection $\tilde{K}^0(X)$ is the set of stable equivalence classes, made into a group by direct sum.

Proof:

Part (1). Suppose first that $E \cong_s F$, so $E \oplus \mathbb{C}^k \cong F \oplus \mathbb{C}^k$ for some k . Passing to classes in $K^0(X)$ and using definition 2.1.1,

$$[E] + k[\mathbb{C}] = [E \oplus \mathbb{C}^k] = [F \oplus \mathbb{C}^k] = [F] + k[\mathbb{C}]$$

Cancelling $k[\mathbb{C}]$ in the abelian group $K^0(X)$ gives $[E] = [F]$.

For the converse, suppose $[E] = [F]$ in $K^0(X)$. By construction of the Grothendieck group, the relation $[E] = [F]$ holds exactly when E and F become isomorphic after adding a common bundle G , that is $E \oplus G \cong F \oplus G$ for some bundle G . This is the defining congruence of definition 2.1.1: equality of $[E]$ and $[F]$ in the group completion of the monoid $\text{Vect}(X)$ means precisely that there exists G with $E \oplus G \cong F \oplus G$. Now apply cofinality (lemma 2.1.6(1)) to G : choose G' with $G \oplus G' \cong \underline{\mathbb{C}}^k$. Adding G' to both sides of $E \oplus G \cong F \oplus G$,

$$E \oplus G \oplus G' \cong F \oplus G \oplus G' \implies E \oplus \underline{\mathbb{C}}^k \cong F \oplus \underline{\mathbb{C}}^k$$

This is exactly stable isomorphism, $E \cong_s F$. The telescoping step (replacing the unknown summand G by a trivial summand $\underline{\mathbb{C}}^k$) is what makes the equivalence concrete.

Part (2). Define Φ on bundles by $\Phi(E) = [E] - \text{rk} E \cdot [\underline{\mathbb{C}}] \in K^0(X)$. Since $\text{rk}([E] - \text{rk} E [\underline{\mathbb{C}}]) = \text{rk} E - \text{rk} E = 0$, the image lies in $\tilde{K}^0(X)$. First, $\Phi(E) = \Phi(F)$ if and only if $E \cong_s F$. Indeed,

$$\Phi(E) = \Phi(F) \iff [E] + \text{rk} F [\underline{\mathbb{C}}] = [F] + \text{rk} E [\underline{\mathbb{C}}] \iff [E \oplus \underline{\mathbb{C}}^{\text{rk} F}] = [F \oplus \underline{\mathbb{C}}^{\text{rk} E}]$$

and by part (1) the last equality means $E \oplus \underline{\mathbb{C}}^{\text{rk} F} \cong_s F \oplus \underline{\mathbb{C}}^{\text{rk} E}$, that is $E \oplus \underline{\mathbb{C}}^{\text{rk} F + k} \cong F \oplus \underline{\mathbb{C}}^{\text{rk} E + k}$ for some k ; setting $m = \text{rk} F + k$ and $n = \text{rk} E + k$, this is precisely $E \cong_s F$. Conversely any stable equivalence $E \oplus \underline{\mathbb{C}}^m \cong F \oplus \underline{\mathbb{C}}^n$ runs the chain backwards, since it gives $[E] + m[\underline{\mathbb{C}}] = [F] + n[\underline{\mathbb{C}}]$ and hence $\Phi(E) = [E] - \text{rk} E [\underline{\mathbb{C}}] = [F] - \text{rk} F [\underline{\mathbb{C}}] = \Phi(F)$ after using $\text{rk} E + m = \text{rk} F + n$. Thus Φ is injective on stable equivalence classes.

For surjectivity, let $\eta \in \tilde{K}^0(X)$. By lemma 2.1.6(2) write $\eta = [E] - n$ for a bundle E ; applying rank gives $0 = \text{rk} \eta = \text{rk} E - n$, so $n = \text{rk} E$ and $\eta = [E] - \text{rk} E [\underline{\mathbb{C}}] = \Phi(E)$. Hence Φ is onto. The group operation $\Phi(E) + \Phi(F) = \Phi(E \oplus F)$ holds by additivity of rank and of the bracket, so the bijection transports $\oplus$ to the group law of $\tilde{K}^0(X)$, completing the proof.

The content of theorem 2.1.8 is that the reduced group $\tilde{K}^0(X)$ is not an abstract quotient but a concrete object: its elements are the stable equivalence classes of bundles, two bundles representing the same element exactly when they agree after stabilizing by trivial summands. This is the form in which K-theory becomes computable, because stable equivalence of bundles over a compact space is governed by maps into the infinite Grassmannian. Indeed the stabilization theorem of the previous chapter (theorem 1.3.2) identifies $\tilde{K}^0(X)$ with $\varinjlim_n [X, \text{Grass}_n] = [X, BU]$, and assembling this with the rank splitting gives a representing space for the full group.

Theorem 2.1.9: Representability of K^0

Let X be a connected compact Hausdorff space. There are natural isomorphisms

$$\tilde{K}^0(X) \cong [X, BU], \quad K^0(X) \cong [X, \mathbb{Z} \times BU]$$

where $BU = \varinjlim_n BU_n$ is the infinite complex Grassmannian and $[X, \mathbb{Z} \times BU]$ denotes homotopy classes of maps into $\mathbb{Z} \times BU$. In the real case the same holds with BU replaced by BO , giving $\widetilde{KO}^0(X) \cong [X, BO]$ and $KO^0(X) \cong [X, \mathbb{Z} \times BO]$.

Proof:

Step 1: the reduced group as maps into BU . By theorem 2.1.8, $\tilde{K}^0(X)$ is the set of stable equivalence classes of bundles over X . By the stabilization theorem (theorem 1.3.2), for X compact the natural maps assemble into an isomorphism

$$\tilde{K}^0(X) \cong \varinjlim_n \text{Vect}^n(X) \cong \varinjlim_n [X, BU_n]$$

where the colimit is taken along the stabilization maps $\text{Vect}^n(X) \rightarrow \text{Vect}^{n+1}(X)$, $E \mapsto E \oplus \mathbb{C}$, which on classifying maps correspond to the inclusions $BU_n \hookrightarrow BU_{n+1}$ given by $V \mapsto V \oplus \mathbb{C}$. Each identification $\text{Vect}^n(X) \cong [X, BU_n]$ is theorem 1.2.13 (in its complex form, with $\text{Grass}_n = BU_n$ the complex Grassmannian of definition 1.2.14). Since X is compact, every map $X \rightarrow BU = \varinjlim_n BU_n$ has image in some BU_n (a compact subset of the colimit $\varinjlim_n BU_n$ of the closed inclusions $BU_n \hookrightarrow BU_{n+1}$ meets only finitely many of the cells adjoined at each stage, hence lies in a single finite stage BU_n), and likewise the compact image of any homotopy $X \times [0, 1] \rightarrow BU$ between two such maps lands in some BU_N . Therefore

$$\varinjlim_n [X, BU_n] \cong [X, \varinjlim_n BU_n] = [X, BU]$$

Combining the displays gives $\tilde{K}^0(X) \cong [X, BU]$. This isomorphism is natural in X : a map $f: X' \rightarrow X$ induces $f^*: \text{Vect}^n(X) \rightarrow \text{Vect}^n(X')$ by pullback (proposition 1.2.1), which corresponds to precomposition with f on classifying maps, hence to $f^*: [X, BU] \rightarrow [X', BU]$.

Step 2: the full group as maps into $\mathbb{Z} \times BU$. Since X is connected, $[X, \mathbb{Z}] \cong \mathbb{Z}$, the homotopy classes of maps into the discrete space $\mathbb{Z}$ being the constant maps. The splitting of corollary 2.1.5 reads $K^0(X) \cong \tilde{K}^0(X) \oplus \mathbb{Z}$. Combining with Step 1,

$$K^0(X) \cong [X, BU] \oplus \mathbb{Z} \cong [X, BU] \times [X, \mathbb{Z}] \cong [X, BU \times \mathbb{Z}] = [X, \mathbb{Z} \times BU]$$

where the last identification uses that homotopy classes of maps into a product are pairs of homotopy classes, $[X, Y \times Z] \cong [X, Y] \times [X, Z]$, and that for connected X the factor $[X, \mathbb{Z}]$ recovers the rank. Naturality is inherited from Step 1 together with naturality of the rank splitting. The real case is identical, replacing BU_n by $BO_n = \text{Grass}_n(\mathbb{R}^\infty)$ throughout and invoking the real form of theorem 1.2.13, completing the proof.

The representability theorem reframes K-theory entirely in homotopy-theoretic terms: computing $K^0(X)$ is computing homotopy classes of maps from X into the fixed space $\mathbb{Z} \times BU$. This is the first sign that K-theory will be a generalized cohomology theory, since cohomology theories are exactly the functors represented by spaces (or spectra). The space $\mathbb{Z} \times BU$ is the degree-zero representing space; Bott periodicity, taken up in the next section, will show that the degree-one space is U and that these two spaces suffice to represent the whole theory. For now the value of the theorem is computational, as the closing examples illustrate.

So far $K^0(X)$ has only an additive structure. The tensor product of definition 1.1.17 supplies a multiplication, turning $K^0(X)$ into a commutative ring. The verification that the tensor product descends to the Grothendieck group is a matter of checking that the bilinearity and distributivity of $\otimes$ over $\oplus$ at the level of bundles survive the passage to formal differences.

Proposition 2.1.10: The Ring Structure on $K^0(X)$

Let X be a compact Hausdorff space. The tensor product of vector bundles (definition 1.1.17) induces on $K^0(X)$ a well-defined product

$$([E_1] - [F_1]) \cdot ([E_2] - [F_2]) = [E_1 \otimes E_2] + [F_1 \otimes F_2] - [E_1 \otimes F_2] - [F_1 \otimes E_2]$$

making $K^0(X)$ a commutative ring with unit $[\mathbb{C}]$, the class of the trivial line bundle. The reduced group $\tilde{K}^0(X)$ is the augmentation ideal of this ring, the kernel of the ring homomorphism $\text{rk}: K^0(X) \rightarrow \mathbb{Z}$. Pullback $f^*: K^0(X) \rightarrow K^0(X')$ along a continuous map is a ring homomorphism.

Proof:

Step 1: the product is well-defined. The assignment $(E_1, E_2) \mapsto [E_1 \otimes E_2]$ is a map $\text{Vect}(X) \times \text{Vect}(X) \rightarrow K^0(X)$. It is additive in each variable: by the distributivity of tensor over direct sum at the bundle level, $(E_1 \oplus E'_1) \otimes E_2 \cong (E_1 \otimes E_2) \oplus (E'_1 \otimes E_2)$ and symmetrically in the second variable (these isomorphisms are recorded in the discussion following definition 1.1.17, where the tensor product is shown to be commutative, associative, unital, and distributive over $\oplus$ fiberwise). Passing to classes,

$$[(E_1 \oplus E'_1) \otimes E_2] = [E_1 \otimes E_2] + [E'_1 \otimes E_2]$$

so the map is biadditive. A biadditive map from $\text{Vect}(X) \times \text{Vect}(X)$ into an abelian group factors through the tensor product of the Grothendieck groups, equivalently extends uniquely to a $\mathbb{Z}$ -bilinear map $K^0(X) \times K^0(X) \rightarrow K^0(X)$. Concretely, this forces the formula in the statement: expanding $([E_1] - [F_1])([E_2] - [F_2])$ by bilinearity gives $[E_1][E_2] - [E_1][F_2] - [F_1][E_2] + [F_1][F_2]$, which is the displayed expression. Independence of the representatives of each virtual bundle is exactly the assertion that the formula respects the relation $[E \oplus F] = [E] + [F]$, which biadditivity guarantees.

Step 2: ring axioms. Commutativity, associativity, and the unit law for the product on $K^0(X)$ follow from the corresponding properties of $\otimes$ on bundles, since the product is determined by its values $[E_1] \cdot [E_2] = [E_1 \otimes E_2]$ on the generating classes and these properties are stated for the bundle-level tensor product. Explicitly, $E_1 \otimes E_2 \cong E_2 \otimes E_1$ gives $[E_1][E_2] = [E_2][E_1]$; $(E_1 \otimes E_2) \otimes E_3 \cong E_1 \otimes (E_2 \otimes E_3)$ gives associativity; and $\mathbb{C} \otimes E \cong E$ gives $[\mathbb{C}][E] = [E]$, so $[\mathbb{C}]$ is the unit. Distributivity of the product over addition in $K^0(X)$ is the bilinearity established in Step 1. Hence $(K^0(X), +, \cdot)$ is a commutative ring with unit $[\mathbb{C}]$.

Step 3: rank is a ring map and $\tilde{K}^0$ is its kernel. For bundles over a connected base, $\text{rk}(E_1 \otimes E_2) = \text{rk} E_1 \cdot \text{rk} E_2$ since the fiber of $E_1 \otimes E_2$ over a point is the tensor product of the fibers, of dimension the product of the dimensions. Thus rk is multiplicative as well as additive, and $\text{rk}[\mathbb{C}] = 1$, so $\text{rk}: K^0(X) \rightarrow \mathbb{Z}$ is a unital ring homomorphism. Its kernel is $\tilde{K}^0(X)$ by definition 2.1.4, which is therefore an ideal, the augmentation ideal of the ring. (Over a base with r components, rk lands in the product ring $\mathbb{Z}^r$.)

Step 4: naturality. For a continuous $f: X' \rightarrow X$, the pullback satisfies $f^*(E_1 \otimes E_2) \cong f^*E_1 \otimes f^*E_2$ and $f^*(E_1 \oplus E_2) \cong f^*E_1 \oplus f^*E_2$ by proposition 1.2.3, and $f^*\mathbb{C} \cong \mathbb{C}$. Hence the induced map $f^*: K^0(X) \rightarrow K^0(X')$ preserves products, sums, and the unit, so it is a ring homomorphism, completing the proof.

With the ring structure in hand, K^0 is a contravariant functor from compact Hausdorff spaces to commutative rings, refining the homotopy-classification of bundles into an algebraic invariant that multiplies. The reduced group sits inside as the augmentation ideal, and one checks readily that the product of two reduced classes is again reduced (the product of two elements of an ideal lies in the ideal), so $\tilde{K}^0(X)$ is a non-unital ring in its own right. The two closing computations below show the smallest nontrivial values of the theory and confirm that the formal machinery agrees with direct calculation.

Example 2.1.11: K^0 of a Contractible Space

Let X be a contractible compact Hausdorff space, for instance a disk D^n or the interval $[0, 1]$. By corollary 1.2.6, every vector bundle over a contractible paracompact base is trivial, so $\text{Vect}^n(X)$ has a single element for each n and $\text{Vect}(X) \cong \mathbb{N}$ as monoids, with $E \mapsto \text{rk} E$. Exactly as in example 2.1.2, the Grothendieck group is

$$K^0(X) \cong \mathbb{Z}, \quad \tilde{K}^0(X) = 0$$

with the rank map an isomorphism onto $\mathbb{Z}$ and the reduced group trivial. This also follows from theorem 2.1.9: a contractible X admits only nullhomotopic maps into BU , so $\tilde{K}^0(X) = [X, BU] = 0$, and $K^0(X) = [X, \mathbb{Z} \times BU] \cong [X, \mathbb{Z}] = \mathbb{Z}$. The ring structure is the unique unital ring structure on $\mathbb{Z}$. Thus K-theory cannot distinguish a contractible space from a point, as a homotopy invariant must.

The contractible case shows that all the content of K-theory is in the reduced group, which vanishes whenever X has the homotopy type of a point. The smallest space with nontrivial reduced K-theory is the 2-sphere, where the canonical line bundle over $\mathbb{CP}^1 = S^2$ generates an infinite cyclic group. This computation is the prototype for every later calculation and the first concrete appearance of Bott periodicity.

Example 2.1.12: Reduced K-Theory of S^2

Let $X = S^2$, identified with the complex projective line $\mathbb{CP}^1$. By theorem 2.1.9,

$$\tilde{K}^0(S^2) \cong [S^2, BU] = \pi_2(BU)$$

the last equality because S^2 is the 2-sphere and homotopy classes of based maps from S^2 agree with free homotopy classes here, BU being simply connected (it is a colimit of the simply connected BU_n , and $\pi_1(BU_n) = \pi_0(U_n) = 0$). The space BU classifies stable bundles, and its homotopy is computed from that of the unitary group by the standard fibration relation $\pi_k(BU) \cong \pi_{k-1}(U)$, valid because BU is the classifying space of the stable unitary group $U = \varinjlim_n U(n)$. Hence

$$\pi_2(BU) \cong \pi_1(U) \cong \mathbb{Z}$$

where $\pi_1(U) = \mathbb{Z}$ is the value $k \equiv 1 \pmod{2}$ entry of the complex Bott table in theorem 1.3.1. Therefore

$$\tilde{K}^0(S^2) \cong \mathbb{Z}$$

An explicit generator is the reduced class of the canonical (tautological) complex line bundle $H \rightarrow \mathbb{CP}^1$ introduced in section 1.2: the element $[H] - [\mathbb{C}] \in \tilde{K}^0(S^2)$ generates. By corollary 2.1.5 the full group is

$$K^0(S^2) \cong \tilde{K}^0(S^2) \oplus \mathbb{Z} \cong \mathbb{Z} \oplus \mathbb{Z}$$

with the rank accounting for the second summand. As a ring, $K^0(S^2) \cong \mathbb{Z}[H]/(H-1)^2$, where H is shorthand for the class $[H]$ and the relation $(H-1)^2 = 0$ records that the reduced generator $h = [H] - 1$ squares to zero. This relation comes from an honest isomorphism of bundles: the canonical line bundle over $\mathbb{CP}^1$ satisfies $(H \otimes H) \oplus \underline{\mathbb{C}} \cong H \oplus H$, recorded in section 1.2. Passing to classes in the ring $K^0(S^2)$ and using proposition 2.1.10, this reads $[H]^2 + 1 = 2[H]$, that is $[H]^2 - 2[H] + 1 = ([H] - 1)^2 = 0$, so $h^2 = 0$. Since $\{1, h\}$ generates $K^0(S^2) \cong \mathbb{Z} \oplus \mathbb{Z}$ additively and $h^2 = 0$, the ring is exactly $\mathbb{Z}[h]/(h^2)$.

The relation $([H] - 1)^2 = 0$ in the last example is worth isolating, because it is the genuinely K-theoretic shadow of the geometric identity $(H \otimes H) \oplus \underline{\mathbb{C}} \cong H \oplus H$ over $\mathbb{CP}^1$. In ordinary cohomology the analogous generator squares to zero for dimensional reasons; here it squares to zero because of an honest isomorphism of vector bundles. This pattern, where a relation among bundles produces a relation in the K-theory ring, recurs in the computation of $K^0(\mathbb{CP}^n)$ in section 2.3, and the squaring-to-zero phenomenon over S^2 is the seed of the Bott periodicity isomorphism studied next.

Exercise 2.1.1

1. Let X be a compact Hausdorff space with r connected components. Show that $K^0(X) \cong \tilde{K}^0(X) \oplus \mathbb{Z}^r$, where $\tilde{K}^0(X)$ is the kernel of the total rank homomorphism $\text{rk}: K^0(X) \rightarrow \mathbb{Z}^r$. Deduce that for the two-point space $X = \{p, q\}$ one has $K^0(X) \cong \mathbb{Z}^2$ and $\tilde{K}^0(X) = 0$.
2. Let A be a commutative monoid. Its Grothendieck group $G(A)$ is the universal abelian group receiving a monoid homomorphism from A . Prove that the canonical map $A \rightarrow G(A)$ is injective if and only if A is cancellative (that is, $a + c = b + c$ implies $a = b$). Using lemma 2.1.6 and theorem 2.1.8, explain why the monoid $\text{Vect}(X)$ need not be cancellative, and identify exactly when two bundle classes $[E], [F]$ become equal in $K^0(X)$.
3. Show that the tangent bundle TS^2 represents the trivial element of the reduced real K-theory $\widetilde{KO}^0(S^2)$, even though TS^2 is not the trivial bundle. (Use that $TS^2 \oplus NS^2 \cong \underline{\mathbb{R}}^3$ for the normal bundle NS^2 , which is trivial, and pass to the appropriate reduced group.)
4. Let $f: X \rightarrow Y$ be a homotopy equivalence of compact Hausdorff spaces. Prove directly from theorem 2.1.9 that $f^*: K^0(Y) \rightarrow K^0(X)$ is a ring isomorphism. Conclude that K^0 is a homotopy-invariant functor.
5. Verify the ring isomorphism $K^0(S^2) \cong \mathbb{Z}[h]/(h^2)$, where $h = [H] - 1$ is the reduced class of the canonical line bundle. Specifically, show that $\{1, h\}$ is a $\mathbb{Z}$ -basis of $K^0(S^2)$ as an abelian group, that $h^2 = 0$, and that the rank homomorphism is the quotient $\mathbb{Z}[h]/(h^2) \rightarrow \mathbb{Z}$ sending $h \mapsto 0$.
6. Let $X = \{*\}$ and recompute $K^0(\{*\})$ as a ring using proposition 2.1.10. Then explain why, for any nonempty compact Hausdorff X , the unique map $X \rightarrow \{*\}$ and any choice of point $* \rightarrow X$ exhibit $\mathbb{Z} = K^0(\{*\})$ as a direct summand (indeed a unital subring) of $K^0(X)$, recovering the rank splitting of corollary 2.1.5 functorially.

2.2 Bott Periodicity and K-Theory as a Cohomology Theory

The group $\tilde{K}^0(X)$ of section 2.1 is a single contravariant homotopy functor on compact spaces. A cohomology theory is more than a single functor: it is a whole graded family $\{h^n\}_{n \in \mathbb{Z}}$ tied together by suspension isomorphisms and by the long exact sequence of a pair. This section assembles topological K-theory into such a theory. The negative groups $K^{-n}(X)$ are defined by iterated reduced suspension, the relative groups $K^0(X, A)$ by collapsing the subspace A , and the entire package is glued together by the deepest input from chapter 1: Bott periodicity. The homotopy statement $\Omega^2(\mathbb{Z} \times BU) \simeq \mathbb{Z} \times BU$ of theorem 1.3.1 becomes, after passing to homotopy classes, the isomorphism $\tilde{K}^0(X) \cong \tilde{K}^0(\Sigma^2 X)$. Periodicity is what collapses the doubly-infinite family $\{K^n\}_{n \in \mathbb{Z}}$ down to two groups, K^0 and K^1 , and it is what lets the suspension axiom of a generalized cohomology theory be verified at all: without it the negative groups would proliferate without control. We work throughout in complex K-theory, writing $K = KU$ and $\tilde{K} = \tilde{KU}$ unless stated otherwise, and we restrict to compact Hausdorff spaces and finite CW pairs, where every bundle has a complement (proposition 1.1.16) and the representability theorem (theorem 2.1.9) applies.

Throughout, X_+ denotes X with a disjoint basepoint adjoined, $\Sigma X = S^1 \wedge X$ is the reduced suspension of a pointed space, and X/A is the quotient of X by a closed subspace A , basepointed at the image of A . The smash product $X \wedge Y = (X \times Y)/(X \vee Y)$ is the reduced product. For a pointed space the reduced group $\tilde{K}^0(X)$ is the kernel of restriction to the basepoint, as in definition 2.1.4.

The passage from a single functor to a graded family rests on one construction: suspension. For ordinary cohomology the suspension isomorphism $\tilde{H}^n(X) \cong \tilde{H}^{n+1}(\Sigma X)$ lets one shift degree by smashing with a circle. K-theory has no a priori notion of degree, so the strategy is reversed: one *defines* the negative groups by suspending the space, declaring $\tilde{K}^{-n}(X)$ to be $\tilde{K}^0$ of the n -fold suspension. The reduced suspension is forced (rather than the unreduced cone-based one) because $\tilde{K}^0$ is a functor on *pointed* spaces.

Definition 2.2.1: Negative K-Groups

Let X be a compact Hausdorff space and let $n \geq 0$ be an integer. The n -th *negative reduced K-group* of a pointed space X is

$$\tilde{K}^{-n}(X) = \tilde{K}^0(\Sigma^n X)$$

where $\Sigma^n X$ is the n -fold reduced suspension and $\Sigma^0 X = X$. For an unpointed compact space X , the n -th *negative (unreduced) K-group* is

$$K^{-n}(X) = \tilde{K}^0(\Sigma^n(X_+)) = \tilde{K}^{-n}(X_+)$$

With $n = 0$ this recovers $K^0(X) = \tilde{K}^0(X_+)$ and $\tilde{K}^0(X)$ for the unreduced and reduced groups respectively.

The two flavors are reconciled by the splitting $X_+ = X \sqcup \{*\}$. Adding a disjoint basepoint converts an unpointed space into a pointed one without losing information, and the reduced theory of X_+ is the unreduced theory of X . Concretely, the inclusion of the added basepoint and the collapse $X_+ \rightarrow S^0$ give a natural splitting $K^{-n}(X) \cong \tilde{K}^{-n}(X) \oplus \tilde{K}^{-n}(S^0)$ when X itself is pointed; since $\tilde{K}^{-n}(S^0) = \tilde{K}^0(\Sigma^n S^0) = \tilde{K}^0(S^n)$, the discrepancy between reduced and unreduced is exactly the K-theory of a sphere. This is the K-theoretic analogue of the $\tilde{H}^*$ versus H^* bookkeeping in ordinary

cohomology, and it is the reason every formula below comes in a reduced and an unreduced version.

Example 2.2.2: Negative Groups of a Point

Take $X = \{*\}$ the one-point space, regarded as pointed. Then $\Sigma^n \{*\} = \{*\}$ for every n , since suspending a point gives a point: $\Sigma \{*\} = S^1 \wedge \{*\} = \{*\}$. Hence

$$\tilde{K}^{-n}(\{*\}) = \tilde{K}^0(\{*\}) = 0$$

for all $n \geq 0$, the reduced group of a point being trivial by definition 2.1.4. For the unreduced groups, $\{*\}_+ = S^0$, so

$$K^{-n}(\{*\}) = \tilde{K}^0(\Sigma^n S^0) = \tilde{K}^0(S^n)$$

The right-hand side is $\mathbb{Z}$ when n is even and 0 when n is odd, as computed for spheres in example 1.2.11 via the clutching correspondence (proposition 1.2.8): every complex bundle over an odd sphere is stably trivial, while $\tilde{K}^0(S^2) \cong \mathbb{Z}$ is generated by the reduced class of the canonical line bundle. Thus the coefficient groups $K^{-n}(\{*\})$ already exhibit the 2-periodic pattern $\mathbb{Z}, 0, \mathbb{Z}, 0, \dots$ that Bott periodicity will impose on every space.

Relative K-theory measures the K-theory of X *rel* a subspace A . The model is the quotient X/A , which collapses A to a single basepoint; classes in the relative group are classes on X that are trivialized over A . This is the geometric content of the cofiber: X/A is the cofiber of the inclusion $A \hookrightarrow X$, and reduced K-theory of the cofiber is the receptacle for relative classes.

Definition 2.2.3: Relative K-Theory

Let (X, A) be a compact pair, meaning X is compact Hausdorff and $A \subseteq X$ is a closed (hence compact) subspace. For $n \geq 0$ the *relative K-groups* are

$$K^{-n}(X, A) = \tilde{K}^{-n}(X/A) = \tilde{K}^0(\Sigma^n(X/A))$$

where X/A is basepointed at the collapsed image of A . When $A = \emptyset$ one collapses the empty set, which by convention adjoins a disjoint basepoint, so $X/\emptyset = X_+$ and $K^{-n}(X, \emptyset) = K^{-n}(X)$ recovers the absolute groups.

The convention $X/\emptyset = X_+$ is what makes the relative theory a genuine generalization of the absolute one. It encodes the principle that “no subspace collapsed” should leave the absolute group unchanged, and adding a disjoint basepoint is precisely the operation that turns the absolute unreduced group into a reduced group on which the machinery operates. With this convention the long exact sequence of the pair $(X, \emptyset)$ degenerates correctly and excision reduces to a tautology.

Example 2.2.4: Relative K-Theory of a Disk Rel its Boundary

Let $X = D^2$ be the closed 2-disk and $A = \partial D^2 = S^1$ its boundary circle. The quotient collapses the boundary to a point, and

$$D^2/S^1 \cong S^2$$

since a disk with its boundary identified to a single point is a 2-sphere. Therefore

$$K^0(D^2, S^1) = \tilde{K}^0(D^2/S^1) = \tilde{K}^0(S^2) \cong \mathbb{Z}$$

the generator being the reduced class $[H] - 1$ of the canonical line bundle $H \rightarrow \mathbb{CP}^1 = S^2$. This is the prototype for the Bott generator: the relative group of (D^2, S^1) houses the class

whose external multiplication implements periodicity. More generally $D^{2k}/\partial D^{2k} \cong S^{2k}$, so $K^0(D^{2k}, \partial D^{2k}) \cong \tilde{K}^0(S^{2k}) \cong \mathbb{Z}$, while collapsing the boundary of an odd-dimensional disk gives an odd sphere and the relative group vanishes.

K-theory carries a multiplication finer than the internal ring structure of proposition 2.1.10: an external product that combines classes on two different spaces into a class on their product. This is what upgrades K-theory from a sequence of abelian groups to a multiplicative cohomology theory, and it is the device that builds the Bott periodicity map.

Definition 2.2.5: External Product

Let X and Y be compact Hausdorff spaces. The *external product* is the homomorphism

$$\mu: K^0(X) \otimes K^0(Y) \longrightarrow K^0(X \times Y)$$

defined on generators by $[E] \otimes [F] \mapsto [p_X^* E \otimes p_Y^* F]$, where $p_X: X \times Y \rightarrow X$ and $p_Y: X \times Y \rightarrow Y$ are the projections and $\otimes$ on the right is the tensor product of bundles (definition 1.1.17). On reduced groups, the external product descends to a map

$$\tilde{\mu}: \tilde{K}^0(X) \otimes \tilde{K}^0(Y) \longrightarrow \tilde{K}^0(X \wedge Y)$$

using that $\tilde{K}^0$ of a smash is the kernel of restriction to the wedge $X \vee Y$.

The external product is well-defined because pullback commutes with tensor product (proposition 1.2.3) and because both operations respect direct sums, so the assignment is additive in each variable and passes to the Grothendieck group of definition 2.1.1. Its reduction to the smash product is the algebraic shadow of the geometric fact that a class trivial on either factor restricts trivially to the wedge $X \vee Y \subseteq X \times Y$, hence factors through the quotient $X \wedge Y = (X \times Y)/(X \vee Y)$. The next definition extracts from this product the single most important construction in topological K-theory.

Definition 2.2.6: The Bott Generator and Bott Map

Let $H \rightarrow \mathbb{CP}^1 = S^2$ be the canonical complex line bundle (example 1.2.11). The *Bott generator* is the reduced class

$$\beta = [H] - 1 \in \tilde{K}^0(S^2)$$

The *Bott map* for a compact pointed space X is the homomorphism

$$\beta_X: \tilde{K}^0(X) \longrightarrow \tilde{K}^0(S^2 \wedge X) = \tilde{K}^0(\Sigma^2 X), \quad \alpha \longmapsto \tilde{\mu}(\beta \otimes \alpha)$$

given by external multiplication with β , using the homeomorphism $S^2 \wedge X \cong \Sigma^2 X$.

The Bott map is the comparison morphism whose invertibility *is* periodicity. Geometrically it takes a virtual bundle on X and tensors it, externally, with the Bott class on S^2 ; the resulting class lives on the double suspension $\Sigma^2 X$ because β is a reduced class on S^2 and the external product of reduced classes lands in the smash. That β_X is an isomorphism is the periodicity theorem proved next.

The homotopy-theoretic Bott periodicity of theorem 1.3.1 states $\Omega^2(\mathbb{Z} \times BU) \simeq \mathbb{Z} \times BU$. The representability theorem (theorem 2.1.9) identifies $K^0(X)$ with $[X, \mathbb{Z} \times BU]$ and $\tilde{K}^0(X)$ with $[X, \mathbb{Z} \times BU]_*$, the based homotopy classes. Combining these two facts gives the K-theoretic periodicity

isomorphism, and the work is to check that the abstract isomorphism coming from the loop-space equivalence agrees with multiplication by the Bott class.

Theorem 2.2.7: Bott Periodicity Isomorphism

For every compact Hausdorff pointed space X the Bott map

$$\beta_X: \tilde{K}^0(X) \longrightarrow \tilde{K}^0(\Sigma^2 X)$$

is an isomorphism of abelian groups, natural in X . Equivalently, for every $n \geq 0$ there are natural isomorphisms

$$\tilde{K}^{-n}(X) \cong \tilde{K}^{-n-2}(X)$$

and in the unreduced setting $K^{-n}(X) \cong K^{-n-2}(X)$.

Proof:

Step 1: The loop-space input. By the representability theorem (theorem 2.1.9), for any compact Hausdorff pointed space Y there is a natural bijection

$$\tilde{K}^0(Y) \cong [Y, \mathbb{Z} \times BU]_*$$

identifying reduced K-theory with based homotopy classes of maps into the classifying space $\mathbb{Z} \times BU$. The adjunction between reduced suspension and based loops gives, for any pointed spaces, a natural bijection $[\Sigma^2 X, Z]_* \cong [X, \Omega^2 Z]_*$. Applying this with $Z = \mathbb{Z} \times BU$,

$$\tilde{K}^0(\Sigma^2 X) \cong [\Sigma^2 X, \mathbb{Z} \times BU]_* \cong [X, \Omega^2(\mathbb{Z} \times BU)]_*$$

Now theorem 1.3.1, in the form $\Omega^2(\mathbb{Z} \times BU) \simeq \mathbb{Z} \times BU$ recorded in section 1.3, provides a homotopy equivalence $b: \mathbb{Z} \times BU \rightarrow \Omega^2(\mathbb{Z} \times BU)$. Composition with b yields a natural bijection

$$[X, \mathbb{Z} \times BU]_* \xrightarrow{\cong} [X, \Omega^2(\mathbb{Z} \times BU)]_*$$

Stringing these together produces a natural isomorphism

$$B_X: \tilde{K}^0(X) \xrightarrow{\cong} \tilde{K}^0(\Sigma^2 X)$$

The maps $[Y, -]_*$ are group homomorphisms because $\mathbb{Z} \times BU$ is an infinite loop space, so B_X is an isomorphism of abelian groups, natural in X since every step is natural.

Step 2: Identifying B_X with multiplication by β . It remains to show $B_X = \beta_X$, the external Bott map. The key is naturality together with the universal example $X = S^0$. The reduced group $\tilde{K}^0(S^0)$ is $\mathbb{Z}$, generated by the class 1 of the trivial line bundle, since S^0 is two points and the reduced group is the kernel of restriction to the basepoint, so that a virtual bundle on S^0 reduced over the basepoint is just an integer rank over the other point. For $X = S^0$ the suspension is $\Sigma^2 S^0 = S^2$, and both B_{S^0} and β_{S^0} are homomorphisms $\mathbb{Z} = \tilde{K}^0(S^0) \rightarrow \tilde{K}^0(S^2) = \mathbb{Z}$.

By construction the external Bott map sends $1 \in \tilde{K}^0(S^0)$ to $\tilde{\mu}(\beta \otimes 1) = \beta$, the generator of $\tilde{K}^0(S^2) \cong \mathbb{Z}$, because external multiplication by the unit class is the identity (the smash $S^2 \wedge S^0 \cong S^2$ and $\beta \otimes 1$ pulls back to β). Thus β_{S^0} is the isomorphism $\mathbb{Z} \rightarrow \mathbb{Z}$ sending 1 to the generator β . On the other hand B_{S^0} is the abstract periodicity isomorphism, which by the normalization of theorem 1.3.1 also carries the generator of $\tilde{K}^0(S^0)$ to the generator of $\tilde{K}^0(S^2)$:

indeed the loop-space equivalence b is chosen so that the resulting class on S^2 is the canonical generator detected by $c_1 = 1$, which is exactly β . Replacing b by its composite with the sign automorphism of $\mathbb{Z} \times BU$ if necessary, we may take $B_{S^0}(1) = \beta = \beta_{S^0}(1)$. Hence $B_{S^0} = \beta_{S^0}$.

Step 3: From the universal example to all X . Both B and β are natural transformations of additive functors on compact pointed spaces, and both are forms of external multiplication by their value on the unit $1 \in \tilde{K}^0(S^0)$. For β this is the definition: $\beta_X(\alpha) = \tilde{\mu}(\beta \otimes \alpha)$ with $\beta = \beta_{S^0}(1)$. For B it is the multiplicativity of loop-space periodicity (the equivalence b of theorem 1.3.1 is one of H -spaces, so B commutes with the external product and is therefore external multiplication by $B_{S^0}(1)$). Since $B_{S^0}(1) = \beta = \beta_{S^0}(1)$ by Step 2, for any $\alpha \in \tilde{K}^0(X)$,

$$B_X(\alpha) = \tilde{\mu}(B_{S^0}(1) \otimes \alpha) = \tilde{\mu}(\beta \otimes \alpha) = \beta_X(\alpha)$$

where the first equality is the multiplicativity of B and the third is the definition of β_X . Therefore $B_X = \beta_X$ for all X , and β_X is the natural isomorphism of Step 1.

Step 4: The periodicity statements. Applying the isomorphism just established to $\Sigma^n X$ in place of X gives

$$\tilde{K}^{-n}(X) = \tilde{K}^0(\Sigma^n X) \xrightarrow[\cong]{\beta_{\Sigma^n X}} \tilde{K}^0(\Sigma^2 \Sigma^n X) = \tilde{K}^0(\Sigma^{n+2} X) = \tilde{K}^{-n-2}(X)$$

which is the reduced periodicity $\tilde{K}^{-n} \cong \tilde{K}^{-n-2}$. Replacing X by X_+ throughout yields the unreduced statement $K^{-n}(X) \cong K^{-n-2}(X)$, completing the proof.

The analytic heart of theorem 1.3.1, that $\Omega^2(\mathbb{Z} \times BU) \simeq \mathbb{Z} \times BU$, is the one input taken from chapter 1 rather than reproved; the proof above is otherwise complete and shows precisely how the homotopy statement becomes the multiplicative statement β_X is an isomorphism. A self-contained K-theoretic proof of periodicity, building the inverse of β_X directly from clutching functions and Laurent-polynomial bundles over $X \times S^2$, is given in [8, Theorem 2.11]; the structure there matches the external-product formulation here, with β_X the same map. The consequence is decisive: instead of a doubly-infinite sequence of groups $\tilde{K}^{-n}$ there are only two distinct ones, $\tilde{K}^0$ and $\tilde{K}^{-1}$, and the latter is conventionally written $\tilde{K}^1$. This is what is meant by saying complex K-theory is $\mathbb{Z}/2$ -graded.

2.2.8 Convention: Periodic Grading By theorem 2.2.7 the groups $\tilde{K}^{-n}(X)$ depend only on the parity of n . One therefore extends the indexing to all $n \in \mathbb{Z}$ by declaring $\tilde{K}^n(X) = \tilde{K}^{n-2}(X) = \tilde{K}^{n+2}(X)$, so that $\tilde{K}^n = \tilde{K}^0$ for n even and $\tilde{K}^n = \tilde{K}^1$ for n odd, where $\tilde{K}^1(X) = \tilde{K}^0(\Sigma X) = \tilde{K}^{-1}(X)$. The same convention applies to the unreduced and relative groups. Henceforth $\tilde{K}^*$ denotes this $\mathbb{Z}/2$ -graded family.

A generalized cohomology theory on compact pairs is, by the axioms developed in the generalized-cohomology chapter of [5], a sequence of contravariant functors h^n together with natural coboundary maps $\delta: h^n(A) \rightarrow h^{n+1}(X, A)$ satisfying homotopy invariance, exactness for pairs, excision, and the suspension isomorphism. The K-groups satisfy all of these, with the suspension isomorphism supplied by periodicity. The next theorem records this; its proof verifies each axiom, drawing the exactness from the cofiber sequence and excision from the homeomorphism type of the relevant quotients.

Theorem 2.2.9: K-Theory is a Generalized Cohomology Theory

The $\mathbb{Z}/2$ -graded family $\{K^n, \tilde{K}^n\}_{n \in \mathbb{Z}}$ of theorem 2.2.7, together with the relative groups $K^n(X, A)$ of definition 2.2.3, forms a multiplicative generalized cohomology theory on compact Hausdorff pairs. That is, it satisfies:

1. (*Homotopy invariance.*) If $f_0 \simeq f_1: (X, A) \rightarrow (Y, B)$ are homotopic maps of pairs, then $f_0^* = f_1^*: K^n(Y, B) \rightarrow K^n(X, A)$.
2. (*Exactness.*) For a compact pair (X, A) with inclusions $i: A \hookrightarrow X$ and $j: (X, \emptyset) \hookrightarrow (X, A)$, the sequence

$$K^n(X, A) \xrightarrow{j^*} K^n(X) \xrightarrow{i^*} K^n(A)$$

is exact at $K^n(X)$.

3. (*Excision.*) If (X, A) is a compact pair and $U \subseteq A$ is open with $\bar{U} \subseteq \text{int} A$, then the inclusion $(X \setminus U, A \setminus U) \hookrightarrow (X, A)$ induces an isomorphism $K^n(X, A) \cong K^n(X \setminus U, A \setminus U)$.
4. (*Suspension.*) There are natural isomorphisms $\tilde{K}^n(X) \cong \tilde{K}^{n+1}(\Sigma X)$, and the periodicity $\tilde{K}^n \cong \tilde{K}^{n+2}$ holds.

Moreover the external product of definition 2.2.5 makes K^* a multiplicative theory.

Proof:

Axiom 1: Homotopy invariance. For absolute groups this is theorem 1.2.5: homotopic maps induce the same map on isomorphism classes of bundles, hence the same map on Grothendieck groups K^0 (definition 2.1.1), and the same on $\tilde{K}^0$. For negative groups, a homotopy f_t induces a homotopy $\Sigma^n f_t$ of the suspended maps, so $f_0^* = f_1^*$ on $\tilde{K}^{-n}(Y) = \tilde{K}^0(\Sigma^n Y)$. For relative groups, a homotopy of pairs $(X, A) \rightarrow (Y, B)$ induces a homotopy of the quotient maps $X/A \rightarrow Y/B$, since the homotopy carries A into B at every time and hence descends to the quotients; homotopy invariance of $\tilde{K}^0$ on the quotients gives the claim.

Axiom 2: Exactness. The inclusion $A \xrightarrow{i} X$ has cofiber X/A , and there is a cofiber sequence

$$A \xrightarrow{i} X \xrightarrow{q} X/A$$

where q is the collapse. The content is that the composite $A \xrightarrow{i} X \xrightarrow{q} X/A$ is the constant map to the basepoint, so $i^*q^* = 0$, and conversely any class on X restricting to zero on A extends over the quotient. Concretely, work with the reduced groups via $K^n(X) = \tilde{K}^n(X_+)$ and $K^n(X, A) = \tilde{K}^n(X/A)$. The collapse $q: X_+ \rightarrow X/A$ (sending A and the added basepoint to the basepoint) and the inclusion $i: A_+ \rightarrow X_+$ give

$$\tilde{K}^n(X/A) \xrightarrow{q^*=j^*} \tilde{K}^n(X_+) \xrightarrow{i^*} \tilde{K}^n(A_+)$$

Exactness at the middle term is the statement that $\ker i^* = \text{im } q^*$. The inclusion $\text{im } q^* \subseteq \ker i^*$ holds because $q \circ i$ is null (it factors through the collapsed point). For the reverse, suppose $\alpha \in \tilde{K}^0(X_+)$ has $i^*\alpha = 0$, so α is represented by a virtual bundle on X that is stably trivial over A . After adding a trivial bundle one may assume $\alpha = [E] - [\dim E]$ with $E|_A$ trivialized by a chosen isomorphism $\varphi: E|_A \cong A \times \mathbb{C}^k$. The pair (E, φ) defines a bundle on the quotient X/A by

gluing E to a trivial bundle over the collapsed point using φ (the clutching construction along A , valid because $A \hookrightarrow X$ is a cofibration of compact spaces and φ extends to a tubular collar), and this descended bundle pulls back to E under q . Hence $\alpha \in \text{im } q^*$, giving $\ker i^* = \text{im } q^*$. The same argument applied to Σ^n yields exactness in every degree.

Axiom 3: Excision. For a compact pair the excision isomorphism is a tautology in this model, because the relative group is defined through the quotient and the quotient is unchanged by excising an open set buried inside A . If $U \subseteq A$ is open with $\overline{U} \subseteq \text{int } A$, then collapsing A in X and collapsing $A \setminus U$ in $X \setminus U$ produce homeomorphic quotients:

$$(X \setminus U)/(A \setminus U) \cong X/A$$

The map is induced by the inclusion $X \setminus U \hookrightarrow X$; it is a continuous bijection of compact Hausdorff spaces (every point of A maps to the single basepoint on both sides, and points outside A are untouched), hence a homeomorphism. Applying $\tilde{K}^n$ to this homeomorphism gives the excision isomorphism $K^n(X, A) \cong K^n(X \setminus U, A \setminus U)$.

Axiom 4: Suspension and periodicity. For a pointed compact space X and any integer $m \geq 0$ the definition of the negative groups gives, by the very meaning of suspension,

$$\tilde{K}^{-(m+1)}(X) = \tilde{K}^0(\Sigma^{m+1}X) = \tilde{K}^0(\Sigma^m(\Sigma X)) = \tilde{K}^{-m}(\Sigma X)$$

which is the suspension isomorphism $\tilde{K}^n(X) \cong \tilde{K}^{n+1}(\Sigma X)$ at $n = -(m+1)$, holding tautologically for all negative degrees. Periodicity (theorem 2.2.7) extends this identity to every integer n by shifting degrees in steps of 2 until they become negative, so the suspension isomorphism holds in the $\mathbb{Z}/2$ -graded family of box 2.2.8. The substantive content beyond the tautology is the periodicity $\tilde{K}^n \cong \tilde{K}^{n+2}$ itself, which is exactly theorem 2.2.7. Together these identify the abstract suspension isomorphism required by the generalized-cohomology axioms of [5] with the maps in hand.

Multiplicativity. The external product μ of definition 2.2.5 is associative and commutative up to natural isomorphism, since the bundle tensor product is, and the diagonal $X \rightarrow X \times X$ pulls μ back to the internal ring product of proposition 2.1.10. The Bott class β is the unit for the periodicity action, so the graded group $K^*(X)$ is a module over the coefficient ring $K^*(\{*\}) = \mathbb{Z}[\beta, \beta^{-1}]$, a Laurent polynomial ring on the invertible degree -2 generator β . This is the multiplicative structure, completing the verification of all axioms.

The theorem places K-theory on the same axiomatic footing as ordinary cohomology, but with one decisive difference: the coefficient ring $K^*(\{*\}) = \mathbb{Z}[\beta, \beta^{-1}]$ is not concentrated in degree 0. The dimension axiom of Eilenberg and Steenrod, which would force $h^n(\{*\}) = 0$ for $n \neq 0$, fails, and it is this failure that makes K-theory a genuinely *generalized* cohomology theory rather than ordinary cohomology in disguise. Every consequence of the axioms valid for an arbitrary generalized theory, the long exact sequence of a pair, Mayer-Vietoris, and the Atiyah-Hirzebruch spectral sequence of [5] computing K^* from ordinary cohomology H^* , now applies to K-theory.

The exactness axiom is local in the sense that it relates three terms; iterating it along the suspensions of a pair produces a long exact sequence running infinitely in both directions. Because of periodicity this long sequence wraps around into a six-term cyclic sequence, the topological precursor of the six-term sequence for C^* -algebras developed in [3]. The next theorem assembles it.

Theorem 2.2.10: The Long Exact Sequence of a Pair

Let (X, A) be a compact Hausdorff pair with inclusions $i: A \hookrightarrow X$ and $j: (X, \emptyset) \hookrightarrow (X, A)$. There is a long exact sequence

$$\cdots \rightarrow K^n(X, A) \xrightarrow{j^*} K^n(X) \xrightarrow{i^*} K^n(A) \xrightarrow{\delta} K^{n+1}(X, A) \xrightarrow{j^*} K^{n+1}(X) \rightarrow \cdots$$

natural in the pair, where δ is the coboundary map. By Bott periodicity (theorem 2.2.7) this sequence is 2-periodic and collapses to the six-term exact sequence

$$\begin{array}{ccccc} K^0(X, A) & \xrightarrow{j^*} & K^0(X) & \xrightarrow{i^*} & K^0(A) \\ \delta \uparrow & & & & \downarrow \delta \\ K^1(A) & \xleftarrow{i^*} & K^1(X) & \xleftarrow{j^*} & K^1(X, A) \end{array}$$

Proof:

Step 1: The Puppe sequence of the inclusion. The cofibration $i: A \hookrightarrow X$ extends to its Puppe (cofiber) sequence of pointed spaces

$$A_+ \xrightarrow{i} X_+ \xrightarrow{q} X/A \xrightarrow{\partial} \Sigma A_+ \xrightarrow{\Sigma i} \Sigma X_+ \xrightarrow{\Sigma q} \Sigma(X/A) \rightarrow \cdots$$

where each space is the cofiber of the preceding map and $\partial: X/A \rightarrow \Sigma A_+$ is the connecting map, identifying the cofiber of q with the suspension ΣA_+ , as constructed in [5]. The construction in [5] establishes that applying any reduced cohomology functor to a cofiber sequence produces an exact sequence: for the reduced functor $\tilde{K}^0$ each consecutive pair of maps in the Puppe sequence gives an exact triple, the exactness being the case $n = 0$ of Axiom 2 in theorem 2.2.9.

Step 2: Applying $\tilde{K}^0$. Apply $\tilde{K}^0$ to the Puppe sequence. Each three-term stretch $W' \rightarrow W \rightarrow W''$ of cofiber and cofiber-of-cofiber yields exactness of $\tilde{K}^0(W'') \rightarrow \tilde{K}^0(W) \rightarrow \tilde{K}^0(W')$, and assembling all of them gives the exact sequence

$$\cdots \rightarrow \tilde{K}^0(\Sigma(X/A)) \rightarrow \tilde{K}^0(\Sigma X_+) \rightarrow \tilde{K}^0(\Sigma A_+) \xrightarrow{\partial^*} \tilde{K}^0(X/A) \rightarrow \tilde{K}^0(X_+) \rightarrow \tilde{K}^0(A_+)$$

Translating with the definitions $\tilde{K}^0(X_+) = K^0(X)$, $\tilde{K}^0(X/A) = K^0(X, A)$, and $\tilde{K}^0(\Sigma X_+) = \tilde{K}^{-1}(X_+) = K^{-1}(X)$, this reads

$$\cdots \rightarrow K^{-1}(X, A) \xrightarrow{j^*} K^{-1}(X) \xrightarrow{i^*} K^{-1}(A) \xrightarrow{\delta} K^0(X, A) \xrightarrow{j^*} K^0(X) \xrightarrow{i^*} K^0(A)$$

where the coboundary δ is induced by the Puppe connecting map ∂ . Iterating the Puppe sequence (each further suspension adds three more terms) extends this to a sequence infinite in both directions, exactly the displayed long exact sequence with K^n written for $\tilde{K}^0 \circ \Sigma^{-n}$ in the periodic sense.

Step 3: Naturality. A map of pairs $(X, A) \rightarrow (Y, B)$ induces a map of cofibration sequences $A \rightarrow X \rightarrow X/A$ to $B \rightarrow Y \rightarrow Y/B$, hence a map of the associated Puppe sequences, hence a commuting ladder after applying $\tilde{K}^0$. Thus the long exact sequence is natural in the pair, and in particular δ is a natural transformation.

Step 4: Collapsing by periodicity. By theorem 2.2.7 the groups satisfy $K^n \cong K^{n-2}$, so the doubly-infinite sequence has only two distinct rows of groups, those in even and odd degree. Identifying $K^{-1} = K^1$ via periodicity folds the long sequence into the cyclic six-term sequence

$$K^0(X, A) \rightarrow K^0(X) \rightarrow K^0(A) \xrightarrow{\delta} K^1(X, A) \rightarrow K^1(X) \rightarrow K^1(A) \xrightarrow{\delta} K^0(X, A)$$

which is the hexagon displayed in the statement. Exactness at each of the six vertices is inherited from the long exact sequence, completing the proof.

The six-term sequence is the computational engine of K-theory. Where ordinary cohomology gives a long exact sequence with infinitely many distinct groups, periodicity compresses the K-theory sequence into six terms arranged in a hexagon, and a single coboundary δ in each parity carries all the connecting information. Two standard specializations deserve to be recorded. The first relates the reduced theory of X , A , and X/A ; the second is the Mayer-Vietoris sequence for a cover.

Corollary 2.2.11: Reduced Cofiber Sequence

For a compact pair (X, A) with A nonempty there is an exact sequence of reduced groups

$$\tilde{K}^0(X/A) \xrightarrow{q^*} \tilde{K}^0(X) \xrightarrow{i^*} \tilde{K}^0(A) \xrightarrow{\delta} \tilde{K}^1(X/A) \xrightarrow{q^*} \tilde{K}^1(X) \xrightarrow{i^*} \tilde{K}^1(A) \xrightarrow{\delta} \tilde{K}^0(X/A)$$

Proof:

Choose the basepoint of X in A . Then $\tilde{K}^n(X) = K^n(X, \text{pt})$ and the splitting $K^n(X) = \tilde{K}^n(X) \oplus K^n(\text{pt})$ is natural in the based maps $A \rightarrow X \rightarrow X/A$, all of which preserve the basepoint. Subtracting the constant-point summand from the six-term sequence of theorem 2.2.10 for (X, A) leaves exactly the displayed sequence of reduced groups, since $K^n(X, A) = \tilde{K}^n(X/A)$ by definition 2.2.3 and the relative term is already reduced, completing the proof.

Corollary 2.2.12: Mayer-Vietoris in K-Theory

Let $X = X_1 \cup X_2$ be a compact Hausdorff space written as the union of two closed subspaces, and set $X_0 = X_1 \cap X_2$. There is a six-term exact Mayer-Vietoris sequence

$$\begin{array}{ccccc} K^0(X) & \longrightarrow & K^0(X_1) \oplus K^0(X_2) & \longrightarrow & K^0(X_0) \\ \delta \uparrow & & & & \downarrow \delta \\ K^1(X_0) & \longleftarrow & K^1(X_1) \oplus K^1(X_2) & \longleftarrow & K^1(X) \end{array}$$

with the horizontal maps the differences of restrictions.

Proof:

Apply the long exact sequence of theorem 2.2.10 to the pairs (X, X_2) and (X_1, X_0) . Under the hypotheses (closed subspaces of a compact space whose union is X , so that $X_1 \hookrightarrow X$ is a cofibration with $X_1 \setminus X_0$ open in $X \setminus X_2$), excision (Axiom 3 of theorem 2.2.9) gives a homeomorphism $X/X_2 \cong X_1/X_0$, hence an isomorphism of relative groups

$$K^n(X, X_2) \cong K^n(X_1, X_0)$$

The Mayer-Vietoris sequence is the standard consequence of this excision isomorphism together with the two long exact sequences of pairs: splicing the pair sequences for (X, X_2) and (X_1, X_0) along the common relative term, and using periodicity to fold the result into six terms, produces the displayed hexagon, completing the proof.

These two corollaries are the tools used in section 2.3 to compute the K-theory of spheres, projective spaces, and other spaces built by gluing. The reduced cofiber sequence handles spaces presented as a quotient, and Mayer-Vietoris handles spaces presented as a union; periodicity guarantees that in either case only two groups, $\tilde{K}^0$ and $\tilde{K}^1$, ever appear.

Exercise 2.2.1

1. Using definition 2.2.1 and the computation of $\tilde{K}^0(S^n)$ in example 2.2.2, determine $\tilde{K}^{-n}(S^1)$ for all $n \geq 0$. Then identify $K^1(\{*\})$ and explain why it vanishes.
2. Let $\beta \in \tilde{K}^0(S^2)$ be the Bott generator and let β_X be the Bott map of definition 2.2.6. Show directly from the external product that $\beta_{S^0}: \tilde{K}^0(S^0) \rightarrow \tilde{K}^0(S^2)$ sends the unit to β , and deduce that β_{S^0} is an isomorphism without invoking theorem 2.2.7.
3. Use the reduced cofiber sequence (corollary 2.2.11) for the pair (D^2, S^1) to recompute $\tilde{K}^0(S^2)$ and $\tilde{K}^1(S^2)$, given that $\tilde{K}^*(D^2) = 0$ (contractibility) and $\tilde{K}^0(S^1) = 0$, $\tilde{K}^1(S^1) \cong \mathbb{Z}$.
4. Verify the excision homeomorphism $(X \setminus U)/(A \setminus U) \cong X/A$ used in the proof of theorem 2.2.9 in the explicit case $X = [0, 2]$, $A = [0, 1]$, $U = [0, \frac{1}{2}]$, by exhibiting the homeomorphism of quotients.
5. Let $X = S^2 \vee S^2$. Using Mayer-Vietoris (corollary 2.2.12) or the reduced cofiber sequence, compute $\tilde{K}^0(S^2 \vee S^2)$ and $\tilde{K}^1(S^2 \vee S^2)$.
6. Show that the coefficient ring $K^*(\{*\})$ of complex K-theory is the Laurent polynomial ring $\mathbb{Z}[\beta, \beta^{-1}]$ with β in degree -2 , and contrast this with the coefficient ring of ordinary cohomology. Explain in one sentence why this difference is exactly the failure of the dimension axiom.

2.3 Computations in K-Theory

The machinery assembled in the previous two sections (the ring $K^0(X)$ of section 2.1, the Bott periodicity isomorphism, and the long exact sequence that turns K^* into a cohomology theory in section 2.2) is worth very little until it produces numbers. This section computes K^* for the spaces a topologist meets first: spheres, complex projective spaces, and products. Every input is exhibited explicitly. The reduced groups $\tilde{K}^0(S^n)$ fall out of the suspension and clutching descriptions together with 2-periodicity; the ring $K^0(\mathbb{CP}^n)$ is computed from the projective-bundle structure and the Leray-Hirsch theorem; and the external product yields a Künneth statement for $K^*(X \times Y)$. The recurring theme is that K-theory, unlike ordinary cohomology, sees the multiplicative geometry of line bundles directly: the generator of $\tilde{K}^0(\mathbb{CP}^n)$ is literally $[H] - 1$ for the tautological line bundle H , and the single relation $([H] - 1)^{n+1} = 0$ encodes the whole ring.

Throughout, $K^0(X) = KU(X)$ denotes complex topological K-theory unless stated otherwise, and $\tilde{K}^0(X)$ is the reduced group, the kernel of the rank map $K^0(X) \rightarrow \mathbb{Z}$ at a chosen basepoint (definition 2.1.4). For a finite CW pair the negative groups $K^{-n}(X)$ and the relative groups

$K^0(X, A)$ are those of definitions 2.2.1 and 2.2.3, and Bott periodicity supplies the isomorphism $\tilde{K}^0(X) \cong \tilde{K}^0(\Sigma^2 X)$ used at every turn (theorem 2.2.7).

The spheres are the natural starting point, since their bundle theory is already in hand. The clutching description of section 1.2 already computed the unreduced bundle sets $\text{Vect}_{\mathbb{C}}^n(S^k)$; what remains is to package them into the K-groups and to read off the ring structure. The cleanest route is through reduced K-theory and the identity $\tilde{K}^0(S^k) \cong \tilde{K}^0(\Sigma^2 S^k) = \tilde{K}^0(S^{k+2})$ supplied by Bott periodicity, which reduces every sphere to one of two base cases. The base cases are S^0 and S^1 , both of which are immediate from the bundle classifications already in hand.

Example 2.3.1: The K-Theory of Spheres

For complex K-theory the reduced groups of the spheres are

$$\tilde{K}^0(S^n) \cong \begin{cases} \mathbb{Z} & n \text{ even} \\ 0 & n \text{ odd} \end{cases} \quad \tilde{K}^{-1}(S^n) \cong \begin{cases} 0 & n \text{ even} \\ \mathbb{Z} & n \text{ odd} \end{cases}$$

and the full unreduced ring of the even sphere is $K^0(S^{2k}) \cong \mathbb{Z}[\eta]/(\eta^2)$, where $\eta = \beta^k \in \tilde{K}^0(S^{2k})$ is the k -fold Bott class and the generator squares to zero.

Step 1: The two base cases. The space S^1 is the suspension ΣS^0 , and every complex vector bundle over it is trivial: by proposition 1.2.8 the isomorphism classes of n -dimensional complex bundles over S^1 are in bijection with $[S^0, \text{GL}_n(\mathbb{C})] = \pi_0(\text{GL}_n(\mathbb{C}))$, which is a single point because $\text{GL}_n(\mathbb{C})$ is path-connected (this is the content of example 1.2.11, item (1), on the complex side). Hence $\text{Vect}_{\mathbb{C}}^n(S^1)$ is a single class for each n , every element of $K^0(S^1)$ is a difference of trivial bundles, and $\tilde{K}^0(S^1) = 0$. For $S^0 = \{+, -\}$ a two-point space, a complex bundle is just a pair of vector spaces, so $K^0(S^0) \cong \mathbb{Z} \oplus \mathbb{Z}$ by rank at each point, and the reduced group (kernel of rank at $+$) is $\tilde{K}^0(S^0) \cong \mathbb{Z}$, generated by the class $[\mathbb{C}_-] - [\mathbb{C}_+]$ supported at the non-basepoint.

Step 2: Suspension and clutching for S^2 . For $k \geq 2$ the correspondence proposition 1.2.8 identifies $\text{Vect}_{\mathbb{C}}^n(S^k)$ with $[S^{k-1}, \text{GL}_n(\mathbb{C})] \cong \pi_{k-1}(\text{GL}_n(\mathbb{C}))$ (based homotopy classes, using path-connectedness to drop the basepoint). In the stable range, and after deformation-retracting $\text{GL}_n(\mathbb{C})$ onto $U(n)$, the reduced group $\tilde{K}^0(S^k)$ is the colimit $\varinjlim_n \pi_{k-1}(U(n)) = \pi_{k-1}(U)$, exactly the stabilization theorem 1.3.2. For $k = 2$ this is $\pi_1(U) \cong \mathbb{Z}$ by Bott periodicity (theorem 1.3.1), so

$$\tilde{K}^0(S^2) \cong \pi_1(U) \cong \mathbb{Z}$$

A generator is the class $\beta = [H] - 1$, where $H \rightarrow \mathbb{CP}^1 = S^2$ is the tautological line bundle and 1 is the trivial line bundle; this β is the *Bott class*, the same element implementing the periodicity isomorphism theorem 2.2.7.

Step 3: Periodicity propagates the answer. Each sphere is a double suspension of one two lower, $S^n = \Sigma^2 S^{n-2}$, so Bott periodicity theorem 2.2.7 gives

$$\tilde{K}^0(S^n) \cong \tilde{K}^0(S^{n-2})$$

Starting from $\tilde{K}^0(S^0) \cong \mathbb{Z}$ and $\tilde{K}^0(S^1) = 0$, induction yields $\tilde{K}^0(S^{2k}) \cong \mathbb{Z}$ and $\tilde{K}^0(S^{2k+1}) = 0$. The statement for $\tilde{K}^{-1}$ follows by the same periodicity from the suspension identity $\tilde{K}^{-1}(S^n) = \tilde{K}^0(\Sigma S^n) = \tilde{K}^0(S^{n+1})$.

Step 4: The ring structure. It remains to see that the generator of $\tilde{K}^0(S^{2k})$ squares to zero. Write $\eta = \beta^k$ for the k -fold external Bott class, a generator of $\tilde{K}^0(S^{2k}) \cong \mathbb{Z}$. The product η^2 lies in

$\tilde{K}^0(S^{2k}) \cdot \tilde{K}^0(S^{2k})$, and the cup product on a reduced group factors through the smash product: multiplication $\tilde{K}^0(S^{2k}) \otimes \tilde{K}^0(S^{2k}) \rightarrow \tilde{K}^0(S^{2k})$ is induced by the diagonal $\Delta : S^{2k} \rightarrow S^{2k} \times S^{2k}$ followed by the projection to the smash $S^{2k} \wedge S^{2k} = S^{4k}$. The composite $S^{2k} \xrightarrow{\Delta} S^{2k} \times S^{2k} \rightarrow S^{2k} \wedge S^{2k}$ is null-homotopic, because the diagonal of a sphere lands in the wedge $S^{2k} \vee S^{2k}$ up to homotopy and the wedge maps to a point in the smash. Hence $\eta^2 = 0$, giving

$$K^0(S^{2k}) \cong \mathbb{Z}[\eta]/(\eta^2)$$

with $\eta = \beta^k$ the generator of the reduced group, completing the computation.

The reduction to S^0 and S^1 is the whole story: K-theory of spheres is $\mathbb{Z}$ in even degrees and 0 in odd degrees, with all higher structure governed by the single Bott class. The vanishing $\eta^2 = 0$ is forced by dimension once one knows that products of reduced classes live on smash products: there is simply no room for β^{2k} in $\tilde{K}^0(S^{2k})$ when $4k > 2k$. This is the K-theoretic shadow of the fact that $H^*(S^n)$ has trivial cup products in positive degrees, and it is the reason the even spheres are “square-zero” extensions of $\mathbb{Z}$. The contrast with $\mathbb{CP}^n$ below is instructive: there the generator does *not* square to zero until the $(n+1)$ -st power, and the difference is exactly that $\mathbb{CP}^n$ has cohomology in many even degrees while S^{2k} has it in only one.

A remark on the real theory rounds out the picture. The same suspension-and-clutching argument applied to KO uses $\pi_{k-1}(O)$ in place of $\pi_{k-1}(U)$, and the 8-fold periodicity of theorem 1.3.1 produces the well-known pattern $\tilde{KO}^0(S^n) \cong \mathbb{Z}, \mathbb{Z}/2, \mathbb{Z}/2, 0, \mathbb{Z}, 0, 0, 0$ for $n \equiv 0, 1, \dots, 7 \pmod{8}$. The two $\mathbb{Z}/2$ ’s in dimensions 1 and 2 are precisely the Möbius-type phenomena recorded in example 1.2.11; they have no analogue in the complex theory, where $\pi_{2j}(U) = 0$ kills the odd spheres outright.

2.3.2 Remark: The Index-Theoretic Meaning of β The generator $\beta \in \tilde{K}^0(S^2)$ is not an arbitrary choice. Under the identification $S^2 = \mathbb{CP}^1$ it is $[H] - 1$, and under the isomorphism $\tilde{K}^0(S^2) \cong \tilde{K}^{-2}(\{*\})$ it is the periodicity element whose inverse multiplication *is* the Bott map. When one later builds the analytic index of [5], β becomes the K-theory class of the symbol of the Dolbeault operator on $\mathbb{CP}^1$, and the equation $\beta^2 = 0$ on the sphere is the statement that this symbol class has no room to interact with itself. The reader should keep $\beta = [H] - 1$ in mind: it is the single generator from which the K-theory of all the spaces in this section is built.

The spheres gave a ring that is essentially trivial ($\mathbb{Z}$ with a square-zero generator). Complex projective space is the first genuinely multiplicative example, and it is the model computation of the subject. The target is the statement that $K^0(\mathbb{CP}^n)$ is a truncated polynomial ring on one generator, with the truncation degree equal to $n+1$. Two independent inputs each prove it: the projective-bundle / Leray-Hirsch computation, which is elementary and self-contained, and the Atiyah-Hirzebruch spectral sequence, which explains *why* the answer has the same shape as $H^*(\mathbb{CP}^n)$. Both are given.

Theorem 2.3.3: K-Theory of Complex Projective Space

Let $H \rightarrow \mathbb{CP}^n$ be the tautological (canonical) complex line bundle and set $t = [H] - 1 \in K^0(\mathbb{CP}^n)$. Then $t^{n+1} = 0$ and

$$K^0(\mathbb{CP}^n) \cong \mathbb{Z}[t]/(t^{n+1})$$

as a ring, a free abelian group of rank $n + 1$ on the basis $1, t, t^2, \dots, t^n$. The odd group vanishes: $K^{-1}(\mathbb{CP}^n) = 0$. The reduced group $\tilde{K}^0(\mathbb{CP}^n)$ is the ideal (t) , free of rank n on $t, \dots, t^n$.

The interpretive content is that the entire ring is generated by the single line bundle H , and that the only relation is the vanishing of $t^{n+1} = ([H] - 1)^{n+1}$. Geometrically, $([H] - 1)^{n+1} = 0$ is a K-theoretic Cayley-Hamilton identity: the line bundle H satisfies a degree- $(n + 1)$ “polynomial” because $\mathbb{CP}^n$ is $2n$ -dimensional and there is no room for a nonzero class in $\tilde{K}^0$ beyond the n -th power. The ring is abstractly isomorphic to the cohomology ring $H^*(\mathbb{CP}^n; \mathbb{Z}) \cong \mathbb{Z}[x]/(x^{n+1})$ with $|x| = 2$, but the isomorphism is not the identity on generators: t corresponds to $e^x - 1 = x + x^2/2 + \dots$ under the Chern character, not to x itself. The two proofs below make the ring structure and the vanishing relation completely explicit.

Proof:

The computation is carried out by the projective-bundle method; an alternative via the spectral sequence is given afterward.

Step 1: $\mathbb{CP}^n$ as a projective bundle and the basic class. Recall that $\mathbb{CP}^n = P(\mathbb{C}^{n+1})$ is the projectivization of the trivial bundle $\mathbb{C}^{n+1} \rightarrow \{*\}$. The tautological line bundle H has fiber over a line $\ell \in \mathbb{CP}^n$ equal to ℓ itself. Its first Chern class $x = c_1(H^*) \in H^2(\mathbb{CP}^n; \mathbb{Z})$ (with H^* the dual, so that x is the standard positive generator) generates the cohomology ring,

$$H^*(\mathbb{CP}^n; \mathbb{Z}) \cong \mathbb{Z}[x]/(x^{n+1})$$

a free abelian group with basis $1, x, \dots, x^n$, concentrated in even degrees ([5]). This is the cohomological input.

Step 2: Leray-Hirsch applied to K-theory. The projection $\mathbb{CP}^n \rightarrow \{*\}$ is the bundle whose fiber is $\mathbb{CP}^n$ itself; to obtain a Leray-Hirsch statement one instead views $\mathbb{CP}^n$ through its even cells. The class $t = [H] - 1 \in \tilde{K}^0(\mathbb{CP}^n)$ restricts on each cell, and the powers $1, t, \dots, t^n$ are the candidate module basis. To see that they form a basis, apply the Leray-Hirsch theorem 3.2.1 in K-theory to the sphere bundle $S(\mathbb{C}^{n+1}) = S^{2n+1} \rightarrow \mathbb{CP}^n$, or more directly run the Atiyah-Hirzebruch comparison of Step 4 below; either shows that $K^0(\mathbb{CP}^n)$ is free of rank $n + 1$ over $K^0(\{*\}) = \mathbb{Z}$ with basis the restrictions of $1, t, \dots, t^n$. We give the self-contained inductive argument.

Step 3: Induction via the cofibration $\mathbb{CP}^{n-1} \hookrightarrow \mathbb{CP}^n$. The pair $(\mathbb{CP}^n, \mathbb{CP}^{n-1})$ has quotient $\mathbb{CP}^n/\mathbb{CP}^{n-1} = S^{2n}$, the top cell. This yields the cofiber sequence and hence the long exact sequence of theorem 2.2.10 (with the relative group definition 2.2.3)

$$\dots \rightarrow \tilde{K}^{-1}(\mathbb{CP}^{n-1}) \rightarrow \tilde{K}^0(S^{2n}) \rightarrow \tilde{K}^0(\mathbb{CP}^n) \rightarrow \tilde{K}^0(\mathbb{CP}^{n-1}) \rightarrow \tilde{K}^1(S^{2n}) \rightarrow \dots$$

By example 2.3.1, $\tilde{K}^0(S^{2n}) \cong \mathbb{Z}$ and $\tilde{K}^{-1}(S^{2n}) = \tilde{K}^1(S^{2n}) = 0$, and by the inductive hypothesis $\tilde{K}^{-1}(\mathbb{CP}^{n-1}) = 0$ and $\tilde{K}^0(\mathbb{CP}^{n-1})$ is free of rank $n - 1$. The sequence therefore collapses to the

short exact sequence

$$0 \rightarrow \tilde{K}^0(S^{2n}) \rightarrow \tilde{K}^0(\mathbb{CP}^n) \rightarrow \tilde{K}^0(\mathbb{CP}^{n-1}) \rightarrow 0$$

with the two outer groups free of ranks 1 and $n - 1$. Since the right-hand group is free, the sequence splits and $\tilde{K}^0(\mathbb{CP}^n)$ is free of rank n ; together with the rank-1 summand $K^0(\{*\}) = \mathbb{Z}$ this gives $K^0(\mathbb{CP}^n)$ free of rank $n + 1$. The base case $n = 1$ is $\tilde{K}^0(\mathbb{CP}^1) = \tilde{K}^0(S^2) \cong \mathbb{Z}$ from example 2.3.1. The vanishing of the odd groups propagates the same way, giving $K^{-1}(\mathbb{CP}^n) = 0$ at every stage.

Step 4: Identifying the basis with powers of t . The class t^n generates the top summand $\tilde{K}^0(S^{2n})$. Indeed, the restriction of H to the top cell $S^{2n} = \mathbb{CP}^n / \mathbb{CP}^{n-1}$ measures, via the Chern character, the class x^n ; concretely, under the splitting of Step 3 the image of t^n in $\tilde{K}^0(\mathbb{CP}^{n-1})$ is 0 (because t^n restricts to a class on $\mathbb{CP}^{n-1}$ which is the n -th power of t there, and $t^n = 0$ on $\mathbb{CP}^{n-1}$ by induction), so t^n comes from $\tilde{K}^0(S^{2n})$, and it is a generator because its Chern character is $x^n \neq 0$. Hence $1, t, \dots, t^n$ is a $\mathbb{Z}$ -basis: each t^k generates the rank-1 subquotient supported on the $2k$ -cell. Consequently t^{n+1} , restricting to $t^{n+1} = 0$ on $\mathbb{CP}^n$ (it is supported beyond the top cell), vanishes. This is the single relation, and

$$K^0(\mathbb{CP}^n) \cong \mathbb{Z}[t]/(t^{n+1})$$

completing the proof.

Proof Alternative proof:

The Atiyah-Hirzebruch spectral sequence explains the answer structurally. For any finite CW complex X the spectral sequence (the Atiyah-Hirzebruch spectral sequence of [5], built from the skeletal filtration and the fact that K^* is a generalized cohomology theory in the sense of the generalized-cohomology axioms of [5]) has

$$E_2^{p,q} = H^p(X; K^q(\{*\})) \implies K^{p+q}(X)$$

where $K^q(\{*\}) = \mathbb{Z}$ for q even and 0 for q odd, by Bott periodicity theorem 2.2.7. For $X = \mathbb{CP}^n$ the cohomology $H^p(\mathbb{CP}^n; \mathbb{Z})$ is $\mathbb{Z}$ for p even with $0 \leq p \leq 2n$ and 0 otherwise. Thus on the E_2 page every nonzero entry $E_2^{p,q}$ has both p and q even, so $p + q$ is always even. Every differential $d_r : E_r^{p,q} \rightarrow E_r^{p+r, q-r+1}$ changes the parity of $p + q$ when r is even and, for r odd, maps between a $(p + q)$ -even entry and a $(p + q)$ -odd entry, which is zero. More precisely, d_r raises total degree by 1, hence sends an even-total-degree class to an odd-total-degree group, and all odd-total-degree groups vanish. Therefore every differential is zero and the spectral sequence collapses at E_2 :

$$E_\infty^{p,q} = E_2^{p,q} = H^p(\mathbb{CP}^n; K^q(\{*\}))$$

The associated graded of $K^0(\mathbb{CP}^n)$ is $\bigoplus_p E_\infty^{p,-p} = \bigoplus_{k=0}^n H^{2k}(\mathbb{CP}^n; \mathbb{Z}) \cong \mathbb{Z}^{n+1}$, and that of $K^{-1}(\mathbb{CP}^n)$ is 0 since all odd cohomology vanishes. As each filtration quotient is free, the filtration of $K^0(\mathbb{CP}^n)$ splits and $K^0(\mathbb{CP}^n) \cong \mathbb{Z}^{n+1}$, with $K^{-1}(\mathbb{CP}^n) = 0$. The multiplicative structure is read off from the filtration: $t = [H] - 1$ has filtration ≥ 2 (it is reduced) and its image in $E_\infty^{2,-2} = H^2(\mathbb{CP}^n; \mathbb{Z})$ is the generator x , so t^k has filtration $\geq 2k$ with leading term x^k . Since $x^{n+1} = 0$ in $H^*(\mathbb{CP}^n)$ and there is nothing in filtration $> 2n$, the products $1, t, \dots, t^n$ are a basis and $t^{n+1} = 0$, recovering $K^0(\mathbb{CP}^n) \cong \mathbb{Z}[t]/(t^{n+1})$, as we sought to show.

The two proofs illuminate different features. The cofibration induction is hands-on: it builds the answer one cell at a time and pins the generator t^k to the $2k$ -cell. The spectral sequence is conceptual: the collapse is forced purely by parity ($K^*(\{*\})$ is concentrated in even degrees and $\mathbb{CP}^n$ has cohomology only in even degrees, so no differential can connect two nonzero groups), and it shows that the abstract shape of $K^0(\mathbb{CP}^n)$ is dictated by $H^*(\mathbb{CP}^n)$. What the spectral sequence does *not* give for free is the precise generator; that $t = [H] - 1$ rather than some other lift of x is what makes the ring genuinely computable, and it is why the multiplicative statement is sharper than “ $K^0(\mathbb{CP}^n)$ is a free group of rank $n+1$ ”. The relation $([H] - 1)^{n+1} = 0$ should be compared with the Chern-class relation $c_1(H)^{n+1} = 0$: the latter holds in H^* , the former in K^0 , and the Chern character intertwines them.

Example 2.3.4: The Chern Character on $\mathbb{CP}^2$

Take $n = 2$, so $K^0(\mathbb{CP}^2) \cong \mathbb{Z}[t]/(t^3)$ with $t = [H] - 1$, free on $1, t, t^2$, and $H^*(\mathbb{CP}^2; \mathbb{Z}) \cong \mathbb{Z}[x]/(x^3)$ with $|x| = 2$. The Chern character $\text{ch} : K^0(X) \rightarrow H^{2*}(X; \mathbb{Q})$ is the ring homomorphism with $\text{ch}(L) = e^{c_1(L)}$ for a line bundle L . Since $c_1(H) = -x$ (the tautological bundle has negative degree; its dual H^* has $c_1 = x$), one computes

$$\text{ch}(H) = e^{-x} = 1 - x + \frac{1}{2}x^2$$

using $x^3 = 0$. Therefore

$$\text{ch}(t) = \text{ch}([H] - 1) = e^{-x} - 1 = -x + \frac{1}{2}x^2$$

and squaring (with $x^3 = 0$),

$$\text{ch}(t^2) = \text{ch}(t)^2 = \left(-x + \frac{1}{2}x^2\right)^2 = x^2$$

Thus on the basis $1, t, t^2$ the Chern character is

$$\text{ch}(1) = 1, \quad \text{ch}(t) = -x + \frac{1}{2}x^2, \quad \text{ch}(t^2) = x^2$$

The matrix of ch whose columns express $\text{ch}(1), \text{ch}(t), \text{ch}(t^2)$ in the basis $\{1, x, x^2\}$ is triangular,

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & \frac{1}{2} & 1 \end{pmatrix}$$

with determinant -1 , so ch is a rational isomorphism. The half-integer entry $\frac{1}{2}$ shows that the integral image $\text{ch}(K^0(\mathbb{CP}^2))$ is a *proper* sublattice of $H^{2*}(\mathbb{CP}^2; \mathbb{Z})$: solving $a \cdot 1 + b(-x + \frac{1}{2}x^2) + c x^2 = x$ forces $b = -1$ and $c = \frac{1}{2} \notin \mathbb{Z}$, so the cohomology generator x is not in the image. This is the integrality phenomenon: K^0 and H^{2*} are abstractly the same free group, but the Chern character identifies them only after inverting 2, and the half-integral discrepancy is the first appearance of the Todd genus that corrects the Riemann-Roch theorem.

The relation $t^{n+1} = 0$ also lets one compute the K-theory of $\mathbb{CP}^\infty$ as a limit, which is the generating case behind the splitting principle and the theory of Chern classes.

2.3.5 Remark: $K^0(\mathbb{CP}^\infty)$ as a Power Series Ring Passing to the colimit $\mathbb{CP}^\infty = \varinjlim_n \mathbb{CP}^n$ requires care: K-theory of an infinite complex is computed by the Milnor exact sequence relating $K^0(\mathbb{CP}^\infty)$ to the inverse limit of the $K^0(\mathbb{CP}^n)$ and a $\lim^1$ term. The transition maps $K^0(\mathbb{CP}^n) \rightarrow K^0(\mathbb{CP}^{n-1})$ are the restrictions $\mathbb{Z}[t]/(t^{n+1}) \rightarrow \mathbb{Z}[t]/(t^n)$ sending $t \mapsto t$, a surjective system, so the $\lim^1$ term vanishes and

$$K^0(\mathbb{CP}^\infty) \cong \varprojlim_n \mathbb{Z}[t]/(t^{n+1}) \cong \mathbb{Z}[[t]]$$

the formal power series ring in $t = [H] - 1$. This is the home of the universal Chern class: a complex line bundle on any space is classified by a map to $\mathbb{CP}^\infty = BU(1)$, and its K-theory invariant is a power series in t . The same computation in cohomology gives $H^*(\mathbb{CP}^\infty; \mathbb{Z}) \cong \mathbb{Z}[[x]]$ truncated to a polynomial ring in each finite degree; the appearance of genuine power series in K-theory (rather than polynomials) reflects that K-theory is not connective in the way ordinary cohomology is. The formal-group-law structure on $\mathbb{Z}[[t]]$ coming from the tensor product of line bundles, $t(L_1 \otimes L_2) = t(L_1) + t(L_2) + t(L_1)t(L_2)$, is the multiplicative formal group, in contrast to the additive formal group $x(L_1 \otimes L_2) = x(L_1) + x(L_2)$ of ordinary cohomology.

The third computation concerns products. Computing K^* of a product reduces, when it works, to the tensor product of the factors' K-theory. The mechanism is the external product, a pairing $K^0(X) \otimes K^0(Y) \rightarrow K^0(X \times Y)$ built from the external tensor product of bundles. After recording it and a worked instance, the section closes with the Künneth statement, proved by comparing the two sides as cohomology theories in one variable.

Example 2.3.6: The External Product on $S^2 \times S^2$

For vector bundles $E \rightarrow X$ and $F \rightarrow Y$, the *external tensor product* is $E \boxtimes F = \pi_X^*(E) \otimes \pi_Y^*(F) \rightarrow X \times Y$, where π_X, π_Y are the two projections. On classes this defines the external product

$$\times: K^0(X) \otimes K^0(Y) \longrightarrow K^0(X \times Y), \quad a \times b = \pi_X^*(a) \cdot \pi_Y^*(b)$$

which is a ring homomorphism out of the tensor-product ring. Take $X = Y = S^2$ with $\tilde{K}^0(S^2) = \mathbb{Z}\beta$, $\beta = [H] - 1$ and $\beta^2 = 0$ (example 2.3.1). Write $\beta_1 = \pi_1^*(\beta)$ and $\beta_2 = \pi_2^*(\beta)$ in $K^0(S^2 \times S^2)$. Since pullback is a ring map and $\beta^2 = 0$ on each factor, $\beta_1^2 = \beta_2^2 = 0$, while the cross term $\beta_1\beta_2 = \beta \times \beta$ is the external square. The external product gives a ring isomorphism

$$K^0(S^2 \times S^2) \cong K^0(S^2) \otimes_{\mathbb{Z}} K^0(S^2) \cong \frac{\mathbb{Z}[\beta_1, \beta_2]}{(\beta_1^2, \beta_2^2)}$$

a free abelian group of rank 4 on the basis $1, \beta_1, \beta_2, \beta_1\beta_2$. The reduced group $\tilde{K}^0(S^2 \times S^2)$ is the rank-3 ideal (β_1, β_2) , and one checks directly that $\beta_1\beta_2$ generates the top class supported on the 4-cell $S^2 \wedge S^2 = S^4$: indeed under the collapse $S^2 \times S^2 \rightarrow S^2 \wedge S^2 = S^4$ the class $\beta_1\beta_2$ pulls back from a generator of $\tilde{K}^0(S^4) \cong \mathbb{Z}$, consistent with $\tilde{K}^0(S^4) \cong \tilde{K}^0(S^2)$ under Bott periodicity. The element $\beta_1\beta_2$ is the external Bott class $\beta \times \beta = \beta^2 \in \tilde{K}^0(S^4)$, the same $\eta = \beta^2$ that appeared as the generator of $\tilde{K}^0(S^4)$ in example 2.3.1; the apparent clash with $\beta^2 = 0$ is resolved by noting that the *internal* square on a single S^2 vanishes while the *external* square on $S^2 \times S^2$ does not, because the diagonal $S^2 \rightarrow S^2 \times S^2$ kills it but the product space retains it.

The example exhibits the general pattern: the K-theory of a product is the tensor product of the

K-theories, provided no torsion or extension obstructions intervene. The precise statement is the Künneth theorem in K-theory, which holds cleanly when one factor has torsion-free, finitely generated K-theory.

Theorem 2.3.7: Künneth Theorem in K-Theory

Let X and Y be finite CW complexes, and suppose at least one of $K^*(X)$, $K^*(Y)$ is a finitely generated free abelian group (for instance whenever that factor is S^n or $\mathbb{CP}^n$). Then the external product is a $\mathbb{Z}/2$ -graded ring isomorphism

$$K^*(X) \otimes_{\mathbb{Z}} K^*(Y) \xrightarrow{\sim} K^*(X \times Y)$$

In particular, if both $K^*(X)$ and $K^*(Y)$ are finitely generated and free, then so is $K^*(X \times Y)$, with rank equal to the product of the ranks.

In words: when the K-theory of one factor is free, computing K^* of a product is pure algebra, the tensor product of graded rings. The freeness hypothesis is what avoids a Tor correction term; in full generality there is a short exact Künneth sequence with a $\text{Tor}_1^{K^*(\{*\})}$ summand, exactly as in ordinary cohomology, but the spaces of this chapter (spheres, projective spaces, and their products) all have free K-theory, so the clean statement applies to every example here.

Proof:

Step 1: The external product is well-defined and natural. For $E \rightarrow X$ and $F \rightarrow Y$ the bundle $E \boxtimes F = \pi_X^* E \otimes \pi_Y^* F$ depends only on the isomorphism classes of E and F by functoriality of pullback (proposition 1.2.3) and bilinearity of the tensor product over $\oplus$ (definition 1.1.17). Hence $a \times b = \pi_X^*(a) \cdot \pi_Y^*(b)$ extends to a homomorphism $K^0(X) \otimes K^0(Y) \rightarrow K^0(X \times Y)$, and using the negative groups (definition 2.2.1) to a graded homomorphism $K^*(X) \otimes K^*(Y) \rightarrow K^*(X \times Y)$. It is a ring map because pullback is a ring map and the cup product on $K^*(X \times Y)$ is graded-commutative.

Step 2: Both sides are cohomology theories in the inducting variable. By symmetry of the external product (the two factors enter on equal footing, since $X \times Y \cong Y \times X$) it costs nothing to label the factors so that Y is the one whose K-theory is finitely generated and free; this is exactly the hypothesis. Fix this free factor Y and regard both sides as cohomology theories in the remaining variable X , over whose cells the induction will run. The functor $X \mapsto K^*(X \times Y)$ is a generalized cohomology theory (it is K^* precomposed with $- \times Y$, and $- \times Y$ preserves cofiber sequences), and $X \mapsto K^*(X) \otimes_{\mathbb{Z}} K^*(Y)$ is a cohomology theory because the fixed factor $K^*(Y)$ is a free $\mathbb{Z}$ -module, so tensoring with it is exact and preserves the long exact sequences (theorem 2.2.10). The external product is a natural transformation between these two theories.

Step 3: The transformation is an isomorphism on a point, hence everywhere. Regard K^* as $\mathbb{Z}/2$ -graded via Bott periodicity, so that the coefficient ring is $K^*(\{*\}) = \mathbb{Z}$ concentrated in even degree. On $X = \{*\}$ the external product is

$$K^*(\{*\}) \otimes_{\mathbb{Z}} K^*(Y) = \mathbb{Z} \otimes_{\mathbb{Z}} K^*(Y) \xrightarrow{\sim} K^*(Y)$$

the unit isomorphism of the tensor product, since $K^*(\{*\}) = \mathbb{Z}$. A natural transformation of cohomology theories on finite CW complexes that is an isomorphism on the point (together with the additivity axiom, which handles disjoint unions and hence S^0) is an isomorphism on all finite CW complexes: this is the standard comparison via the five lemma and induction over cells, run on the skeletal filtration. Concretely, suppose the transformation is an isomorphism

for all complexes with fewer cells than X . Attaching a cell to a subcomplex X' gives a cofiber sequence $X' \hookrightarrow X \rightarrow X/X' = S^m$, and naturality produces a map of long exact sequences (theorem 2.2.10) in which the maps for X' and for S^m are isomorphisms (the latter because $K^*(S^m) \otimes K^*(Y) \rightarrow K^*(S^m \times Y)$ is an iterated suspension of the point case, using Bott periodicity theorem 2.2.7). The five lemma then forces the map for X to be an isomorphism.

Step 4: Conclusion. The induction starts from $X = \{*\}$ (Step 3) and covers every finite CW complex X , so for the fixed free factor Y the external product

$$K^*(X) \otimes_{\mathbb{Z}} K^*(Y) \xrightarrow{\sim} K^*(X \times Y)$$

is an isomorphism for *every* finite CW complex X , with no freeness assumed on $K^*(X)$. The freeness that was used is that of the fixed factor $K^*(Y)$, which is what made $X \mapsto K^*(X) \otimes K^*(Y)$ a cohomology theory in Step 2; by the relabelling at the start of that step this covers the stated hypothesis that at least one of $K^*(X)$, $K^*(Y)$ is finitely generated and free. When *both* factors are finitely generated and free, the rank and freeness statements follow since a tensor product of finitely generated free abelian groups is finitely generated and free of the product rank, completing the proof.

With the Künneth theorem the K-theory of every product of spheres and projective spaces is now mechanical. For instance $K^0((\mathbb{CP}^n)^{\times m}) \cong \mathbb{Z}[t_1, \dots, t_m]/(t_1^{n+1}, \dots, t_m^{n+1})$, the tensor power of $\mathbb{Z}[t]/(t^{n+1})$, with each $t_i = \pi_i^*([H] - 1)$ the pulled-back generator from the i -th factor. This is the K-theoretic analogue of the cohomology of a product of projective spaces, and it is the computation that underlies the splitting principle: a sum of line bundles over a base B is classified by a map to $(\mathbb{CP}^\infty)^{\times m} = BU(1)^m$, and its K-theory class is a polynomial in the pulled-back generators.

Exercise 2.3.1

1. Using only the suspension identity $\tilde{K}^{-1}(X) = \tilde{K}^0(\Sigma X)$ and the values $\tilde{K}^0(S^0) \cong \mathbb{Z}$, $\tilde{K}^0(S^1) = 0$, together with Bott periodicity, compute $\tilde{K}^0(S^n)$ and $\tilde{K}^{-1}(S^n)$ for $n = 3, 4, 5, 6$. State the unreduced groups $K^0(S^n)$ and $K^{-1}(S^n)$ in each case.
2. Let $t = [H] - 1 \in K^0(\mathbb{CP}^2)$. Compute, in the basis $1, t, t^2$ of $\mathbb{Z}[t]/(t^3)$, the products $t \cdot t$, $t \cdot t^2$, and $(1+t)^2$. Deduce the K-theory class of $H^{\otimes 2} = H^2$ as an element of $\mathbb{Z}[t]/(t^3)$.
3. Use the Atiyah-Hirzebruch spectral sequence to compute $K^*(S^2 \times S^4)$. State the E_2 page explicitly, explain why all differentials vanish, and give the ranks of K^0 and K^{-1} . Confirm the answer agrees with the Künneth theorem applied to $K^*(S^2) \otimes K^*(S^4)$.
4. Let $H \rightarrow \mathbb{CP}^1 = S^2$ be the tautological line bundle and $\beta = [H] - 1$. Show directly, without invoking the smash-product argument, that $\beta^2 = 0$ in $K^0(\mathbb{CP}^1)$ by using the relation $t^{n+1} = 0$ of theorem 2.3.3 for $n = 1$. Then explain why the external square $\beta \times \beta$ is nonzero in $K^0(S^2 \times S^2)$.
5. Compute the rank of $\tilde{K}^0(\mathbb{CP}^n \times \mathbb{CP}^m)$ and write down a $\mathbb{Z}$ -basis in terms of the pulled-back generators $t_1 = \pi_1^*([H] - 1)$ and $t_2 = \pi_2^*([H] - 1)$. Identify the single class supported on the top cell.
6. The reduced suspension formula gives $\tilde{K}^0(\Sigma X) \cong \tilde{K}^{-1}(X)$. Using $\mathbb{CP}^2/\mathbb{CP}^1 = S^4$ and the cofiber long exact sequence, recompute $\tilde{K}^0(\mathbb{CP}^2)$ and verify it is free of rank 2, matching theorem 2.3.3.

Characteristic Classes

In this chapter, we shall explore the relation between vector bundles and cohomology rings, in particular there shall be many vector bundles that correspond to certain generators of cohomology rings, sometimes even generating them. The images of these vector bundles into cohomology rings shall be called *characteristic classes*, or sometimes more rarely *cohomology classes*. As cohomology rings represent some non trivial twisting, so do vector bundles. This connection between “twisting” or some obstruction to triviality shall be made precise in this section by showing how many linearly independent sections can exist for certain characteristic classes.

This section shall cover 4 important characteristic classes;

1. Stiefel-Whitney classes: $w_i(E) \in H^i(B; \mathbb{Z}/2\mathbb{Z})$ for real vector bundles $E \rightarrow B$
2. Chern classes: $c_i(E) \in H^{2i}(B; \mathbb{Z})$ for complex vector bundles $E \rightarrow B$
3. Pontryagin classes: $p_i(E) \in H^{4i}(B; \mathbb{Z})$ for oriented real vector bundles $E \rightarrow B$
4. Euler class: $e(E) \in H^n(B; \mathbb{Z})$ for an oriented n -dimensional real vector bundle $E \rightarrow B$ (note there is only one)

We shall explore each of these characteristic classes throughout this chapter. The idea behind all of these classes is to measure how far away a vector bundle $E \rightarrow B$ is from being trivial. By theorem 1.2.5, if $f : B \rightarrow \text{Grass}_n$ is nullhomotopic, $f : E \rightarrow B$ is trivial. Measuring the failure of this map being nullhomotopic would then use the tools of homotopy and cohomology theory.

3.1 Definition and Intuitions

As (co)homology classes can be defined over general groups $H^i(X; G)$, it is worth-while being a little more general in our definition:

Definition 3.1.1: Characteristic Classes

Let G be a topological group, and $b_G(-)$ be the contravariant functor associating to each topological space B the set (sometimes group, sometimes ring)^a $b_G(B)$ and f to the pullback f^* . Then a *characteristic class* is a natural transformation

$$c : b_G \Rightarrow H^*$$

In particular, a characteristic class c associated to each principal G -bundle $P \rightarrow X$ to an element $c(P) \in H^*(X)$ where

$$c(f^*P) = f^*c(P)$$

where the f^* on the left hand side is the pullback map and the f^* on the right hand side is the image of the class of P (i.e. $c(P)$) by the induced map in cohomology. For a specific H^i , we shall denote the characteristic class as c_i .

^adepending on the choice of G

For the remainder of this section, we shall focus on the case of $G = V$ is a vector space, either real or complex, and stick to the convention that $\text{Vect}^n(B)$ means $\text{Vect}_{\mathbb{R}}^n(B)$. Then we have that $\text{Vect}(B)$ is associated to $[B, \text{Grass}_n]$. In particular, if $f : B \rightarrow \text{Grass}_n$ is a classifying map and $E = f^*(E_n)$, then

$$c_i(E) = c_i(f^*(E_n)) = f^*(c_i(E_n))$$

Thus, if f induces the trivial map on G -cohomology, then the characteristic classes $c_i(E)$ are trivial. Let us now think of the specific classes:

Definition 3.1.2: Stiefel-Whitney Classes

If $G = \mathbb{Z}/2\mathbb{Z}$ and $b_G(B) = \text{Vect}^n(B)$ for some $n \in \mathbb{N}_{>0}$, then the characteristic class is called the *Stiefel-Whitney Class*, and will be denoted $w(E)$ (or $w_i(E)$ for the $H^i(B; \mathbb{Z}/2\mathbb{Z})$ cohomology)

These classes shall be of interest to us as they shall generate the cohomology ring $H^*(\text{Grass}_n; \mathbb{Z}/2\mathbb{Z})$, in case:

$$H^*(\text{Grass}_n; \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}_2[w_1(E_n), \dots, w_n(E_n)]$$

In fact we shall prove in [ref:HERE](#). This is not the case if $\mathbb{Z}$ -coefficients are used, we will need the Pontryagin class to characterize these:

Definition 3.1.3: Pontryagin Classes

If $G = \mathbb{Z}$ and $b_G(B) = \text{Vect}^n(B)$ for some $n \in \mathbb{N}_{>0}$, then the characteristic class is called the *Pontryagin-Whitney Class*, and will be denoted $w(E)$ (or $w_i(E)$ for the $H^{4i}(B; \mathbb{Z})$ cohomology)

As hinted in the above, the values of interest are in H^{4i} as these are the only classes with nonzero values. We shall show that f induces the trivial map in $\mathbb{Z}$ coefficient if the Stiefel-Whitney and Pontryagin classes of E are all zero. If E is orientable, then we can take $f : E \rightarrow \widetilde{\text{Grass}}_n$ where $\widetilde{\text{Grass}}_n$ has the orientation structure defined earlier. Then we get:

Definition 3.1.4: Euler Class

If $G = \mathbb{Z}$ and $b_G(B) = \widetilde{\text{Vect}}^n(B)$ of orientable real vector bundles for some $n \in \mathbb{N}_{>0}$, then the characteristic class is called the *Euler Class*, and will be denoted $e(E)$.

We shall show that f induces the trivial map for an oriented real vector bundle if the Stiefel-Whitney, Pontryagin, and Euler classes are trivial.

Finally, as the complex Grassmannian has a different structure do to how eliminating 0 from $\mathbb{C}$ leaves a path-connected component, we shall have the following:

Definition 3.1.5: Chern Class

If $G = \mathbb{Z}$ and $b_G(B) = \text{Vect}_{\mathbb{C}}^n(B)$ of complex bundles for some $n \in \mathbb{N}_{>0}$, then the characteristic class is called the *Chern Class*, and will be denoted $c(E)$ (or $c_i(E)$ for the $H^{2i}(B; \mathbb{Z})$ cohomology)

We shall show that

$$H^*(\text{Grass}_n(\mathbb{C}); \mathbb{Z}) \cong \mathbb{Z}[c_1, \dots, c_n]$$

where $c_i = c_i(\text{Grass}_n(\mathbb{C}))$

Note how in many of the above we required to use many types of characteristic classes to determine triviality. This is because trivial characteristic classes are necessary but not sufficient conditions to determine triviality. The simplest non-trivial example is S^5 , which we established in section 1.3 is non-trivial, but it shall have trivial characteristic classes for all but the Euler class. This is because the first three characteristic classes have the property of being *stably invariant* meaning that direct sums with trivial bundles does not change the characteristic class. As we shall see, all spheres are stably trivial, meaning they direct sum to the trivial bundle, and hence the first 3 classes cannot distinguish odd dimensional spheres.

Let us now think about for a moment what characteristic classes are detecting. The first “obstruction” it detects is orientability. If $E \rightarrow B$ is a vector bundle where B is path-connected, then we may define a map:

$$\pi_1(B) \rightarrow \mathbb{Z}/2\mathbb{Z}$$

where it assigns 0 or 1 to each loop according to whether orientation of fibers is preserved or reversed (as one goes around the loop). As $\mathbb{Z}/2\mathbb{Z}$ is abelian, the homomorphism factors through the abelianization $H_1(B)$ of $\pi_1(B)$, and the homomorphisms $H_1(B) \rightarrow \mathbb{Z}/2\mathbb{Z}$ are elements of $H^1(B; \mathbb{Z}/2\mathbb{Z})$. From this, we have elements of $H^1(B; \mathbb{Z}/2\mathbb{Z})$ associated to E which when E is orientable. This shall be the interpretation of the first Stiefel-Whitney class $w_1(E)$.

To generalize to higher Stiefel-Whitney classes, let us re-interpret this result when B is a CW complex. Then E is orientable if and only if the restriction of the 1-skeleton B^1 of B is orientable, as all loops in B can be deformed to lie in B^1 . Then a vector bundle over a 1-dimensional CW complex is trivial if and only if it is orientable, so we see that $w_1(E)$ measures whether E is trivial over the 1-skeleton B^1 . Notice that the fibers can be higher dimensional, namely on each loop we are looking at vector bundles on S^1 . Now, we can ask the following: given E is trivial over the 1-skeleton, can we extend it to be trivial over the 2-skeleton. Assuming E is trivial over B^1 and its fibers are n -dimensional, we can choose n orthonormal sections over B^1 and ask whether these sections extend to orthonormal sections over each 2-cell. As each 2-cell is given by the characteristic map $D^2 \rightarrow B$, we can

pull back E over the disk D^2 , where the sections over B^1 pullback to sections over ∂D^2 . As D^2 is contractible, this bundle is trivial. Choosing a trivialization, the sections over ∂D^2 determine a map:

$$\partial D^2 \rightarrow O(n)$$

Choosing a trivialization that has the same orientation as the trivialization of E over B^1 , we have the image is in $SO(n)$. Such a map is exactly an element of $\pi_1(SO(n))$, hence we have that each 2-cell is identified with an element of $\pi_1(SO(n))$. Then B^1 extends to orthonormal sections over B^2 if and only if this element of $\pi_1(SO(n))$ is zero for each 2-cell.

This groups is also relatively easy to compute : when $n = 1$ it is 0, when $n = 2$ it is $\mathbb{Z}$ (as $SO(2)$ is homeomorphic to S^1). For $n \geq 3$, it is $\mathbb{Z}/2\mathbb{Z}$ (exercise). Thus, the inclusion $SO(2) \hookrightarrow SO(n)$ induces a surjection $\pi_1(SO(2)) \twoheadrightarrow \pi_1(SO(n))$, so the generator of $\pi_1(SO(n))$ can be represented by the loop given by rotating the plane through 2π degrees, fixing the orthogonal complement of the plane. From this, we can define a function which assigns each 2-cell (regarded as a 2-dimensional cellular cochain) with coefficients in $\pi_1(SO(n))$. This shall be a cellular cocycle, and so an element of $H^2(B; \pi_1(SO(n)))$. This element shall be called $w_2(E)$, and will not depend on the choice of sections over B^1 . We can thus see that E is trivial over B^2 if and only if $w_2(E)$ is zero in $H^2(B; \pi_1(SO(n)))$. Note that if $n = 1$, clearly $w_2(E) = 0$. If $n = 2$, then $w_2(E) \in H^2(B; \mathbb{Z})$ and is equal to the Euler class $e(E)$ which shall be covered later. If $n \geq 3$, then $w_2(E)$ is in $H^2(B; \mathbb{Z}/2\mathbb{Z})$.

The natural next question to ask if if E is trivial over B^2 , can it be extended to be trivial over B^3 which again is given by looking at elements of $\pi_2(SO(n))$. Fortunately for us, $\pi_2(SO(n)) = 0$ for all $n \in \mathbb{N}_{>0}$ and so the sections automatically extend over B^3 . For extended to 4-cells, we have that $\pi_3(SO(n))$ is relatively simple to compute (see [8, p.75]). For higher dimensional extension B^k for $k \geq 5$, things become a good deal more complicated as the groups $\pi_{k-1}(SO(n))$ become increasingly difficult to compute. We know they are finitely generated abelian groups whose free part is fairly easy to compute, but the torsion part is complicated¹. In the range of $n > k$, then Bott periodicity gives the value of $\pi_{k-1}(SO(n))$. For $k = 4i$, we get:

$$\pi_{4i-1}(SO(n)) \cong \mathbb{Z}$$

which is why the Pontryagin class $p_i(E) \in H^{4i}(B; \mathbb{Z})$, and from the above we see it detects the non-triviality of bundles with obstruction:

$$w_{4i}(E) \in H^{4i}(B; \pi_{4i-1}(SO(n)))$$

Given the dimension $n > 4i$, we get that the other nontrivial stable groups are:

$$\pi_{k-1}(SO(n)) \cong \mathbb{Z}/2\mathbb{Z} \quad k = 8i + 1, 8i + 2$$

However, the obstructions $w_k(E)$ in these cases shall not be detected by Stiefel-Whitney or Pontryagin classes except when $i = 0$. (see [ref:HERE](#)).

Another way of measuring triviality will lead to the Euler class. Consider an n -dimensional vector bundle $E \rightarrow B$. Let's say we are looking for an obstruction to the existence of n orthonormal sections. Let us first consider the case where we are looking for an obstruction of there being 1 orthonormal section, or more particularly a section of unit vectors. As before we can proceed by induction over the skeleta of B . Assuming we have section over B^{k-1} , to extend over the k -cell we can pull back E

¹Interestingly, this contrasts with elliptic curves over rationals $E(\mathbb{Q})$ for which it is relatively easy to compute the torsion but a million dollar problem (literally) to compute the free part

as before to the trivial bundle over D^k via the characteristic map so that we are looking at the unit vector section ∂D^k . But this time we were not looking for orientability, this is *exactly* the what we tried to accomplish. Hence, if $k < n$, such an extension always is possible, so the first (possibly) nontrivial obstruction occurs when $k = n$ where

$$\pi_{k-1}(S^{n-1}) \cong \mathbb{Z}$$

Note that to identify $\mathbb{Z}$ to $\pi_{n-1}(S^{n-1})$ in a consistent way, it is *necessary* be that E is orientable. Thus class is the Euler class and shall be denoted $e(E)$. Dropping the orientability condition, we must work with $\mathbb{Z}/2\mathbb{Z}$ coefficients and we get the Stiefel-Whitney class $w_n(E) \in H^n(B; \mathbb{Z}/2\mathbb{Z})$.

3.2 Stiefel-Whitney and Chern Classes

The intuition presented above is computationally difficult. The approach we shall take here will be require more algebraic topology, but will be equivalent (see 3.4 to for the comparisons). As usual, we shall assume the base space is paracompact. The following two results will be needed for the proof of the existence of the characteristic classes:

Theorem 3.2.1: Leray-Hirsch Theorem

Let R be a commutative ring and $p : E \rightarrow B$ be a fiber bundle. In [5] it was shown $H^*(E; R)$ is a $H^*(B; R)$ -module where

$$\alpha\beta = p^*(\alpha) \smile \beta$$

with $\alpha \in H^*(B)$ and $\beta \in H^*(E)$. If

1. for each fiber F the inclusion $F \hookrightarrow E$ induces a surjection on $H^*(-; R)$
2. $H^n(F; R)$ is a free R -module of finite rank for each n .

Then $H^*(E; R)$ is a free $H^*(B; R)$ -module. A basis for the $H^*(B; R)$ -module $H^*(E; R)$ can be obtained by choosing any set of elements in $H^*(E; R)$ that map to a basis for $H^*(F; R)$ under the map induced by the inclusion.

Proof:

This is the Leray-Hirsch theorem, proved via the collapse of the Serre spectral sequence in [5] (spectral-sequences chapter), for a fibration whose fiber cohomology is free of finite rank and restricts from the total space.

Proposition 3.2.2: Splitting Principle

Let $\pi : E \rightarrow B$ be a vector bundle. Then there exists a space $F(E)$ and a map $p : F(E) \rightarrow B$ such that the pullback $p^*(E) \rightarrow F(E)$ splits as a direct sum of line bundles, and

$$p^* : H^*(B; \mathbb{Z}/2\mathbb{Z}) \hookrightarrow H^*(F(E); \mathbb{Z}/2\mathbb{Z})$$

is injective

Proof:

This is the splitting principle, proved in full in [5]: the flag bundle $F(E) \rightarrow B$ is built as a tower of projectivizations, and the Leray-Hirsch theorem (theorem 3.2.1) makes each stage injective on cohomology, so p^* is injective and p^*E splits into a sum of line bundles.

Theorem 3.2.3: Existence of Stiefel-Whitney Classes

Let $E \rightarrow B$ be a real vector bundle. Then there exists a sequence of functions $w_1, w_2, \dots$ where $w_i(E) \in H^i(B; \mathbb{Z}/2\mathbb{Z})$ that depends only on the isomorphism type of E such that

1. $w_i(f^*(E)) = f^*(w_i(E))$ for a pullback $f^*(E)$
2. $w(E_1 \oplus E_2) = w(E_1) \smile w(E_2)$ where $w = 1 + w_1 + w_2 + \dots \in H^*(B; \mathbb{Z}/2\mathbb{Z})$
3. $w_i(E) = 0$ if $i > \dim E$
4. Given $E \rightarrow \mathbb{RP}^\infty$ is the canonical line bundle, $w_1(E)$ is a generator of $H^1(\mathbb{RP}^\infty; \mathbb{Z}/2\mathbb{Z})$.

The Stiefel-Whitney class w is called the *total Stiefel-Whitney class*.

Note that by (3), w only has finitely many nonzero terms, showing it is well-defined in $H^i(B; \mathbb{Z}/2\mathbb{Z})$. The $\oplus$ to $\smile$ is shorthand for the Whitney sum formula:

$$w_n(E_1 \oplus E_2) = \sum_{i+j=n} w_i(E_1) \smile w_j(E_2)$$

where $w_0 = 1$. Property (4) is a non triviality condition, as we can just say $w_i(E) = 0$ for all $i > 0$. As proving the result for Chern classes is ostensibly the same, we shall in fact do both:

Theorem 3.2.4: Existence of Chern Classes

Let $E \rightarrow B$ be a complex vector bundle. Then there exists a sequence of functions $c_1, c_2, \dots$ where $c_i(E) \in H^{2i}(B; \mathbb{Z})$ that depends only on the isomorphism type of E such that

1. $c_i(f^*(E)) = f^*(c_i(E))$ for a pullback $f^*(E)$
2. $c(E_1 \oplus E_2) = c(E_1) \smile c(E_2)$ where $c = 1 + c_1 + c_2 + \dots \in H^*(B; \mathbb{Z})$
3. $c_i(E) = 0$ if $i > \dim E$
4. Given $E \rightarrow \mathbb{CP}^\infty$ is the canonical line bundle, $c_1(E)$ is a generator of $H^1(\mathbb{CP}^\infty; \mathbb{Z})$ specified in advance.

The extra condition for (4) is a non triviality *and* a normalization condition, as each function c_i can be replaced by $k^i c_i$ for a fixed integer k which gives new functions satisfying (1)-(3).

Proof:

Both existence statements are proved in full in [5]. There the Stiefel-Whitney and Chern classes are constructed as the coefficients of the defining relation $x^n + w_1(E)x^{n-1} + \dots + w_n(E) = 0$ (respectively $x^n - c_1(E)x^{n-1} + \dots + (-1)^n c_n(E) = 0$) in the free $H^*(B)$ -module $H^*(P(E))$ supplied by the Leray-Hirsch theorem (theorem 3.2.1), with the axioms (1)-(4) verified there. This is the bundle-theoretic projective-bundle route; [5] also computes the same classes as the

polynomial generators of $H^*(BU(n))$ and $H^*(BO(n); \mathbb{Z}/2\mathbb{Z})$ by the Serre spectral sequence and shows the two constructions agree.

Note that for property (4), the canonical line bundle over $\mathbb{RP}^1$ could have been used instead of $\mathbb{RP}^\infty$ as $\mathbb{RP}^1 \hookrightarrow \mathbb{RP}^\infty$ induces an isomorphism

$$H^1(\mathbb{RP}^\infty; \mathbb{Z}/2\mathbb{Z}) \cong H^1(\mathbb{RP}^1; \mathbb{Z}/2\mathbb{Z})$$

and similarly for Chern classes.

Example 3.2.5: Stiefel-Whitney and Chern Classes

1. By property (1), the trivial bundle $B \times \mathbb{R}^n$ has

$$w_i(E) = 0 \quad \forall i > 0$$

as the product is the pullback of the bundle over a point.

2. By property (b), the direct sum of a bundle with a product bundle does not change its Stiefel-Whitney classes; this property is called *stable*. Using this, we see that the tangent bundle TS^n to S^n is stably trivial, and hence

$$w_i(TS^n) = 0 \quad \forall i > 0$$

More generally, as :

$$(1 + w_1 + \cdots)(1 + w'_1 + \cdots) = 1 + (w_1 + w'_1) + (w_2 + w_1w'_2 + w'_2) + \cdots$$

if we have $w(E_1)$ and $w(E_1 \oplus E_2)$, we can determine $w(E_2)$ as:

$$\begin{aligned} w_1 + w'_1 &= a_1 \\ w_2 + w_1w'_1 + w'_2 &= a_2 \\ &\vdots \\ \sum_i w_{n-1}w'_i &= a_n \end{aligned}$$

If $E_1 \oplus E_2$ is the trivial bundle, $a_i = 0$ for all $i > 0$, so $w(E_1)$ determines $w(E_2)$ uniquely by the above formulas. The same applies to Chern classes.

3. We can prove there is no bundle $E \rightarrow \mathbb{RP}^\infty$ whose sum with the line bundle $E_1(\mathbb{R}^\infty)$ is trivial. For

$$w(E_1(\mathbb{R}^\infty)) = 1 + \omega$$

where ω is a generator of $H^1(\mathbb{RP}^\infty; \mathbb{Z}/2\mathbb{Z})$. Then we have (as we are using $\mathbb{Z}/2\mathbb{Z}$ coefficients):

$$w(E) = (1 + \omega)^{-1} = 1 + \omega + \omega^2 + \cdots$$

Thus we have nonzero $w_i(E) = \omega^i$ for all i , but that contradicts $w_i(E) = 0$ for all $i > \dim E$. We finally showed the necessity for compactness for proposition 1.1.16.

4. (Hatcher construct an n -dimensional vector bundle $E \rightarrow B$ where $w_i(E)$ is nonzero for each $i \leq E$, [8, p.82]).

Proposition 3.2.6: Relating Chern and Stiefel-Whitney Classes

Let $E \rightarrow B$ be a complex n -dimensional vector bundle, and regard it as a $2n$ -dimensional real vector bundle. Then

$$w_{2i+1}(E) = 0$$

and $w_{2i}(E)$ is the image of $c_i(E)$ under the coefficient homomorphism:

$$H^{2i}(B; \mathbb{Z}) \rightarrow H^{2i}(B; \mathbb{Z}/2\mathbb{Z})$$

Proof:

Proved in [5] by comparing the real and complex projectivizations $\mathbb{RP}(E) \rightarrow \mathbb{CP}(E)$: the mod-2 reduction of the Chern defining relation pulls back to the Stiefel-Whitney defining relation, giving $w_{2i+1}(E) = 0$ and $w_{2i}(E) = \bar{c}_i(E)$.

The cohomology rings of the Grassmannians, $H^*(\text{Grass}_n(\mathbb{C}); \mathbb{Z}) = \mathbb{Z}[c_1, \dots, c_n]$ and $H^*(\text{Grass}_n(\mathbb{R}); \mathbb{Z}/2\mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}[w_1, \dots, w_n]$, are computed in [5] (there as $H^*(BU(n))$ and $H^*(BO(n); \mathbb{Z}/2\mathbb{Z})$), realizing the Chern and Stiefel-Whitney classes as the polynomial generators.

3.3 Euler and Pontryagin Classes

The Stiefel-Whitney and Chern classes of the previous section live in $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}$ cohomology and are built uniformly from the projective bundle of E . The two remaining classes from definition 3.1.4 and definition 3.1.3 are different in character. The Euler class $e(E)$ is an integral class attached to an *oriented* real bundle of rank exactly equal to its top degree; it is the integral refinement of the top Stiefel-Whitney class and the precise obstruction to a nowhere-zero section. The Pontryagin classes $p_i(E)$ are integral classes of an oriented real bundle obtained by complexifying E and reading off its Chern classes. This section develops both from the existence theorems already proved (theorem 3.2.4), relates the Euler class to the Thom class and to the top Chern class, establishes the Whitney-sum and naturality behaviour of each, and ends by attaching to a closed smooth manifold its Pontryagin and Stiefel-Whitney *numbers*, the characteristic numbers that govern cobordism.

Throughout, $E \rightarrow B$ is a vector bundle over a paracompact base, and cohomology is singular cohomology with the indicated coefficients. The orientation of a real bundle is a choice of generator of $H^n(E, E_0; \mathbb{Z}) \cong \mathbb{Z}$ over each fiber, where E_0 denotes the complement of the zero section, fitting together continuously; this is the data already packaged in the functor $\widetilde{\text{Vect}}^n$ of definition 3.1.4.

The cleanest construction of the Euler class passes through the Thom isomorphism, which was established in [5]. For an oriented real vector bundle $\pi : E \rightarrow B$ of rank n , let $E_0 \subseteq E$ be the complement of the zero section. The orientation determines a *Thom class*

$$u(E) \in H^n(E, E_0; \mathbb{Z})$$

restricting on each fiber $(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\})$ to the preferred generator of $H^n(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; \mathbb{Z}) \cong \mathbb{Z}$, and cup product with $u(E)$ induces the Thom isomorphism $H^k(B; \mathbb{Z}) \xrightarrow{\sim} H^{k+n}(E, E_0; \mathbb{Z})$ ([5]). The zero section $z : B \rightarrow E$ of definition 1.1.9 is a homotopy equivalence, so $H^*(E; \mathbb{Z}) \cong H^*(B; \mathbb{Z})$, and one may pull the Thom class back along the inclusion $(E, \emptyset) \hookrightarrow (E, E_0)$ to obtain a class in $H^n(E; \mathbb{Z}) \cong H^n(B; \mathbb{Z})$.

Proposition 3.3.1: Euler Class as Restriction of the Thom Class

Let $\pi : E \rightarrow B$ be an oriented real vector bundle of rank n , with Thom class $u(E) \in H^n(E, E_0; \mathbb{Z})$. Write $j^* : H^n(E, E_0; \mathbb{Z}) \rightarrow H^n(E; \mathbb{Z})$ for the map induced by the inclusion of pairs $(E, \emptyset) \hookrightarrow (E, E_0)$, and let $z : B \rightarrow E$ be the zero section. Then the Euler class of definition 3.1.4 is

$$e(E) = z^*(j^*(u(E))) \in H^n(B; \mathbb{Z})$$

Equivalently, under the identification $\pi^* : H^n(B; \mathbb{Z}) \xrightarrow{\sim} H^n(E; \mathbb{Z})$, the class $j^*(u(E))$ corresponds to $e(E)$.

The two formulations agree because z and π are mutually inverse homotopy equivalences, so $z^* = (\pi^*)^{-1}$ on cohomology. The content of the proposition is that the Euler class is what remains of the Thom class once the relative information (the behaviour near the zero section) is forgotten and only the absolute class on B is retained.

Proof:

This is the construction of the Euler class in [5], as the restriction of the Thom class to the zero section; the Thom-isomorphism details are given there.

This description makes two features immediate. First, the degree of $e(E)$ equals the rank of E , in contrast to the Chern and Pontryagin classes which live in twice or four times the index. Second, the Euler class is genuinely an orientation-dependent invariant: reversing the orientation of E negates the Thom class, hence negates $e(E)$. The reduction of $e(E)$ modulo 2 loses the orientation data and recovers the top Stiefel-Whitney class.

Proposition 3.3.2: Mod-2 Reduction of the Euler Class

Let $E \rightarrow B$ be an oriented real vector bundle of rank n . Then the image of $e(E)$ under the coefficient reduction $\rho : H^n(B; \mathbb{Z}) \rightarrow H^n(B; \mathbb{Z}/2\mathbb{Z})$ is the top Stiefel-Whitney class:

$$\rho(e(E)) = w_n(E)$$

Proof:

Proved in [5]: the mod-2 reduction of the integral Euler class is the top Stiefel-Whitney class.

The Euler class comes into its own for complex bundles, where it recovers a Chern class. A complex vector bundle of complex rank n has a canonical orientation as a real bundle of rank $2n$: a complex basis $v_1, iv_1, \dots, v_n, iv_n$ of a fiber is an oriented real basis, and any two complex bases are related by an element of $\mathrm{GL}_n(\mathbb{C})$, which is connected, so the induced real orientation is independent of the choice. Thus every complex bundle has a well-defined Euler class of its underlying oriented real bundle, and one expects it to coincide with the top Chern class, which is the only Chern class living in the top real degree $2n$. The next theorem confirms this.

Theorem 3.3.3: Euler Class Equals Top Chern Class

Let $E \rightarrow B$ be a complex vector bundle of complex rank n , and let $E_{\mathbb{R}}$ denote its underlying real vector bundle of rank $2n$, with the canonical orientation described above. Then

$$e(E_{\mathbb{R}}) = c_n(E)$$

in $H^{2n}(B; \mathbb{Z})$, where $c_n(E)$ is the top Chern class of definition 3.1.5.

Proof:

Proved in [5] by the splitting principle: for a complex bundle the top Chern class equals the Euler class of the underlying canonically-oriented real bundle.

The theorem identifies the top Chern class with a section-theoretic invariant: $c_n(E)$ is exactly the obstruction to a nowhere-zero section of the complex bundle E . This is the algebraic-topology incarnation of the fact that a generic section of a rank- n complex bundle over a $2n$ -manifold vanishes at a finite set of points whose signed count is $\langle c_n(E), [B] \rangle$. The identification also gives a uniform formula for the Euler class of any complex bundle, which is the route by which Euler classes are computed in practice.

Example 3.3.4: Euler Class of the Tangent Bundle of S^2

The tangent bundle TS^2 is an oriented real rank-2 bundle. Its Euler class lies in $H^2(S^2; \mathbb{Z}) \cong \mathbb{Z}$, so it is determined by the Euler number $\langle e(TS^2), [S^2] \rangle$. Two routes compute it.

First, $S^2 = \mathbb{CP}^1$ and $T\mathbb{CP}^1$ is a complex line bundle. As recorded in section 1.2, TS^2 is the oriented plane bundle $E_2 \rightarrow S^2$ with clutching function $z \mapsto z^2$ of degree 2. The Euler number of $E_m \rightarrow S^2$ is m relative to the standard orientation, so

$$\langle e(TS^2), [S^2] \rangle = 2$$

By theorem 3.3.3 this also equals $\langle c_1(T\mathbb{CP}^1), [\mathbb{CP}^1] \rangle$.

Second, the Euler number of the tangent bundle of any closed oriented surface equals its Euler characteristic, by the Poincaré-Hopf theorem ([5]). Since $\chi(S^2) = 2$, this agrees: $\langle e(TS^2), [S^2] \rangle = 2$. In particular $e(TS^2) \neq 0$, which is the cohomological statement of the Hairy Ball theorem: TS^2 has no nowhere-zero section.

The Euler class shares the formal structure of the other characteristic classes: naturality under pullback and multiplicativity under Whitney sum. Both were used inside the proof of theorem 3.3.3; the following collects them as a standalone statement, together with the section-theoretic vanishing criterion that motivates the obstruction-theoretic viewpoint of the next section.

Proposition 3.3.5: Properties of the Euler Class

Let $E, E_1, E_2 \rightarrow B$ be oriented real vector bundles over a paracompact base, with E of rank n . Then:

1. (Naturality) For $f : B' \rightarrow B$ continuous, $e(f^*E) = f^*(e(E))$, with f^*E given the pulled-back orientation.
2. (Whitney sum) $e(E_1 \oplus E_2) = e(E_1) \smile e(E_2)$, with $E_1 \oplus E_2$ given the direct-sum orientation.
3. (Odd rank torsion) If n is odd then $2e(E) = 0$.
4. (Vanishing) If E admits a nowhere-zero section, then $e(E) = 0$.

Proof:

Each property is established in [5], where the Euler class is developed from the Thom isomorphism and its naturality, Whitney-sum behaviour, and the 2-torsion of the odd-rank Euler class are proved.

Property (4) is the bridge to the obstruction-theoretic reading of section 3.4: a nonzero Euler class is an obstruction to the existence of a nowhere-zero section, and it is in fact the *primary* such obstruction, the first one encountered as one extends a section over the skeleta of B . Property (3) explains why Euler classes of odd-rank bundles carry only 2-torsion information and are therefore usually studied for even-rank (in particular complex) bundles. The Whitney formula (2) should be contrasted with the additive index behaviour of the top Chern class: it is consistent with theorem 3.3.3 because the top Chern class of a sum is the product of the top Chern classes only when one accounts for the degree shift, exactly as in the proof above.

The Pontryagin classes of definition 3.1.3 are defined for oriented real bundles and refine the obstruction information in degrees divisible by 4. The efficient definition realizes them through Chern classes of the *complexification*. Given a real vector bundle $E \rightarrow B$, its complexification is the complex bundle

$$E \otimes \mathbb{C} := E \otimes_{\mathbb{R}} \underline{\mathbb{C}}$$

of complex rank equal to the real rank of E , where $\underline{\mathbb{C}}$ is the trivial complex line bundle; fiberwise this tensors each real fiber with $\mathbb{C}$. The complexification carries a canonical conjugation, and the central computational fact is that $E \otimes \mathbb{C}$ is isomorphic to its own conjugate.

Theorem 3.3.6: Pontryagin Classes via Chern Classes

Let $E \rightarrow B$ be a real vector bundle. Then the complexification $E \otimes \mathbb{C}$ is isomorphic to its conjugate bundle $\overline{E} \otimes \mathbb{C}$, so its odd Chern classes are 2-torsion:

$$2c_{2i+1}(E \otimes \mathbb{C}) = 0$$

The Pontryagin class of definition 3.1.3 is given by

$$p_i(E) = (-1)^i c_{2i}(E \otimes \mathbb{C}) \in H^{4i}(B; \mathbb{Z})$$

and the total Pontryagin class $p(E) = 1 + p_1(E) + p_2(E) + \cdots$ satisfies, modulo elements of order 2,

$$p(E_1 \oplus E_2) \equiv p(E_1) \smile p(E_2) \pmod{2\text{-torsion}}$$

Proof:

Proved in [5]: the complexification $E \otimes \mathbb{C}$ is self-conjugate, so its odd Chern classes are 2-torsion and the Pontryagin classes are its even Chern classes with the sign $(-1)^i$, giving $p_i(E) = (-1)^i c_{2i}(E \otimes \mathbb{C})$.

The theorem reduces every question about Pontryagin classes to a question about Chern classes of the complexification, where the full machinery of the splitting principle and the Whitney formula is available. The 2-torsion caveat in the Whitney formula is genuine and not an artifact: the failure of strict multiplicativity is detected precisely by the odd Chern classes of the complexifications, which are 2-torsion but need not vanish. Over a base with no 2-torsion in its integral cohomology, or after tensoring with $\mathbb{Z}[\frac{1}{2}]$ or $\mathbb{Q}$, the Pontryagin classes are strictly multiplicative.

A first consequence relates the Pontryagin classes of a complex bundle to its own Chern classes, which is the form in which Pontryagin classes of tangent bundles of complex manifolds are computed.

Corollary 3.3.7: Pontryagin Classes of a Complex Bundle

Let $F \rightarrow B$ be a complex vector bundle of complex rank n , regarded as an oriented real bundle $F_{\mathbb{R}}$. Then $F_{\mathbb{R}} \otimes \mathbb{C} \cong F \oplus \overline{F}$, and the Pontryagin classes of $F_{\mathbb{R}}$ are the degree- $(4i)$ components of the product $c(F) \smile c(\overline{F})$ up to sign: for each i ,

$$p_i(F_{\mathbb{R}}) = (-1)^i \sum_{r+s=2i} (-1)^s c_r(F) \smile c_s(F)$$

in $H^{4i}(B; \mathbb{Z})$. Equivalently, the equality of total classes

$$1 - p_1(F_{\mathbb{R}}) + p_2(F_{\mathbb{R}}) - \cdots = (1 + c_1(F) + \cdots + c_n(F)) \smile (1 - c_1(F) + c_2(F) - \cdots)$$

holds in the part of $H^*(B; \mathbb{Z})$ of degrees divisible by 4; the degree- $(4i+2)$ components of the right-hand side are 2-torsion and are discarded.

Proof:

Proved in [5]: for a complex bundle V the complexification of its underlying real bundle splits as $V_{\mathbb{R}} \otimes \mathbb{C} \cong V \oplus \bar{V}$, from which the Pontryagin classes are read off.

Example 3.3.8: Pontryagin Class of the Tangent Bundle of $\mathbb{CP}^2$

Let $a \in H^2(\mathbb{CP}^2; \mathbb{Z})$ be the generator with $\langle a^2, [\mathbb{CP}^2] \rangle = 1$, normalized so that $a = c_1$ of the hyperplane bundle. The cohomology ring is $H^*(\mathbb{CP}^2; \mathbb{Z}) = \mathbb{Z}[a]/(a^3)$. The total Chern class of the tangent bundle is

$$c(T\mathbb{CP}^2) = (1 + a)^3 = 1 + 3a + 3a^2$$

using the Euler sequence $T\mathbb{CP}^2 \oplus \underline{\mathbb{C}} \cong (H^*)^{\oplus 3}$ and theorem 3.2.4(2), where H^* is the dual of the tautological bundle with $c_1(H^*) = a$. Thus $c_1 = 3a$ and $c_2 = 3a^2$.

To compute p_1 , complexify the real tangent bundle. By corollary 3.3.7, $(T\mathbb{CP}^2)_{\mathbb{R}} \otimes \mathbb{C} \cong T\mathbb{CP}^2 \oplus \overline{T\mathbb{CP}^2}$, with

$$c(\overline{T\mathbb{CP}^2}) = 1 - 3a + 3a^2$$

since $c_k(\bar{F}) = (-1)^k c_k(F)$. Multiplying,

$$c((T\mathbb{CP}^2)_{\mathbb{R}} \otimes \mathbb{C}) = (1 + 3a + 3a^2)(1 - 3a + 3a^2) = 1 + (3 + 3 - 9)a^2 = 1 - 3a^2$$

where the degree-6 and higher terms vanish since $a^3 = 0$. Hence $c_2((T\mathbb{CP}^2)_{\mathbb{R}} \otimes \mathbb{C}) = -3a^2$, and by theorem 3.3.6,

$$p_1(T\mathbb{CP}^2) = (-1)^1 c_2((T\mathbb{CP}^2)_{\mathbb{R}} \otimes \mathbb{C}) = 3a^2$$

So the total Pontryagin class is $p(T\mathbb{CP}^2) = 1 + 3a^2$.

The payoff of the whole theory of characteristic classes is the assignment of numerical invariants to manifolds. When the base is a closed smooth manifold, characteristic classes of its tangent bundle can be evaluated against the fundamental class to produce integers (or elements of $\mathbb{Z}/2\mathbb{Z}$). These characteristic numbers are the invariants through which cobordism theory is governed: they are the complete set of invariants detecting when a manifold bounds.

Recall that a closed smooth m -manifold M has a fundamental class $[M] \in H_m(M; \mathbb{Z}/2\mathbb{Z})$ always, and $[M] \in H_m(M; \mathbb{Z})$ when M is oriented ([5]). A partition of an integer m is a sequence $I = (i_1, \dots, i_r)$ with $i_1 + \dots + i_r = m$; for partitions into the relevant degrees one evaluates the corresponding monomial in characteristic classes on $[M]$.

Definition 3.3.9: Stiefel-Whitney Number

Let M be a closed smooth m -manifold with tangent bundle TM and fundamental class $[M] \in H_m(M; \mathbb{Z}/2\mathbb{Z})$. For a partition $I = (i_1, \dots, i_r)$ of m , the associated *Stiefel-Whitney number* is

$$w_I[M] = \langle w_{i_1}(TM) \smile \dots \smile w_{i_r}(TM), [M] \rangle \in \mathbb{Z}/2\mathbb{Z}$$

using the Stiefel-Whitney classes of definition 3.1.2. Only partitions I of m give possibly-nonzero numbers, since the relevant cup product must land in $H^m(M; \mathbb{Z}/2\mathbb{Z})$.

Definition 3.3.10: Pontryagin Number

Let M be a closed oriented smooth manifold of dimension $m = 4k$ with tangent bundle TM and fundamental class $[M] \in H_m(M; \mathbb{Z})$. For a partition $I = (i_1, \dots, i_r)$ of k , the associated *Pontryagin number* is

$$p_I[M] = \langle p_{i_1}(TM) \smile \dots \smile p_{i_r}(TM), [M] \rangle \in \mathbb{Z}$$

using the Pontryagin classes of definition 3.1.3. Since $p_{i_j}(TM) \in H^{4i_j}(M; \mathbb{Z})$, the cup product lands in $H^{4k}(M; \mathbb{Z}) = H^m(M; \mathbb{Z})$ exactly when I partitions k ; Pontryagin numbers are defined only when $\dim M \equiv 0 \pmod{4}$.

The Pontryagin and Stiefel-Whitney numbers are diffeomorphism invariants of M , being built functorially from the tangent bundle and the fundamental class. Their significance is that they are unchanged under the much coarser equivalence of cobordism: two closed manifolds are cobordant if their disjoint union bounds a compact manifold with boundary. Characteristic numbers are cobordism invariants because the relevant classes extend over the bounding manifold while the fundamental class of a boundary is itself a boundary, so the pairing vanishes; this is the content of the next example's computation pattern and is proved in full in [5] via the long exact sequence of the pair $(W, \partial W)$.

Example 3.3.11: Pontryagin Number of $\mathbb{CP}^2$

The manifold $\mathbb{CP}^2$ is closed, oriented, of real dimension $4 = 4 \cdot 1$, so it has a single Pontryagin number, indexed by the partition $I = (1)$ of $k = 1$. From example 3.3.8, $p_1(T\mathbb{CP}^2) = 3a^2$ with $\langle a^2, [\mathbb{CP}^2] \rangle = 1$. Therefore

$$p_1[\mathbb{CP}^2] = \langle p_1(T\mathbb{CP}^2), [\mathbb{CP}^2] \rangle = \langle 3a^2, [\mathbb{CP}^2] \rangle = 3$$

This is consistent with the Hirzebruch signature theorem, which in dimension 4 reads $\sigma(M) = \frac{1}{3}p_1[M]$: the signature of $\mathbb{CP}^2$ is $\sigma(\mathbb{CP}^2) = 1$ (its intersection form on H^2 is the rank-one form $\langle a, a \rangle = 1$), and indeed $\frac{1}{3} \cdot 3 = 1$. Since $p_1[\mathbb{CP}^2] = 3 \neq 0$, the manifold $\mathbb{CP}^2$ is not the boundary of any compact oriented 5-manifold: it represents a nonzero class in the oriented cobordism group $\Omega_4^{SO} \cong \mathbb{Z}$.

The two systems of characteristic numbers control the two basic cobordism theories. Thom's theorems make this precise: unoriented cobordism is detected completely by Stiefel-Whitney numbers, and oriented cobordism is detected rationally by Pontryagin (and Stiefel-Whitney) numbers. The proofs require the computation of the cobordism rings as homotopy groups of Thom spectra and are beyond the scope of this book; complete accounts are in the cited sources.

Theorem 3.3.12: Characteristic Numbers Determine Cobordism

Let M, N be closed smooth manifolds of the same dimension.

1. (Unoriented cobordism, Thom) M and N are cobordant (over the unoriented cobordism relation) if and only if they have the same Stiefel-Whitney numbers $w_I[M] = w_I[N]$ for every partition I . In particular, M bounds a compact manifold if and only if all its Stiefel-Whitney numbers vanish. The unoriented cobordism ring $\mathfrak{N}_*$ is a polynomial algebra over $\mathbb{Z}/2\mathbb{Z}$.
2. (Oriented cobordism, Thom) If M and N are oriented and all their Pontryagin numbers and Stiefel-Whitney numbers agree, then M and N are cobordant in the rational oriented cobordism group; equivalently, some disjoint union of copies of M is oriented-cobordant to the corresponding union of copies of N . The oriented cobordism ring tensored with $\mathbb{Q}$ is the polynomial algebra $\Omega_*^{SO} \otimes \mathbb{Q} \cong \mathbb{Q}[\mathbb{CP}^2, \mathbb{CP}^4, \dots]$ generated by the even complex projective spaces, and Pontryagin numbers give a complete set of rational invariants.

Proof Key ideas:

The full proof identifies the cobordism groups with stable homotopy groups of Thom spectra (MO for the unoriented case, MSO for the oriented case) via the Pontryagin-Thom construction, then computes those homotopy groups. This machinery (Thom spectra, the Thom isomorphism in generalized cohomology, and the computation of $\mathfrak{N}_*$ and $\Omega_*^{SO} \otimes \mathbb{Q}$) is developed in full in [5], where the substantive converse (vanishing characteristic numbers imply bounding) is proved.

The forward implication, that characteristic numbers are cobordism invariants, is within scope: if W is a compact manifold with $\partial W = M$, then $TW|_M \cong TM \oplus \mathbb{R}$, so $w(TM)$ and $p(TM)$ are restrictions of classes on W ; the fundamental class $[M]$ is the image of the relative class under $\partial : H_{m+1}(W, \partial W) \rightarrow H_m(\partial W)$, and naturality of the cap product forces the pairing $\langle w_I(TM), [M] \rangle$ to vanish. Hence a manifold that bounds has all characteristic numbers zero, completing the in-scope direction.

The theorem closes the circle opened at the start of the chapter: characteristic classes were introduced as measures of the twisting of a bundle, and characteristic numbers turn that local twisting data into global invariants strong enough to classify manifolds up to the bounding relation. The unoriented statement is a complete invariant; the oriented statement is complete only rationally, because Ω_*^{SO} has torsion (detected by Stiefel-Whitney numbers and by more refined invariants) that Pontryagin numbers, being rational, cannot see. The appearance of $\mathbb{CP}^2, \mathbb{CP}^4, \dots$ as rational generators is why example 3.3.11 matters: $\mathbb{CP}^2$ is the first generator, and its nonzero Pontryagin number $p_1[\mathbb{CP}^2] = 3$ certifies that it is not rationally null-cobordant.

Exercise 3.3.1

1. Let $E \rightarrow B$ be an oriented real vector bundle of odd rank n over a base B with $H^n(B; \mathbb{Z})$ free of torsion (for instance $B = S^n$ with n odd). Show that $e(E) = 0$.
2. Using the Euler sequence $T\mathbb{CP}^n \oplus \mathbb{C} \cong (H^*)^{\oplus(n+1)}$, compute the Euler number $\langle e(T\mathbb{CP}^n), [\mathbb{CP}^n] \rangle$ of the tangent bundle of $\mathbb{CP}^n$, and check that it equals $\chi(\mathbb{CP}^n)$.
3. Let $E \rightarrow B$ be a real vector bundle that is the underlying real bundle of a complex line bundle L . Show that $p_1(E) = c_1(L)^2$.

4. Show that the Pontryagin classes are stable: $p_i(E \oplus \mathbb{R}^k) = p_i(E)$ for the trivial bundle $\mathbb{R}^k$. Deduce that $p_i(TS^n) = 0$ for all $i > 0$, using that $TS^n \oplus NS^n \cong \mathbb{R}^{n+1}$ with NS^n trivial. Then explain why this argument does *not* extend to an arbitrary stably trivial bundle in the sense of definition 1.1.12 (a bundle E with $E \oplus F \cong \mathbb{R}^N$ for *some* F , not necessarily trivial), and exhibit a counterexample using $T\mathbb{CP}^2$ from example 3.3.8.
5. Compute the total Pontryagin class $p(T\mathbb{CP}^3)$ and all Pontryagin numbers of $\mathbb{CP}^3$. Reconcile your answer with the fact that $\mathbb{CP}^3$ has odd complex dimension.
6. Let M be a closed oriented 4-manifold all of whose Stiefel-Whitney numbers and Pontryagin numbers vanish. What does theorem 3.3.12 allow you to conclude about M , and what can it not conclude? Illustrate with the distinction between M being null-cobordant and some multiple of M being null-cobordant.

3.4 Characteristic Classes as Obstructions

The algebraic construction of section 3.2 produces the Stiefel-Whitney and Chern classes through the Leray-Hirsch theorem, with no visible connection to the geometry of sections. The intuition sketched in section 3.2, by contrast, read every characteristic class as the obstruction to extending a partial section from one skeleton of the base to the next. This section makes that reading precise. The machinery is the primary-obstruction theory developed in [5] (the chapter on homotopy theory): given a fibration over a CW complex whose fiber is $(m-1)$ -connected, a section defined over the m -skeleton extends over the $(m+1)$ -skeleton precisely when a single cohomology class, the *primary obstruction* living in $H^{m+1}(B; \pi_m(F))$, vanishes. Applied to the sphere bundle of a vector bundle and to the various reductions of its structure group, this single principle recovers the Euler class as the obstruction to a nowhere-zero section, the first Stiefel-Whitney class as the obstruction to orientability, the second as the obstruction to a spin structure, and the top Chern class as the complex analogue of the Euler class. The uniform statement is that each characteristic class measures the failure of a reduction of the structure group, and we make this the organizing theme.

We begin by recording the form of the obstruction theory we will use, so that every later proof can cite a single fixed apparatus. Let $p: E \rightarrow B$ be a fibration over a CW complex B , with fiber F , and suppose F is path-connected and $(m-1)$ -connected (so $\pi_i(F) = 0$ for $i < m$ and $\pi_m(F) \neq 0$ is the first nonvanishing homotopy group). A section over the m -skeleton B^m always exists when the lower obstructions vanish, and the question of extending it over B^{m+1} is governed as follows. For each $(m+1)$ -cell e_α^{m+1} , with attaching map $\varphi_\alpha: S^m \rightarrow B^m$, the section restricted to $\varphi_\alpha(S^m)$ together with a trivialization of E over the cell determines an element of $\pi_m(F)$; assigning this element to e_α^{m+1} defines a cellular cochain

$$\mathfrak{o}(s) \in C^{m+1}(B; \pi_m(F))$$

which is in fact a cocycle. Its class

$$[\mathfrak{o}(s)] \in H^{m+1}(B; \pi_m(F))$$

is independent of the choice of section over the m -skeleton up to the choices made on the $(m-1)$ -skeleton; when F is m -simple (so $\pi_1(F)$ acts trivially on $\pi_m(F)$, automatic when $\pi_1(F) = 0$) this class depends only on the bundle, and is called the *primary obstruction*. The section extends over B^{m+1} if and only if $[\mathfrak{o}(s)] = 0$, and the primary obstruction is natural with respect to pullback. All of these statements are proved in [5]; we treat the chapter there as a black box and refer to it as the *obstruction machinery*.

The geometric problem “does E admit a nowhere-zero section” is, after normalizing lengths, the problem of finding a section of the unit sphere bundle. This reduction is recorded once for later use.

Definition 3.4.1: Sphere Bundle

Let $p: E \rightarrow B$ be a real vector bundle of rank n equipped with an inner product (which exists by proposition 1.1.14). The *sphere bundle* $S(E) \rightarrow B$ is the subspace of E consisting of unit vectors,

$$S(E) = \{v \in E \mid \langle v, v \rangle = 1\}$$

with the restricted projection. It is a fiber bundle with fiber the unit sphere $S^{n-1} \subset \mathbb{R}^n$.

A nowhere-zero section $s: B \rightarrow E$ of the vector bundle is the same datum as a section of $S(E)$, since dividing s by its (nowhere-zero, continuous) length $|s|$ lands in $S(E)$, and conversely a section of $S(E)$ is in particular a nowhere-zero section of E (see definition 1.1.8 for the notion of section). The fiber S^{n-1} is $(n-2)$ -connected with $\pi_{n-1}(S^{n-1}) \cong \mathbb{Z}$, so the obstruction machinery places the first obstruction to a section of $S(E)$ in $H^n(B; \pi_{n-1}(S^{n-1}))$. The remaining input is that this group is $H^n(B; \mathbb{Z})$ once an orientation has been fixed, since identifying $\pi_{n-1}(S^{n-1})$ with $\mathbb{Z}$ canonically is exactly a choice of orientation of the fiber. This is the geometric content of the orientability hypothesis in the definition of the Euler class.

The Euler class $e(E) \in H^n(B; \mathbb{Z})$ of an oriented rank- n bundle was introduced in definition 3.1.4 as a characteristic class, defined for the universal oriented bundle and pulled back. The obstruction-theoretic description identifies it with the primary obstruction to a section of the sphere bundle, and this is the statement that makes the name “Euler class” meaningful: when E is the tangent bundle, the obstruction to a nowhere-zero vector field is the Euler characteristic.

Theorem 3.4.2: Euler Class as Primary Obstruction

Let $p: E \rightarrow B$ be an oriented real vector bundle of rank n over a CW complex B . The primary obstruction to the existence of a nowhere-zero section of E is the Euler class:

$$[\sigma(s)] = e(E) \in H^n(B; \pi_{n-1}(S^{n-1})) \cong H^n(B; \mathbb{Z})$$

where the identification $\pi_{n-1}(S^{n-1}) \cong \mathbb{Z}$ is the one fixed by the orientation of E . In particular, if E admits a nowhere-zero section, then $e(E) = 0$.

The theorem says two things at once: that the obstruction cocycle of the sphere bundle, an a priori sections-theoretic invariant, agrees with the Euler class, an a priori cohomological invariant; and that the orientation is exactly the bookkeeping needed to read the obstruction in integral cohomology rather than mod 2.

Proof:

Step 1: The obstruction lives in the asserted group and the lower obstructions vanish. By definition 3.4.1 the sphere bundle $S(E) \rightarrow B$ has fiber S^{n-1} , which is path-connected for $n \geq 2$ and satisfies $\pi_i(S^{n-1}) = 0$ for $i < n-1$ and $\pi_{n-1}(S^{n-1}) \cong \mathbb{Z}$. (For $n = 1$ the fiber is S^0 and the statement is the orientability discussion of theorem 3.4.5 below; assume $n \geq 2$.) Thus S^{n-1} is $(n-2)$ -connected, and the obstruction machinery of [5] gives, first, that a section over the $(n-1)$ -skeleton B^{n-1} exists with no obstruction, since all the groups $H^{i+1}(B; \pi_i(S^{n-1}))$ vanish for $i+1 \leq n-1$; and second, that the first potentially nonzero obstruction to extending such a

section over B^n is the primary obstruction

$$[\mathfrak{o}(s)] \in H^n(B; \pi_{n-1}(S^{n-1}))$$

Because E is oriented, the fibers of $S(E)$ are coherently oriented spheres, so π_1 of the base acts trivially on $\pi_{n-1}(S^{n-1})$ through the fiber; the bundle of coefficient groups is the constant system $\mathbb{Z}$, and the orientation provides the canonical isomorphism $\pi_{n-1}(S^{n-1}) \cong \mathbb{Z}$. Hence the primary obstruction is well-defined and lies in $H^n(B; \mathbb{Z})$.

Step 2: Both invariants are natural transformations, hence given by universal classes. The assignment $E \mapsto [\mathfrak{o}(s)]$ is natural: if $f: B' \rightarrow B$ is cellular and $E' = f^*(E)$, then a section of $S(E)$ over a skeleton pulls back to a section of $S(E')$ over the corresponding skeleton, and the obstruction cochain pulls back cell by cell, so $[\mathfrak{o}(f^*E)] = f^*[\mathfrak{o}(E)]$. This is the naturality clause of the obstruction machinery. The Euler class e is also natural by definition 3.1.4 (it is a characteristic class, hence a natural transformation $\widetilde{\text{Vect}}^n \Rightarrow H^n(-; \mathbb{Z})$). Thus both $E \mapsto [\mathfrak{o}(s)]$ and $E \mapsto e(E)$ are natural transformations $\widetilde{\text{Vect}}^n(-) \rightarrow H^n(-; \mathbb{Z})$ on the homotopy category of CW complexes. Since $\widetilde{\text{Vect}}^n(-) \cong [-, \widetilde{\text{Grass}}_n]$ is represented by the oriented Grassmannian (theorem 1.2.13, oriented variant) and $H^n(-; \mathbb{Z}) \cong [-, K(\mathbb{Z}, n)]$ is represented by an Eilenberg-MacLane space, the Yoneda lemma identifies each natural transformation with a single *universal class* in $H^n(\widetilde{\text{Grass}}_n; \mathbb{Z})$, namely its value on the universal oriented bundle $E_n \rightarrow \widetilde{\text{Grass}}_n$. Two such transformations agree if and only if their universal classes agree. So it suffices to prove $[\mathfrak{o}(E_n)] = e(E_n)$ in $H^n(\widetilde{\text{Grass}}_n; \mathbb{Z})$.

Step 3: Compute both universal classes on the bottom sphere. The two universal classes are compared by pulling back along the maps $S^n \rightarrow \widetilde{\text{Grass}}_n$ classifying clutched bundles, which is enough because the Euler class normalization of definition 3.1.4 is fixed precisely by its restriction to such spheres: by construction e is the natural transformation whose value on the clutched bundle $E_g \rightarrow S^n$ (built via proposition 1.2.8 and its oriented variant from $g: S^{n-1} \rightarrow SO(n)$) is the degree of $\text{ev} \circ g$, where $\text{ev}: SO(n) \rightarrow S^{n-1}$ evaluates at a fixed unit vector. It therefore suffices to show $[\mathfrak{o}(E_g)]$ equals this same degree for every such g , since a natural transformation into $H^n(-; \mathbb{Z})$ agreeing with e on all clutched bundles over S^n has the same restriction as e to the image of $\pi_n(\widetilde{\text{Grass}}_n)$, and the universal class $e(E_n)$ is by its very definition determined by these restrictions. Now compute $[\mathfrak{o}(E_g)]$. A section of $S(E_g)$ over S^n is a choice, over each hemisphere $D_\pm^n$ where E_g is trivial as $D_\pm^n \times S^{n-1}$, of a map $D_\pm^n \rightarrow S^{n-1}$, compared over the equator. The obstruction cocycle evaluated on the single top cell of S^n is the difference class in $\pi_{n-1}(S^{n-1})$ of the two boundary maps $S^{n-1} \rightarrow S^{n-1}$ after applying the clutching identification; taking the constant section on each hemisphere, this difference is the homotopy class of the composite

$$S^{n-1} \xrightarrow{g} SO(n) \xrightarrow{\text{ev}} S^{n-1}$$

By definition of the degree, this homotopy class is the integer assigned to the top cell, so $[\mathfrak{o}(E_g)] \in H^n(S^n; \mathbb{Z}) \cong \mathbb{Z}$ is the degree of $\text{ev} \circ g$.

Step 4: Match the two integers. For every clutched bundle $E_g \rightarrow S^n$ the obstruction integer of Step 3 equals the degree of $\text{ev} \circ g$, which by the normalization recalled in Step 3 is exactly $e(E_g)$. Hence $[\mathfrak{o}(E_g)] = e(E_g)$ for all g , so the universal classes of $[\mathfrak{o}(-)]$ and e have the same restriction along every map $S^n \rightarrow \widetilde{\text{Grass}}_n$ classifying a clutched bundle; as these restrictions determine $e(E_n)$, the two universal classes coincide, $[\mathfrak{o}(E_n)] = e(E_n)$. By Step 2, $[\mathfrak{o}(s)] = e(E)$ for all oriented rank- n bundles.

Step 5: Conclusion. If E admits a nowhere-zero section, then $S(E)$ admits a global section, so in particular the section over B^{n-1} extends over B^n , forcing the primary obstruction to vanish; therefore $e(E) = [\sigma(s)] = 0$. This gives the displayed identification and the stated consequence, as we sought to show.

The vanishing of the Euler class is necessary but not sufficient for a nowhere-zero section, because $e(E) = 0$ only guarantees that the section extends over the n -skeleton; further obstructions, the secondary obstructions of [5], may block extension over higher skeleta. For $\text{rk} E = \dim B$ the n -skeleton is everything and the Euler class is a complete obstruction, which is the case relevant to vector fields on closed manifolds.

Corollary 3.4.3: Euler Class of the Tangent Bundle

Let M be a closed oriented smooth n -manifold and let TM be its tangent bundle, oriented by the orientation of M . Then evaluating the Euler class on the fundamental class gives the Euler characteristic,

$$\langle e(TM), [M] \rangle = \chi(M)$$

Consequently, if M admits a nowhere-zero vector field then $\chi(M) = 0$.

The corollary is the Poincaré-Hopf theorem in the language of characteristic classes: the obstruction to combing a manifold (finding a nowhere-zero tangent vector field) is a single integer, the Euler characteristic, and that integer is the Euler class evaluated on the fundamental class.

Proof:

Proved in [5] via the Poincaré-Hopf theorem: the Euler class of the tangent bundle pairs with the fundamental class to give the Euler characteristic.

Example 3.4.4: The Tangent Bundle of S^{2n} Has No Nowhere-Zero Section

Let $M = S^{2n}$ with its standard orientation, $n \geq 1$. The Euler characteristic of an even-dimensional sphere is

$$\chi(S^{2n}) = 1 + (-1)^{2n} = 2$$

(its homology is $\mathbb{Z}$ in degrees 0 and $2n$ and 0 otherwise). By corollary 3.4.3,

$$\langle e(TS^{2n}), [S^{2n}] \rangle = \chi(S^{2n}) = 2 \neq 0$$

so $e(TS^{2n})$ is twice a generator of $H^{2n}(S^{2n}; \mathbb{Z}) \cong \mathbb{Z}$ and in particular is nonzero. By theorem 3.4.2 the tangent bundle TS^{2n} admits no nowhere-zero section, that is, S^{2n} carries no nowhere-vanishing vector field. For $n = 1$ this is the hairy-ball theorem recalled after definition 1.1.9, now obtained from the Euler class.

The contrast with odd spheres is sharp: $\chi(S^{2n+1}) = 1 + (-1)^{2n+1} = 0$, consistent with the standard nowhere-zero field $x \mapsto (-x_2, x_1, -x_4, x_3, \dots)$ on $S^{2n+1} \subset \mathbb{R}^{2n+2}$. Here the vanishing of χ , hence of the integral pairing, is necessary for the field to exist, and on spheres it is also sufficient because the only obstruction is the primary one.

The Euler class required an orientation even to be defined. The obstruction that decides whether an orientation exists at all is the first Stiefel-Whitney class. Orientability of a rank- n bundle is a

reduction of the structure group from $O(n)$ to $SO(n)$, equivalently a reduction of the associated fiber datum from the two-point set $O(n)/SO(n) = \{\pm 1\}$ to a point; the obstruction to such a reduction is a class in $H^1(B; \mathbb{Z}/2\mathbb{Z})$.

Theorem 3.4.5: First Stiefel-Whitney Class as Orientability Obstruction

Let $p: E \rightarrow B$ be a real vector bundle of rank n over a path-connected CW complex B . Then E is orientable if and only if $w_1(E) = 0$ in $H^1(B; \mathbb{Z}/2\mathbb{Z})$. More precisely, $w_1(E)$ is the primary obstruction to reducing the structure group of E from $O(n)$ to $SO(n)$, and the orientations of E , when they exist, form a torsor under $H^0(B; \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$.

The first Stiefel-Whitney class is thus the complete and only obstruction to orientability, with no higher obstructions, because the relevant fiber $O(n)/SO(n)$ is a single point set with a free $\mathbb{Z}/2\mathbb{Z}$ -action and has no higher homotopy. The class records, loop by loop, whether transporting an orientation of the fiber around the loop returns it to itself or reverses it.

Proof:

Proved in [5]: $w_1(E)$ is the primary obstruction to orientability, the class in $H^1(B; \mathbb{Z}/2\mathbb{Z})$ of the orientation double cover, which admits a section (equivalently E is orientable) exactly when $w_1(E) = 0$.

Example 3.4.6: The Möbius Bundle Is Non-Orientable

Let $E \rightarrow S^1$ be the Möbius bundle (example 1.1.4), the rank-1 bundle $I \times \mathbb{R}/\sim$ with (s, t) for $s \in I = [0, 1]$ the base coordinate and $t \in \mathbb{R}$ the fiber coordinate, glued by $(0, t) \sim (1, -t)$. Transporting a chosen fiber orientation once around the base circle reverses it: an orientation is a choice of sign on $\mathbb{R}$, and traversing the base from $s = 0$ to $s = 1$ identifies the fiber at $s = 1$ with the fiber at $s = 0$ through $t \mapsto -t$, which carries the orientation $+$ to $-$. Hence the homomorphism $\pi_1(S^1) = \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ of theorem 3.4.5 sends the generator to 1, so it is the nonzero homomorphism and

$$w_1(E) \neq 0 \quad \text{in} \quad H^1(S^1; \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$$

Therefore E is non-orientable. Equivalently, the orientation double cover $\text{Or}(E) \rightarrow S^1$ is the connected double cover $S^1 \xrightarrow{z \mapsto z^2} S^1$, which has no section. This recovers, through the orientability obstruction, the fact recorded after definition 1.1.9 that the Möbius bundle is not isomorphic to the trivial bundle $S^1 \times \mathbb{R}$: a trivial bundle is orientable, while $w_1(E) \neq 0$.

By contrast, the trivial line bundle has $w_1 = 0$ and the cylinder $S^1 \times \mathbb{R}$ admits the constant orientation; the two bundles are distinguished by $w_1 \in H^1(S^1; \mathbb{Z}/2\mathbb{Z})$, which takes both of its values.

Once a bundle is orientable, so $w_1(E) = 0$, the next reduction of the structure group is from $SO(n)$ to its connected double cover $\text{Spin}(n)$. A spin structure is exactly a lift of the structure group along $\text{Spin}(n) \rightarrow SO(n)$, and the obstruction to such a lift is the second Stiefel-Whitney class. The relevant standard fact (proved in [5], and for the Lie-theoretic input [7]) is that for $n \geq 3$ the group $SO(n)$ has $\pi_1(SO(n)) \cong \mathbb{Z}/2\mathbb{Z}$, and $\text{Spin}(n)$ is its universal (double) cover.

Definition 3.4.7: Spin Structure

Let $E \rightarrow B$ be an oriented real vector bundle of rank $n \geq 3$, with principal $SO(n)$ -bundle of oriented orthonormal frames $P_{SO} \rightarrow B$. A *spin structure* on E is a principal $\text{Spin}(n)$ -bundle $P_{\text{Spin}} \rightarrow B$ together with a map $P_{\text{Spin}} \rightarrow P_{SO}$ over B that restricts on each fiber to the double cover $\text{Spin}(n) \rightarrow SO(n)$.

A spin structure is a coherent, globally consistent way of taking the “square root” of the rotation group acting on the fibers; the obstruction to assembling the local square roots into a global one is a mod-2 class in degree two, and the content of the next theorem is that this class is $w_2(E)$.

Theorem 3.4.8: Second Stiefel-Whitney Class as Spin Obstruction

Let $E \rightarrow B$ be an oriented real vector bundle of rank $n \geq 3$ over a CW complex B . Then E admits a spin structure if and only if $w_2(E) = 0$ in $H^2(B; \mathbb{Z}/2\mathbb{Z})$. When spin structures exist, they form a torsor under $H^1(B; \mathbb{Z}/2\mathbb{Z})$.

Proof:

Proved in [5]: $w_2(E)$ is the primary obstruction to a spin structure, the class in $H^2(B; \mathbb{Z}/2\mathbb{Z})$ of the lifting problem along $\text{Spin}(n) \rightarrow SO(n)$, and the spin structures form a torsor under $H^1(B; \mathbb{Z}/2\mathbb{Z})$ when it vanishes.

The pattern is now visible: orientability removes w_1 , a spin structure removes w_2 , and each removal is a reduction of the structure group whose obstruction is the corresponding Stiefel-Whitney class. The Lie-theoretic ceiling on this tower is set by the homotopy of $SO(n)$: since $\pi_2(SO(n)) = 0$ there is no obstruction in degree three, and the next genuine obstruction appears in degree four, governed by $\pi_3(SO(n)) \cong \mathbb{Z}$, which is where the first Pontryagin-type class lives, consistent with the discussion preceding theorem 3.2.3.

Example 3.4.9: Spin Structures on Surfaces and on $\mathbb{CP}^2$

1. Let Σ be a closed orientable surface and $E \rightarrow \Sigma$ an oriented rank- n bundle with $n \geq 3$. Since $H^2(\Sigma; \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$, the bundle admits a spin structure if and only if the single class $w_2(E) \in \mathbb{Z}/2\mathbb{Z}$ vanishes. For the (stabilized) tangent bundle, $w_2(T\Sigma)$ is the mod-2 reduction of the Euler class, and $\langle e(T\Sigma), [\Sigma] \rangle = \chi(\Sigma) = 2 - 2g$ is even, so $w_2(T\Sigma) = 0$ and every orientable surface is spin. The spin structures then form a torsor under $H^1(\Sigma; \mathbb{Z}/2\mathbb{Z}) \cong (\mathbb{Z}/2\mathbb{Z})^{2g}$, giving 2^{2g} of them, matching the classical count.
2. Consider $\mathbb{CP}^2$, whose tangent bundle (as a rank-4 real bundle) has total Stiefel-Whitney class computed from the complex structure. The mod-2 reduction of the total Chern class of $T\mathbb{CP}^2$ is $(1+a)^3 = 1 + 3a + 3a^2 = 1 + a + a^2$ in $H^*(\mathbb{CP}^2; \mathbb{Z}/2\mathbb{Z})$, where a is the generator of $H^2(\mathbb{CP}^2; \mathbb{Z}/2\mathbb{Z})$. By proposition 3.2.6, $w_2(T\mathbb{CP}^2)$ is the mod-2 reduction of $c_1(T\mathbb{CP}^2) = 3a$, namely $a \neq 0$. Hence $w_2(T\mathbb{CP}^2) \neq 0$ and $\mathbb{CP}^2$ admits no spin structure on its tangent bundle, a closed orientable 4-manifold that is not spin.

The complex story parallels the real one with the simplification that $\text{GL}_n(\mathbb{C})$, hence $U(n)$, is connected: there is no orientability obstruction, every complex bundle is canonically oriented as a real bundle,

and the first interesting obstruction to a section appears at the top. For a complex bundle of rank n the unit sphere bundle has fiber S^{2n-1} , which is $(2n-2)$ -connected with $\pi_{2n-1}(S^{2n-1}) \cong \mathbb{Z}$, so the primary obstruction lives in $H^{2n}(B; \mathbb{Z})$, the degree of the top Chern class.

Theorem 3.4.10: Top Chern Class as Section Obstruction

Let $E \rightarrow B$ be a complex vector bundle of rank n over a CW complex B . The primary obstruction to the existence of a nowhere-zero section of E is the top Chern class

$$c_n(E) \in H^{2n}(B; \pi_{2n-1}(S^{2n-1})) \cong H^{2n}(B; \mathbb{Z})$$

In particular, if E admits a nowhere-zero section, then $c_n(E) = 0$. Equivalently, $c_n(E) = e(E_{\mathbb{R}})$, the Euler class of the underlying oriented real rank- $2n$ bundle $E_{\mathbb{R}}$.

Proof:

Proved in [5]: the top Chern class is the primary obstruction to a nowhere-zero section of a complex bundle, equal to the Euler class of the underlying oriented real bundle.

The top Chern class therefore plays, for complex bundles, exactly the role the Euler class plays for oriented real bundles, and the identity $c_n(E) = e(E_{\mathbb{R}})$ is the precise form of the slogan that the top Chern class is the complex Euler class. Lower Chern classes $c_k(E)$ for $k < n$ are, in the same spirit, the obstructions to finding $n - k + 1$ linearly independent sections, governed by the homotopy of complex Stiefel manifolds, though the higher obstructions there are no longer always primary.

Example 3.4.11: The Top Chern Class of a Bundle With a Section

Let $E \rightarrow B$ be a complex rank- n bundle that splits off a trivial line summand, $E \cong E' \oplus \mathbb{C}$ for some complex rank- $(n-1)$ bundle E' . The trivial summand supplies a nowhere-zero section (the constant section $b \mapsto (0, 1)$), so by theorem 3.4.10,

$$c_n(E) = 0$$

This also follows directly from the Whitney formula: $c(\mathbb{C}) = 1$, so $c(E) = c(E') \smile 1 = c(E')$, and $c(E')$ has no component in degree $2n$ because E' has complex rank $n-1$, forcing $c_n(E) = 0$. The two computations agree, illustrating that the obstruction-theoretic vanishing and the multiplicative vanishing are the same phenomenon.

A concrete instance is the tangent bundle of $\mathbb{CP}^n$ via the Euler sequence $0 \rightarrow \mathbb{C} \rightarrow H^{\oplus(n+1)} \rightarrow T\mathbb{CP}^n \rightarrow 0$, where H is the hyperplane bundle: here $c_n(T\mathbb{CP}^n)$ need not vanish (indeed $\langle c_n(T\mathbb{CP}^n), [\mathbb{CP}^n] \rangle = \chi(\mathbb{CP}^n) = n+1 \neq 0$), and correspondingly $T\mathbb{CP}^n$ has no nowhere-zero section, recovering that $\mathbb{CP}^n$ has nonzero Euler characteristic.

Exercise 3.4.1

1. Let $E \rightarrow B$ be an oriented real rank- n bundle over a CW complex with $\dim B < n$. Using theorem 3.4.2, show that E admits a nowhere-zero section and that $e(E) = 0$. Explain why this is consistent with $e(E) \in H^n(B; \mathbb{Z})$.
2. Compute w_1 of the Möbius bundle pulled back along the degree-2 map $f: S^1 \rightarrow S^1$, $z \mapsto z^2$, and decide whether the pullback is orientable. (Use theorem 3.4.5 and naturality.)

3. Let M be a closed oriented 3-manifold. Show that $\chi(M) = 0$ and conclude that M admits a nowhere-zero vector field. Where does the argument use corollary 3.4.3, and where does it use that the only obstruction is primary?
4. For an oriented real rank-2 bundle $E \rightarrow B$, show directly from theorem 3.4.2 and theorem 3.4.5 that the mod-2 reduction of $e(E)$ equals $w_2(E)$.
5. Let $E \rightarrow B$ be a complex rank- n bundle and suppose $c_n(E) \neq 0$. Prove that E is not isomorphic to a bundle of the form $E' \oplus \mathbb{C}$, and that the underlying real bundle $E_{\mathbb{R}}$ has no nowhere-zero section. Which theorem of this section gives each conclusion?
6. Explain why $\mathbb{CP}^2$ admits no spin structure on its tangent bundle while S^4 does, using theorem 3.4.8. Compute the number of spin structures on S^4 .